

Lecture notes on

Quantum Field Theory

(Based on the lectures by *Dr. M Arshad Momen*)

These lecture notes are based on the *Quantum Field Theory* course taught
by

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Please note that these are working drafts and may contain typographical
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free to contact us.

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(Class-1) Lecture (-2) : Thrs, Apr 17, 2025

Introductory class**1.1 Statistical Mechanics****Ising Model: Insight**

The expression:

$$H = -J \sum_{\langle i,j \rangle} S_i S_j$$

is the **Hamiltonian** of a physical system.
Here,

- $H \rightarrow$ Represents the total energy of the system.
- $J \rightarrow$ This is called the **coupling constant**, that determines the interaction strength between spins.
 1. If $J > 0$, H represents a system of **ferromagnetic** material ($\uparrow\uparrow\uparrow$) $\rightarrow$ Aligned in the same direction.
 2. If $J < 0$, it's **antiferromagnetic** ($\uparrow\downarrow\uparrow$) $\rightarrow$ Alternately aligned.

Question: How to know whether a system is ferro or antiferromagnetic?

Ground State
 $\downarrow$ (Excited)
 Exhibit collective excitations
 $\downarrow$
 Behave Like Quasiparticles

Quasiparticle

It is a collective emergent excitation in a many-body system that behaves like a particle. They are not real particles but they have energy, momentum, and spin.

Magnons

1. Magnons are often considered quasiparticles, which are the excitation of a many-body system in ferromagnetic materials.
2. When a ferromagnetic system is disturbed from its equilibrium position, the aligned magnetic moments can wobble or oscillate around their equilibrium direction, creating a wave-like pattern of spin changes.
3. The spin waves are quantized, meaning they represent the existence of discrete packets of energy. These packets are called **Magnons**.

Dispersion Relation

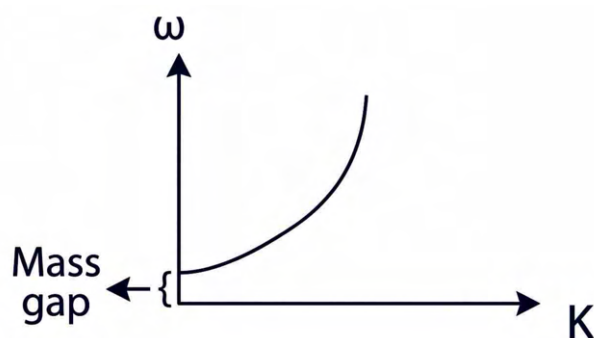
Describes the relationship between the frequency (ω) and the wave vector (k) of a wave.

$$p = \hbar k, \quad E = \hbar \omega$$

Massive

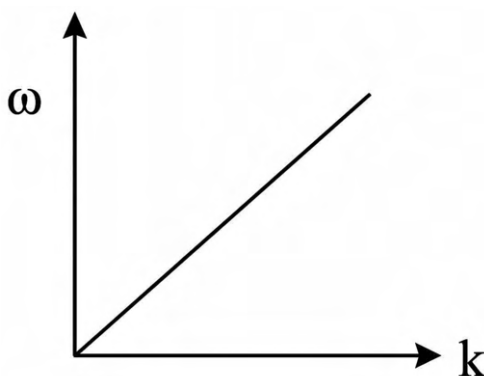
- Minimum energy required for an entity to exist comes from the equation:

$$E = mc^2, \quad \rightarrow \quad \text{This denotes the **mass gap**}$$



Massless

- The relationship is linear.
- For photon: $\omega = ck$
- True for every relativistic particle.



1.2 Quantum Mechanics

Density matrix is a more general property than the wave function ψ to represent or describe the state of a quantum system. It arises from an alternative interpretation of the first postulate of quantum mechanics, which asserts that ψ describes the state of a system.

Fundamental Postulates of Quantum Mechanics:

1. Physical observables are represented by operators. Operators map to real numbers.
2. States are mathematical objects that assign real expectation values to those operators.

Conditions to be operators:

$$\omega(\mathbb{I}) = 1 \quad , \quad \omega(A^\dagger A) \geq 0$$

Question: What is algebra?

An algebra is a vector space equipped with a product operation between the elements, and the vector space is closed under that product operation.

Angular Momentum Operators:

- $[S_1, S_2] = i\hbar S_3$ (finite-dimensional representation)
- $[L_1, L_2] = i\hbar L_3$ (infinite-dimensional representation)

These are different representations of the same underlying algebraic structures, i.e., representation of a bigger space acting on different vector spaces.

Probability Interpretation

$\psi \rightarrow$ Probability Density

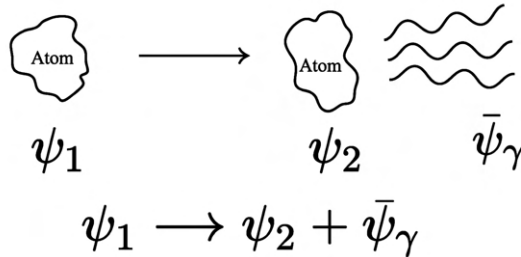
$$\int |\psi|^2 d\tau = \text{constant} = 1$$

Not always right, especially when particle number is not conserved.

This condition is violated when an atom undergoes a decay process, such as:

$$\text{Atom} \rightarrow \text{Atom} + \bar{\psi}_\gamma$$

$$\psi_I \rightarrow \psi_{II} + \bar{\psi}_\gamma$$



The conservation of total probability in such transitions implies:

$$\int |\psi_I|^2 d\tau = \int (|\psi_{II}|^2 + |\bar{\psi}_\gamma|^2) d\tau$$

Two Cornerstones of 20th Century Physics:

- **Special Relativity:** $E = mc^2$
- **Quantum Mechanics:** $\Delta p \Delta x \geq \hbar/2$

However, these two principles are not inherently compatible. For instance, taking the uncertainty principle to the limit:

$$\Delta x \rightarrow 0 \Rightarrow \Delta p \rightarrow \infty$$
$$\text{As, } \Delta E = \frac{(\Delta p)^2}{2m} \rightarrow \infty$$

Energy uncertainty generates multi-particles. This leads to new phenomena, such as particle creation, which cannot be fully explained by either theory alone — highlighting the need for **Quantum Field Theory**.

Reading Assignments:

Reading Assignments:

- Von Neumann — *Mathematical Foundations of Quantum Mechanics*,
(for mathematical formalism and axiomatic treatment.)
- Jackson — *Classical Electrodynamics*, Chapters 11 and 12

(Class-2) Lecture (-1): Sat, Apr 19, 2025

Review of Special Relativity

- Collection of events: $\{x\} = \mathcal{M} \rightarrow$ set of events in spacetime
- An event can be labelled by 4 coordinates: x^a , $a = 0, 1, 2, 3$

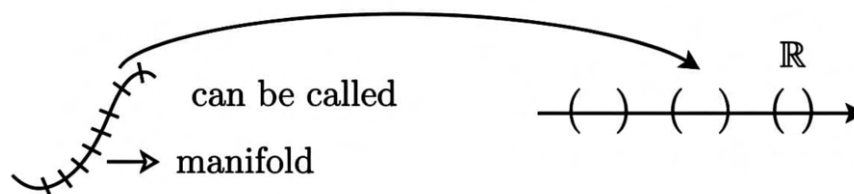
2.1 Manifold

Spacetime is endowed with a manifold structure $\Rightarrow$ means that: spacetime, although it can be curved and complex on a large scale, is treated mathematically as a **manifold** — a space that, in small patches, looks like regular, flat space and time.

Definition

A mathematical space on which calculus can be consistently defined. It is typically covered by multiple coordinate charts (or ‘patches’) that together form an “atlas”.

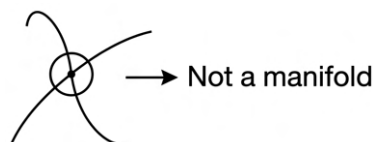
Spacetime $\rightarrow$ Manifold ($\mathbb{R}^4$)



Each of the local neighborhoods in a manifold is isomorphic to line segments in Euclidean space.

- **Each of the local neighborhoods:** refers to small regions around each point on the manifold.
- **Isomorphic to line segments:** means that they have the same structure, in the sense that we can map one onto the other without losing information.

On the other hand,



Because if there are ‘cuts’ in spacetime — like tears, edges or points where the structure breaks down — then spacetime isn’t a proper manifold at those points.

Manifold Requirements

A manifold must satisfy:

1. **Locally Euclidean:** Locally looks like flat space ($\mathbb{R}^4$).
2. **Smooth or differentiable:** Calculus can be done.

3. **No breaks or cuts:** The space must be continuous and connected in a smooth way.

Therefore, when there are ‘cuts’, then at those points, the spacetime fails to be a manifold because smooth coordinates can’t be defined or calculus can’t be done nicely there.

Difference between SR and GR

- **SR** needs only one single patch. One single patch can cover everything.

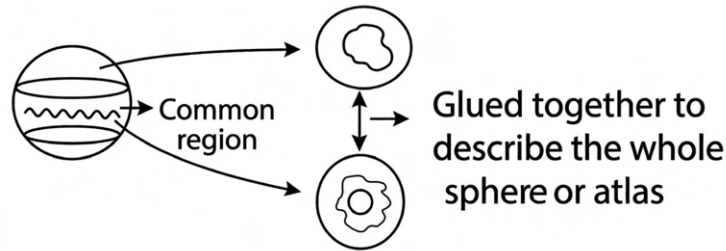
(A single patch = one coordinate system that works in a specific region of spacetime)

- **GR** needs multiple patches to cover everything.

Example: Coordinate Patches on a Sphere

To describe a sphere (specifically a 2D sphere like the surface of the Earth), at least 2 coordinate patches (charts) are needed to smoothly cover the entire surface.

A single coordinate system works everywhere except at the poles.



Extra:

In N dimensions, we need ${}^N C_2$ angles to describe the coordinate system.

Example: In 10 dimensions, ${}^{10} C_2 = 45$ angles are needed.

- ${}^{10} C_2$ because we need 2 vectors to define a plane in space.
- Rotation is defined in a plane.

Now, the spacetime interval (or “distance-squared”) between two events: Any distance-squared between two points in a space with some metric can be written as a bilinear form:

$$\begin{aligned} d^2(x, y) &= (x - y)^T H (x - y) \\ &= (x - y)^a \eta_{ab} (x - y)^b \\ &= [(x - y)^0]^2 - (\vec{x} - \vec{y}) \cdot (\vec{x} - \vec{y}) \end{aligned}$$

We use **East-coast convention**:

$$H \equiv \eta_{ab} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

In Special Relativity (SR), η is fixed and globally defined. However, in General Relativity (GR), the metric becomes a function of position.

The speed of information transfer is limited by the speed of light.

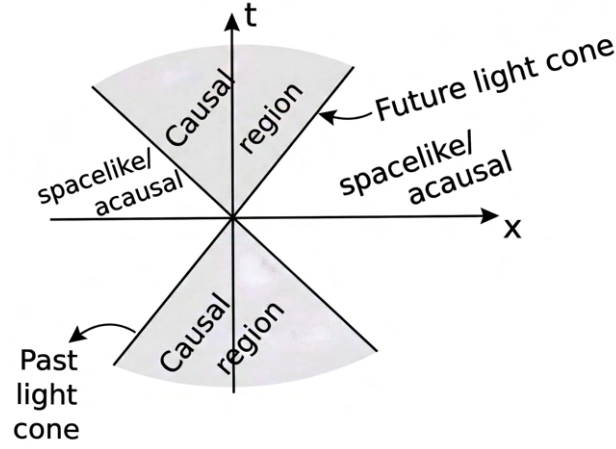


Figure: Light Cone

Constants and Units:

$$c = 1 \Rightarrow [L] = [T]$$

$$E = mc^2 \Rightarrow E = m$$

$$\hbar = 1 \Rightarrow \lambda = \frac{\hbar}{mc} \sim \frac{1}{E}$$

$$\therefore [M], [L], [T] \equiv E, \frac{1}{E}, \frac{1}{E}$$

By convention, we will measure everything in energy scales.

Postulate 1 :

Laws of nature must be such that all (...^a) observers^b will experience the same causal phenomena.

^aSomething else will sit here

^bOne who records— always tied to the coordinate system.

This postulate implies that laws must be written using **tensors** (they are independent of the coordinate system).

Galilean Transformations:

$$x' = x - vt$$

- In frame $S \Rightarrow m \frac{d^2x}{dt^2} = f = ma$
- In frame $S' \Rightarrow m \frac{d^2x'}{dt'^2} = f'$

Under the assumptions: $t' = t, \quad m' = m$

And, since $x' = x - vt$,

$$\begin{aligned} f' &= m \frac{d^2x'}{dt'^2} \\ \Rightarrow f' &= m \frac{d^2(x - vt)}{dt^2} \\ \Rightarrow f' &= m \frac{dv}{dt} \quad [\because \frac{d^2}{dt^2}(-vt) = 0] \\ \Rightarrow f' &= ma \\ \therefore f' &= f \end{aligned}$$

This invariance is a “symmetry” of the Newtonian laws.

Lorentz Transformations

Lorentz transformations are linear transformations that keep the “distance squared” unchanged.

$$d^2(x) = x^T H x$$

$$x^a \rightarrow x'^a = \Lambda^a_b x^b \quad [\text{not } \Lambda^a_b x^b]$$

$$\begin{pmatrix} x'^0 \\ x'^1 \\ x'^2 \\ x'^3 \end{pmatrix} = \begin{pmatrix} \Lambda^0_0 & \Lambda^0_1 & \Lambda^0_2 & \Lambda^0_3 \\ \Lambda^1_0 & \Lambda^1_1 & \Lambda^1_2 & \Lambda^1_3 \\ \Lambda^2_0 & \Lambda^2_1 & \Lambda^2_2 & \Lambda^2_3 \\ \Lambda^3_0 & \Lambda^3_1 & \Lambda^3_2 & \Lambda^3_3 \end{pmatrix} \begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix}$$

Rotations

$$\text{If } x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, \text{ then: } x^T = (x_1 \quad x_2 \quad x_3)$$

Hence:

$$x^T x = (x_1 \quad x_2 \quad x_3) \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = x_1^2 + x_2^2 + x_3^2 = r^2$$

Now, under rotation:

$$x \rightarrow x' = R x$$

Then,

$$\begin{aligned}(x')^T x' &= (Rx)^T (Rx) \\ &= x^T R^T R x \\ &= x^T x\end{aligned}$$

Therefore:

$$\therefore \boxed{R^T R = R^T \mathbb{I} R = \mathbb{I}}$$

Similarly, under Lorentz transformation:

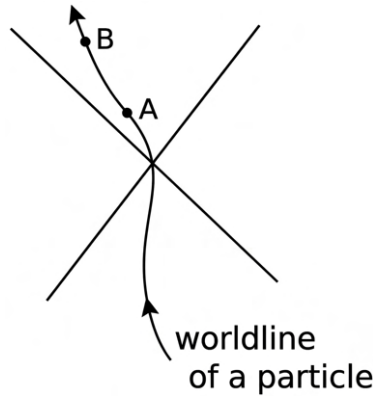
$$\begin{aligned}d^2(x') &= x'^T H x' \\ &= (\Lambda x)^T H (\Lambda x) \\ &= x^T \Lambda^T H \Lambda x \\ &= x^T H x\end{aligned}$$

$$\therefore \boxed{\Lambda^T H \Lambda = H}$$

Just as $R^T R = \mathbb{I}$ defines orthogonal transformations, $\Lambda^T H \Lambda = H$ defines Lorentz transformations.

$$\boxed{[R^T R = I] \subset [\Lambda^T H \Lambda = H] \text{ are the examples of classical groups.}}$$

Dynamics



- Let $A \rightarrow x = (x^0, x^1, x^2, x^3)$
- $B \rightarrow x + dx = (x^0 + dx^0, \dots, x^3 + dx^3)$

Hence,

$$d^2(A, B) = (dx^0)^2 - d\vec{x} \cdot d\vec{x} = d\tau^2$$

For a comoving observer¹, we have

$$d\vec{x} = 0$$

$$\text{thus: } d\tau^2 = (dx^0)^2$$

(Time dilation simply arises from this equation)

¹An observer who is at rest relative to the object or system being studied — meaning their relative velocity is zero.

A numerical explanation:

- If $d^2(A, B) \rightarrow \text{constant}$ (let it be 10), then:

$$d\tau^2 \rightarrow 10 \quad (\text{comoving}) \quad [\because d\vec{x} \cdot d\vec{x} = 0]$$

- If not comoving, say $d\vec{x} \cdot d\vec{x} = 3$, then to keep $d^2(A, B) = 10$, we need to have:

$$(dt)^2 = d^2(A, B) + d\vec{x} \cdot d\vec{x} = 10 + 3 = 13$$

$$\text{Thus, } \boxed{(dt)^2 \neq (d\tau)^2}$$

Proper Time (τ)

τ = Time measured by an observer moving along the worldline
 = Time measured by a co-moving observer

$$\begin{aligned} d\tau^2 &= dt^2 - d\vec{x} \cdot d\vec{x} \\ \Rightarrow 1 &= \left(\frac{dt}{d\tau}\right)^2 - \left(\frac{d\vec{x}}{d\tau}\right)^2 \\ &\quad \left[\text{here, } u^a = \left(\frac{dt}{d\tau}, \frac{d\vec{x}}{d\tau}\right) \right] \\ \Rightarrow u^a u_a &= u \cdot u = 1 \end{aligned}$$

As we know,

$$\boxed{u \cdot u \begin{cases} > 0 & ; \text{ Timelike} \\ < 0 & ; \text{ Spacelike} \\ = 0 & ; \text{ Lightlike} \end{cases}}$$

Then:

$$u \cdot u = 1 \quad \therefore \text{4-velocity is timelike}$$

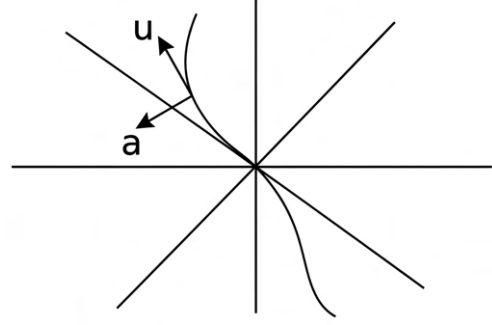
Significance

1. The worldline of a massive object always stays inside the light cone — it can never go faster than light.
2. The object is always moving forward in time according to its own clock.
3. If it were spacelike, that would imply faster-than-light motion.

$$\begin{aligned} \text{Now, } u \cdot u &= 1 \\ \Rightarrow \frac{du^a}{dt} u_a + \frac{du_a}{dt} u^a &= 0 \\ \Rightarrow \frac{du^a}{dt} u_a &= 0 \\ \Rightarrow a^b u_b &= 0 \end{aligned}$$

$\Rightarrow$ 4-acceleration $\perp$ 4-velocity

$\therefore$ 4-acceleration is spacelike



Significance:

1. It is not contributing to your time flow.
2. It changes spatial direction through spacetime.
3. The 4-acceleration only bends the path, it doesn't speed up or slow down in the time direction.

Lorentz Force

Relativistic generalization of Newton's second law:

$$f^b = ma^b \quad (\text{Four-force})$$

In an electromagnetic field:

$$\begin{aligned}
 a^b &= F^b_c u^c \\
 \Rightarrow a_b &= F_{bc} u^c && [\text{Lowering the index}] \\
 \Rightarrow a_b u^b &= F_{bc} u^b u^c && [\text{Contracting with } u^b] \\
 \Rightarrow F_{bc} u^b u^c &= 0 && [\text{Since } a^b u_b = 0] \\
 \Rightarrow F_{bc} u^b u^c + F_{bc} u^b u^c &= 0 \\
 \Rightarrow F_{bc} u^b u^c + F_{cb} u^c u^b &= 0 \\
 \Rightarrow F_{bc} u^b u^c + F_{cb} u^b u^c &= 0 \quad \{ \because [u^b, u^c] = 0 \} \\
 \Rightarrow (F_{bc} + F_{cb}) u^b u^c &= 0 \\
 \therefore F_{bc} &= -F_{cb}
 \end{aligned}$$

The field strength tensor is antisymmetric.

The classical (3-vector) form of the Lorentz force law:

$$m\vec{a} = q(\vec{E} + \vec{v} \times \vec{B})$$

Relativistic form:

$$m \frac{du^a}{d\tau} = q F^a_b u^b$$

Field strength tensor F^{ab} :

$$F^{ab} \equiv \begin{bmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & -B_z & B_y \\ E_y & B_z & 0 & -B_x \\ E_z & -B_y & B_x & 0 \end{bmatrix}$$

Action Principle

Action, $S \Rightarrow \delta S = 0$ [minimum]

(Observer independent)

$\Downarrow$ consequence

Euler-Lagrange (EL) equations.

The action should be invariant under coordinate transformation.

$$\begin{aligned} S &\propto \text{proper length of worldline} \\ \Rightarrow S &\propto \int d\tau \\ \Rightarrow S &= m \int d\tau \end{aligned}$$

(where m is not relativistic mass, no relativistic mass exists.)

Action must be scalar, no change will happen under transformation.

Dimensional Analysis of Action:

- S is dimensionless
- $d\tau$ has units of length
- m has units of inverse length

$$\begin{aligned} S[x] &= m \int \sqrt{\eta_{ab} dx^a dx^b} = m \int d\tau \\ &= m \int \sqrt{\eta_{ab} \frac{dx^a}{d\lambda} \frac{dx^b}{d\lambda}} d\lambda \rightarrow (\text{Reparameterization invariant}) \end{aligned}$$

($d\lambda$ can be any time, not necessarily just proper time)

HW: Derive the Euler-Lagrange equations corresponding to this action (Note: The Hamiltonian of this system is zero.)

(Class-3) Lecture 0: Thrs, Apr 24, 2025

Preliminary QFT**3.1 Action for a Point Particle (Massive)**

The action is given by

$$S = m \int d\tau \quad (1)$$

where $d\tau$ can be anything, just with respect to a comoving observer.

Expanding,

$$\begin{aligned} S &= m \int \sqrt{\eta_{ab} dx^a dx^b} \\ &= m \int \sqrt{\eta_{ab} \frac{dx^a}{d\lambda} \frac{dx^b}{d\lambda}} d\lambda \quad \text{where, } \lambda : \text{ personal time (subjective)} \\ &= \int \mathcal{L}(x, \dot{x}) d\lambda \quad \tau : \text{ objective time} \end{aligned} \quad (2)$$

Now,

$$\int_a^b dy = y(b) - y(a)$$

represents coordinates dy can also be written as

$$dy = \frac{df}{dy} dy$$

Therefore it is geometrical in nature and hence reparameterization invariant.

From equation (2), the conjugate momenta will be -

$$\begin{aligned} p_a &= \frac{\partial \mathcal{L}}{\partial \dot{x}^a} \\ \Rightarrow p_a &= \frac{\partial}{\partial \dot{x}^a} \left[m \sqrt{\eta_{ab} \dot{x}^a \dot{x}^b} \right] \\ \Rightarrow p_a &= m \frac{(\eta_{ab} \dot{x}^b + \eta_{ab} \dot{x}^a \delta_a^b)}{2\sqrt{\dot{x}^b \dot{x}_b}} \end{aligned} \quad (3)$$

$$\begin{aligned} &[\because \frac{\partial \dot{x}^b}{\partial \dot{x}^a} = \delta_a^b] \\ \Rightarrow p_a &= m \frac{(\dot{x}_a + \dot{x}_a)}{2\sqrt{\dot{x}^b \dot{x}_b}} \\ \therefore p_a &= \frac{m \dot{x}_a}{\sqrt{\dot{x}^b \dot{x}_b}} \end{aligned} \quad (4)$$

Note: How are p and v related?

The trivial answer is $p = mv$, but more generally,

$$p_a = \frac{\partial \mathcal{L}}{\partial \dot{x}^a}.$$

Here, p_a is the cotangent vector, i.e., the dual of the tangent vector $(\dot{x}^a)$.

Now, the Hamiltonian is:

$$\begin{aligned}
 H &= p_a \dot{x}^a - \mathcal{L} \\
 &= \frac{m \dot{x}_a}{\sqrt{\dot{x}^b \dot{x}_b}} \dot{x}^a - m \sqrt{\dot{x}_c \dot{x}^c} \\
 &= \frac{m r^2}{\sqrt{r^2}} - m \sqrt{r^2} \quad [\because \dot{x}_a \dot{x}^a = r^2] \\
 &= m \sqrt{r^2} - m \sqrt{r^2} \\
 &= 0
 \end{aligned}$$

Significance:

As the Hamiltonian is the generator of time, and for this system it is zero implies that the worldline is fixed. The history of the particle (i.e., the trajectory) is known. There is no evolution of the system, no dynamics of the system. Only spacetime exists. Dynamics is the construction of spacetime itself in our head.

Equation of Motion and Phase Space

The hallmark difference between Hamilton's and Lagrangian's equations of motion is that Hamilton's equations are first order in time, whereas those of Lagrange's are second order in time.

Example:

1.

$$\frac{d\vec{S}}{d\tau} = \mu \vec{S} \times \vec{B} \longrightarrow \text{Hamilton's equation of motion, **first order in time.**}$$

And the Hamiltonian is:

$$H = \frac{1}{2\mu} S_a S^a$$

Here, $\{s\} \rightarrow$ The set of all spin components constitute the phase space. In the cross-product, there are 3 components and so, the phase space is 3-dimensional.

2. Similarly, the phase space constituted by x and p is 2-dimensional because

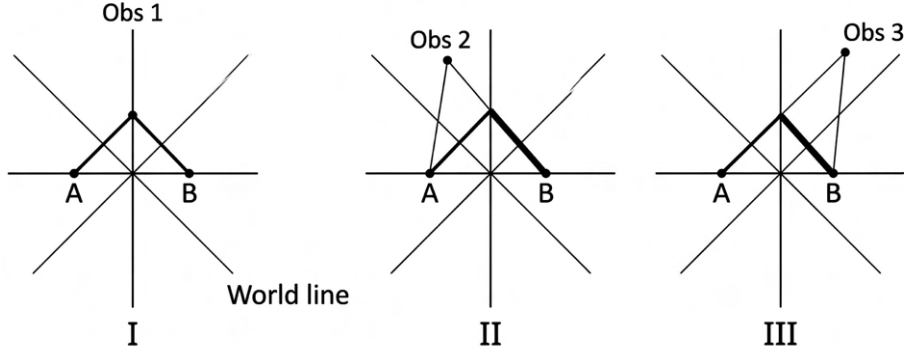
$$\frac{dx}{dt} = \{x, H\}_{\text{PB}} \quad \text{and} \quad \frac{dp}{dt} = \{p, H\}_{\text{PB}}$$

3. Also, in the Heisenberg picture,

$$\frac{dA}{dt} = \frac{1}{i\hbar} [H, A] \quad (\text{three such equations})$$

Fun Fact: Relativity of Simultaneity

It is the concept that distant simultaneity—whether two spatially separated events occur at the same time—is not absolute but depends on the observer's reference frame.



- The physical laws are always observer independent.
- But the order of events, i.e., the **history** is observer dependent
- In picture-I: For observer 1, both events A and B happen simultaneously.
- In picture-II: A happens first, then B.
- In picture-III: B happens first, then A.

$\therefore$ **Simultaneity is not a covariant statement.**

Back to the point particle

As we know,

$$p_a = \frac{\partial \mathcal{L}}{\partial \dot{x}^a} = \frac{m \dot{x}_a}{\sqrt{\dot{x}^b \dot{x}_b}}$$

It might seem that, as in $\dot{x}^a$ there are 4 components, so will be p_a . But it's not. The system may seem unconstrained, but it's not!

$$p^a p_a = \frac{m \dot{x}^a}{\sqrt{\dot{x}^b \dot{x}_b}} \times \frac{m \dot{x}_a}{\sqrt{\dot{x}^b \dot{x}_b}} = \frac{m^2 \dot{x}^a \dot{x}_a}{(\dot{x}^b \dot{x}_b)} = \frac{m^2 r^2}{(\sqrt{r^2})^2} = m^2$$

Thus,

$$p^a p_a = m^2$$

$$\therefore p^2 = m^2 \quad (5)$$

How to quantize a system:

In an unconstrained (and Cartesian) system,

$$p_i \leftrightarrow -i \partial_i, \quad [\hbar = 1] \quad (\text{In Schrödinger picture})$$

This is intimately connected to the coordinate system and works only in Cartesian. Moreover, this correspondence is not unique: Adding $G(x)$ doesn't change the system, as $[\hat{x}_i, \hat{p}_j] = i$ still works.

So, to satisfy $[\hat{x}_i, \hat{p}_j] = i$, it's a realization that

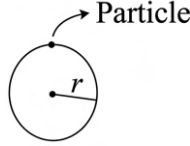
$$p_i \rightarrow -i \partial_i$$

Dirac's quantization procedure for constrained systems

1. Start with classical system, with constraints, e.g., $\phi(q, p) = 0$.
2. Promote variables to operators: $q \rightarrow \hat{q}, p \rightarrow \hat{p}$.
3. Promote constraints too: $\phi(q, p) \rightarrow \hat{\phi}(\hat{q}, \hat{p})$
4. Impose constraints on states:

$$\hat{\phi}|\psi\rangle = 0$$

This will tell which quantum systems are physical.

Example 1: Particle on a Circle

For a particle on a circle, we can immediately write the Lagrangian of the system:

$$L = T - V(x, y) = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) - V(x, y) \quad (6)$$

Now, following the procedures:

1. Constraint: $\phi(x, y) = x^2 + y^2 - r^2 = 0$
2. Promoting variables to operators:

$$x \rightarrow \hat{x}, \quad y \rightarrow \hat{y}, \quad p \rightarrow \hat{p}_i = -i\partial_i$$

3. Similar with constraint:

$$\phi(x, y) \rightarrow \hat{\phi}(\hat{x}, \hat{y}) = \hat{x}^2 + \hat{y}^2 - r^2 = 0$$

Also, to incorporate the constraint into the Lagrangian, we need Lagrange's multiplier, i.e.,

$$L = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) - V(x, y) + \lambda(\neq x, y)(\dot{x}^2 + \dot{y}^2 - r^2)$$

(**Note:** λ is not a function in this case.)

- If λ is a function of (x, y) , then the derivative of it needs to be considered too. But we want no dynamics of the system due to this multiplier. In that case,

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 0$$

Hence, $p = 0$, no evolution due to λ exists!

4. Therefore, Hamiltonian will be:

$$\begin{aligned} H &= \frac{1}{2m}(p_x^2 + p_y^2) + V(x, y) \\ \hat{H} &= \frac{1}{2m}(\hat{p}_x^2 + \hat{p}_y^2) + \hat{V}(x, y) \\ \Rightarrow \hat{H} &= \frac{-1}{2m}(\partial_x^2 + \partial_y^2) + \hat{V}(x, y) \end{aligned} \quad (7)$$

Now, imposing constraints on the state:

$$\hat{\phi}(\hat{x}, \hat{y}) \psi(x, y) = 0$$

$$\Rightarrow (\hat{x}^2 + \hat{y}^2 - r^2) \psi(x, y) = 0$$

i.e., if the constraint is satisfied, the allowed states (or values) form a subset of the full space.

We can use this formalism to quantize the system of a point particle.

Klein Gordon (KG) equation:

Now, using equation (5), i.e.,

$$p^2 = m^2 \quad \Rightarrow \quad \text{constraint} \quad \hat{\phi} = \hat{p}^2 - m^2 = 0$$

a.k.a. Schrödinger correspondence.

Imposing constraint on the state:

$$\begin{aligned} (\hat{p}^2 - m^2) \psi(x) &= 0 \\ \Rightarrow -(\partial^2 + m^2) \psi(x) &= 0 \end{aligned}$$

The KG (Klein-Gordon) equation!

3.2 Action for Massless Particle

From equation (1), we get:

$$S = m \int d\tau$$

(It wasn't first recognized as mass, but later leveled as mass from the constraint $p^2 = m^2$.)

Now, if $m = 0$, $S = 0$; action vanishes!

Then, the equivalent way to get around this problem is to define a new action:

$$S = \frac{1}{2} \int \left[\frac{1}{e} \dot{x}^2 + em^2 \right] d\tau = \int \mathcal{L} d\tau \quad (8)$$

where e is a Lagrange multiplier and does not propagate to the equations of motion.

Equation of motion for e :

$$\begin{aligned} \frac{d}{d\tau} \frac{\partial \mathcal{L}}{\partial \dot{e}} - \frac{\partial \mathcal{L}}{\partial e} &= 0 \\ \Rightarrow -\frac{\partial}{\partial e} \left[\frac{1}{e} \dot{x}^2 + em^2 \right] &= 0 \\ \Rightarrow -\left(-\frac{1}{e^2} \dot{x}^2 + m^2 \right) &= 0 \\ \therefore e &= \frac{\sqrt{\dot{x}^2}}{m} \end{aligned} \quad (9)$$

Putting in equation (8),

$$\begin{aligned}
 S &= \frac{1}{2} \int \left(\frac{m}{\sqrt{\dot{x}^2}} \dot{x}^2 + \frac{\sqrt{\dot{x}^2}}{m} m^2 \right) d\tau \\
 &= \frac{1}{2} \int 2m\sqrt{\dot{x}^2} d\tau \\
 &= m \int \sqrt{\dot{x}^2} d\tau
 \end{aligned} \tag{10}$$

and

$$p_a = \frac{\partial \mathcal{L}}{\partial \dot{x}^a} = \frac{\dot{x}}{e} \tag{11}$$

Hamiltonian

$$\begin{aligned}
 H &= p_a \dot{x}^a - \mathcal{L} \\
 &= \frac{\dot{x}}{e} \dot{x} - \frac{1}{2} \left(\frac{\dot{x}^2}{e} + em^2 \right) \\
 &= \frac{\dot{x}^2}{e} - \frac{\dot{x}^2}{2e} - \frac{em^2}{2} \\
 &= \frac{1}{2} e \left(\frac{\dot{x}^2}{e^2} - m^2 \right) \\
 \therefore H &= \frac{1}{2} e (p_a^2 - m^2)
 \end{aligned} \tag{12}$$

i.e., Hamiltonian is proportional to the constraint of the system:

$$H \propto (p_a^2 - m^2)$$

It means that the Hamiltonian vanishes “on-shell,” that is, when the constraints are satisfied. That is when real time evolution disappears.

Extra: Generic form of Schrödinger equation is:

$$i\hbar \frac{\partial \psi}{\partial t} = H\psi$$

- H will change according to the system.
- In Cartesian coordinates, one particular form is

$$\left(-\frac{\hbar^2}{2m} \nabla^2 + V \right) \psi = H\psi = i\hbar \frac{\partial \psi}{\partial t}$$

- But in spin systems, the Hamiltonian will be

$$H = \mu \vec{B} \cdot \vec{S}$$

(Class-4) Lecture 1: Sat, May 17, 2025

Poincaré Algebra

Previously we've seen the Lagrangian for the massive and massless particles are -

$$L = M \sqrt{\dot{x}^p \dot{x}^p} \quad (\text{massive})$$

$$L = \frac{1}{2} \left(\frac{1}{e} \dot{x}^2 - e m^2 \right) \quad (\text{massless})$$

These two are equivalent “on-shell” condition . **On-shell** means a particle or field satisfies its equation of motion - usually the relativistic energy momentum relation: $E^2 = \vec{p}^2 + m^2$

So a particle is on shell if: $p^\mu p_\mu = m^2$ This is also called the **mass-shell condition**.

Question: What is a particle?

Particle is something that carries an irreducible unitary representation of the Poincaré group.

Question: What is Lorentz transformation?

A Lorentz transformation is a linear transformation that preserves the Minkowski spacetime interval-

$$\eta_{\mu\nu} x^\mu x^\nu = \eta_{\rho\sigma} x^\rho x^\sigma$$

That is,

$$x'^\mu = \Lambda^\mu{}_\nu x^\nu \quad ; \quad \text{such that, } \Lambda^\top \eta \Lambda = \eta$$

Notes -

- The origin is invariant in the transformation.
- Rotation is also a linear transformation where origin is kept invariant

4.1 Poincaré transformation

Here the requirement that the origin remain invariant is relaxed, allowing transformations that preserve the interval between any two points including translation also.

Linear transformation which leaves the distance-square between any two points invariant constitute Poincaré transformation. It is also a group.

Question: What are groups? (set + binary operation)

A group is denoted by $(G, *)$ is set of objects denoted G and some operation on those objects denoted $*$, subject to the following -

- Associative
- Inverse
- Close Set
- Existence of Identity

4.2 Poincaré/Affined group

Poincaré group = Lorentz group + Translation = Rotation + Boosts + Translation

$$x^m \rightarrow x'^m = (\Lambda x)^m + a^m = \Lambda^m_n x^n + a^m$$

Here, $(\Lambda, a) \rightarrow$ set of elements $\therefore x' = (\Lambda, a)x$

Λ = Lorentz matrix

a = Translation vector

(i) Associativity

We want to check associativity of the group law

$$(\Lambda_1, a_1) \circ (\Lambda_2, a_2) = (\Lambda_1 \Lambda_2, \Lambda_1 a_2 + a_1).$$

Left side:

$$\begin{aligned} [(\Lambda_3, a_3) \circ (\Lambda_2, a_2)] \circ (\Lambda_1, a_1) &= (\Lambda_3 \Lambda_2, \Lambda_3 a_2 + a_3) \circ (\Lambda_1, a_1), \\ &= (\Lambda_3 \Lambda_2 \Lambda_1, \Lambda_3 \Lambda_2 a_1 + \Lambda_3 a_2 + a_3). \end{aligned}$$

Right side:

$$\begin{aligned} (\Lambda_3, a_3) \circ [(\Lambda_2, a_2) \circ (\Lambda_1, a_1)] &= (\Lambda_3, a_3) \circ (\Lambda_2 \Lambda_1, \Lambda_2 a_1 + a_2), \\ &= (\Lambda_3 \Lambda_2 \Lambda_1, \Lambda_3 (\Lambda_2 a_1 + a_2) + a_3), \\ &= (\Lambda_3 \Lambda_2 \Lambda_1, \Lambda_3 \Lambda_2 a_1 + \Lambda_3 a_2 + a_3). \end{aligned}$$

Thus, associativity holds.

Note:

Is the space-time position operator (x^a) a well-defined operator or observable in relativistic quantum theory?

No, it is not. Therefore, relativistic quantum mechanics in the usual sense does not exist — only *relativistic quantum field theory (QFT)* exists. This is because:

1. In non-relativistic quantum mechanics (NQRМ), the position operator $\hat{x}$ is well-defined. A particle can be localized in space and the wavefunction $\psi(x, t)$ gives the probability amplitude for the particle's position.
2. In relativistic quantum mechanics (RQM), trying to promote space-time coordinates, $x^a = (t, \vec{x})$, to operators runs into problems due to:
 - Time and space do not enter symmetrically as operators — time is treated as a parameter, not as an operator.

Therefore, only **relativistic quantum fields** exist, where the fundamental objects are fields $\hat{\phi}(x)$ — operator-valued functions over space-time.

Observables are constructed from fields, not from position operators.

(ii) Identity

Let $(I, 0)$ be the identity:

$$(I, 0) \circ (\Lambda, a) = (I\Lambda, Ia + 0) = (\Lambda, a),$$

$$(\Lambda, a) \circ (I, 0) = (\Lambda I, \Lambda 0 + a) = (\Lambda, a).$$

Hence, $(I, 0)$ is the identity element.

(iii) Inverse

By definition,

$$x' = \Lambda x + a \equiv (\Lambda, a) x.$$

Thus,

$$x = \Lambda^{-1}(x' - a) = \Lambda^{-1}x' - \Lambda^{-1}a = (\Lambda^{-1}, -\Lambda^{-1}a) x'.$$

Therefore,

$$(\Lambda, a)^{-1} = (\Lambda^{-1}, -\Lambda^{-1}a).$$

Verification

$$\begin{aligned} (\Lambda, a)^{-1} \circ (\Lambda, a) &= (\Lambda^{-1}, -\Lambda^{-1}a) \circ (\Lambda, a), \\ &= (\Lambda^{-1}\Lambda, \Lambda^{-1}a - \Lambda^{-1}a), \\ &= (I, 0) \equiv e \in O(1, 3) \ltimes \mathbb{R}^4. \end{aligned}$$

$$\therefore O(1, 3) \ltimes \mathbb{R}^4 \text{ forms a group.}$$

4.2.1 Group Composition

In Poincaré group, group composition refers to how two Poincaré transformations combine when applied one after the other. Now, the Lorentz transformation is,

$$x'^{\mu} = \Lambda^{\mu}_{\nu} x^{\nu} + a^{\mu} ; \Lambda^{\mu}_{\nu} \in \text{SO}(1, 3)$$

Poincaré transformation is denoted as a pair, (Λ, a) . Meaning,

1. Apply Lorentz transformation, Λ
2. Then translate by a

Law: Given two transformations,

$$(\Lambda_1, a_1) = (\Lambda, a) ; (\Lambda_2, a_2) = (\Lambda', a')$$

Their composition [do (Λ', a') after (Λ, a)] is,

$$\begin{aligned} (\Lambda', a')(\Lambda, a)x &= (\Lambda', a')(\Lambda x, a) \\ &= \Lambda'(\Lambda x + a) + a' \\ &= \Lambda'\Lambda x + \Lambda'a + a' \\ \therefore (\Lambda', a')(\Lambda, a) &= (\Lambda'\Lambda, \Lambda'a + a') \end{aligned} \tag{4.2}$$

Applying matrices R:

$$R(\Lambda', a')R(\Lambda, a) = R(\Lambda'\Lambda, \Lambda'a + a') \quad (4.3)$$

Question: Why Unitary representation

Eugene Wigner classified all possible elementary particles in terms of irreducible unitary representations of the Poincaré group. These representations are labeled by:

- Particle's mass • Particle's spin

We need unitary representations because of *the preservation of probability*. In quantum mechanics, we don't perform transformations on x directly, but on a wavefunction $\psi(x)$:

$$\psi'(x) = U(x)\psi(x) = \psi(x + a)$$

Then, if $\langle \psi | \psi \rangle = \langle \psi' | \psi' \rangle$, transformations are required to be unitary.

Therefore the unitary representation must satisfy (4.3) -

$$U(\Lambda', a')U(\Lambda, a) = U(\Lambda'\Lambda, \Lambda'a + a') \quad (4.4)$$

Note: Dimension of Unitary Transformation in Hilbert Space

In Hilbert space, unitary transformations are not restricted to infinite dimensions. They can be finite or infinite dimensional depending on the physical system:

- Finite-dimensional unitary representations arise in systems with discrete bases — for example, spin systems where spin states form a finite-dimensional Hilbert space.
- Infinite-dimensional unitary representations are necessary for systems with continuous degrees of freedom — such as position and momentum representations in quantum mechanics.

Now, $U^\dagger U = \hat{\mathbb{I}}$; If isomorphic and acts on Hilbert space.

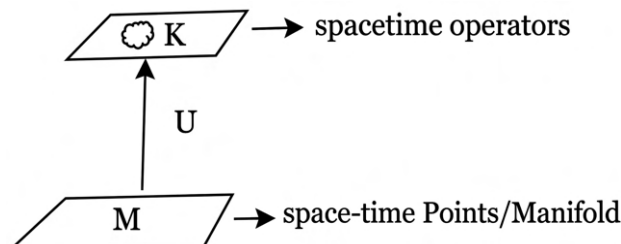
$$\text{If } gg' = g'' \text{ and } gg^{-1} = \mathbb{I}$$

$$\text{Then } U(g)U(g') = U(g'') \text{ and } U(g)U(g^{-1}) = U(\mathbb{I}) = \hat{\mathbb{I}}$$

$$\therefore U(g^{-1}) = U^\dagger(g)$$

$$\therefore U^\dagger(\Lambda, a) = U((\Lambda, a)^{-1}) = U(\Lambda^{-1}, -\Lambda^{-1}a) \quad (4.5)$$

This transformation acts on space time.



Notes:**Group Representation**

When studying symmetry transformations, we represent them using matrices that act on a vector space. The collection of such matrices that preserves the group structure is called a *representation* of the group. The space on which these matrices act is known as the *representation space*.

Irreducible Representation: A representation is called irreducible if it has no proper, non-trivial invariant subspace.

Invariant Subspace: Let $T : V \rightarrow V$ be a linear operator on a vector space V . A subspace $W \subseteq V$ is called **invariant** under T if:

$$\forall w \in W, \quad T(w) \in W$$

That is, applying T to any vector in W keeps it inside W .

Similarity Transformation

A similarity transformation rewrites a matrix A in a different basis using an invertible matrix S :

$$A' = SAS^{-1}$$

Key properties:

- (i) A and A' are similar matrices; they represent the same linear transformation in different bases.
- (ii) They have identical eigenvalues, trace, and determinant.

Example: If $AB = C$, then under similarity transformation:

$$\begin{aligned} A' &= SAS^{-1} \\ B' &= SBS^{-1} \\ \therefore A'B' &= (SAS^{-1})(SBS^{-1}) = S(AB)S^{-1} = SC S^{-1} = C' \end{aligned}$$

Thus, C' is the similarity transformation of C .

Conjugation of a Group Element:

If $B' = SBS^{-1}$, we call B' the *conjugate* of B by S . This does not imply a group product, it's a transformation under a change of basis.

Back to Wigner, recall (4.4) and (4.5) ,

$$U(\Lambda', a')U(\Lambda, a) = U(\Lambda'\Lambda, \Lambda'a + a') \quad (4.4)$$

$$U^\dagger(\Lambda, a) = U\left((\Lambda, a)^{-1}\right) = U(\Lambda^{-1}, -\Lambda^{-1}a) \quad (4.5)$$

Using both the equations we get,

$$\begin{aligned}
 U(\Lambda, a)U(\Lambda', a')U^\dagger(\Lambda, a) &= U(\Lambda\Lambda', \Lambda a' + a)U(\Lambda^{-1}, -\Lambda^{-1}a) \\
 &= U(\Lambda\Lambda'\Lambda^{-1}, \Lambda\Lambda'(-\Lambda'^{-1}a) + \Lambda a' + a) \\
 &= U(\Lambda\Lambda'\Lambda^{-1}, \Lambda a' + (\mathbb{I} - \Lambda\Lambda'\Lambda^{-1})a)
 \end{aligned} \tag{4.6}$$

Then by combining (??) and (4.2) we get,

$$\begin{aligned}
 (\Lambda, a)(\Lambda', a')(\Lambda, a)^{-1} &\equiv (\Lambda\Lambda', \Lambda a' + a)(\Lambda', -\Lambda'^{-1}a) \\
 &= \Lambda\Lambda'\Lambda^{-1}, \Lambda\Lambda'(-\Lambda'^{-1}a) + \Lambda a' + a \\
 &= \Lambda\Lambda'\Lambda^{-1}, \Lambda a' + a - \Lambda\Lambda'\Lambda^{-1}a \\
 &= (\Lambda\Lambda'\Lambda^{-1}, \Lambda a' + (\mathbb{I} - \Lambda\Lambda'\Lambda^{-1})a)
 \end{aligned} \tag{4.7}$$

This result means:

- Conjugating a Lorentz transformation by another Lorentz + translation produces a new Lorentz transformation plus a shifted translation.
- In other words, translations and Lorentz transformations do not commute independently — the Lorentz transformation acts nontrivially on translations.
- This is the hallmark of the semi-direct product structure of the Poincaré group.

So the calculation is demonstrating explicitly that the Poincaré group is a semi-direct product

$$\mathbb{R}^{1,3} \rtimes SO(1,3).$$

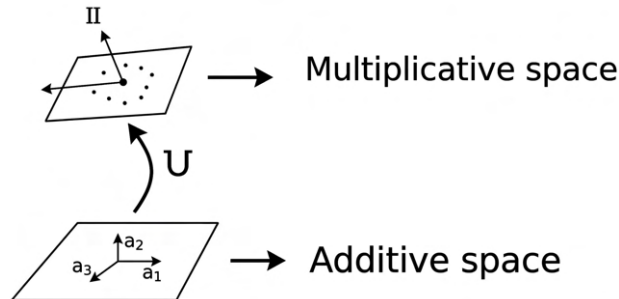
It shows how translations transform under Lorentz transformations inside the group law, which is essential in Wigner's classification of particles.

Now, the theory of Lie groups tells us that we can write any unitary matrix in the form:

$$U(\theta) = e^{i\vec{\theta} \cdot \vec{H}} \quad , \quad \theta = (\theta_1, \theta_2, \dots), \quad H = (H_1, H_2, \dots)$$

where $\vec{\theta}$ and $\vec{H}$ are vectors of parameters and Hermitian operators, respectively. This form holds under the assumption: $U(0) = \hat{\mathbb{I}}$

Unitary transformation transforms the additive space to multiplicative space.



Note

It is also possible for $U = -\hat{\mathbb{I}}$, since

$$U^\dagger U = \mathbb{I}$$

However, this does **not** satisfy the condition $U(0) = \hat{\mathbb{I}}$ and therefore cannot be expressed in the exponential form $e^{i\vec{\theta} \cdot \vec{H}}$.

Now for small θ , $U(\theta)$ can be Taylor expanded-

$$U(\theta) = \hat{\mathbb{I}} + i \vec{\theta} \cdot \vec{H} + \dots$$

Now evaluation of the wave function can be written as the form-

$$\begin{aligned} \psi(x+a) &= e^{a \frac{d}{dx}} \psi(x) = \left(1 + a \frac{d}{dx} + \dots\right) \psi(x) \\ \Rightarrow \psi(x+a) &= e^{\frac{ia}{\hbar} (-i\hbar \frac{d}{dx})} \psi(x) \end{aligned}$$

Also for rotations we know that,

$$R^\top R = R^\top \mathbb{I} R = \mathbb{I}$$

and for Lorentz transformation,

$$\Lambda^\top \eta \Lambda = \eta$$

$$\begin{aligned} \text{Let, } R &= \mathbb{I} + \epsilon W \\ R^\top &= \mathbb{I} + \epsilon W^\top \end{aligned}$$

for small transformations where, ϵ = number, W = matrix that generates the rotation. It belongs to the Lie algebra of the rotation group $\text{SO}(n)$.

$$\begin{aligned} \therefore R^\top R &= (\mathbb{I} + \epsilon W^\top)(\mathbb{I} + \epsilon W) \\ &= \mathbb{I} + \epsilon(W^\top + W) + \epsilon^2 W^\top W \end{aligned}$$

Lie theory says that the substantial structure of group is encoded in its first order infinitesimals. So, $\epsilon^2 \rightarrow 0$. And $R^\top R = \mathbb{I}$ implies that-

$$W = -W^\top \quad (\text{Antisymmetric})$$

In d dimensions, there are $d \times d = d^2$ parameters, and main diagonal constitute of 0. Therefore independent $d^2 - d$, lower and upper triangular part have the same input except the minus sign. Hence, number of the independent parameters are-

$$\frac{d^2 - d}{2} = \frac{d(d-1)}{2} = {}^d\mathbf{C}_2$$

Then the affine or, Poincaré group has

$$(\Lambda, a) \rightarrow \frac{d(d-1)}{2} + d = \frac{d(d+1)}{2} \quad \text{dimensions}$$

This looks like rotation in $(d + 1)$ dimensions!

Now for Poincaré transformation,

$$x' = \Lambda x + a \quad ;$$

$$[\Lambda \rightarrow \text{Multiplicative(only for rotations)}, a \rightarrow \text{additive}]$$

Therefore, for the $(d + 1)$ dimensional rotation it can be written as in $2D$:

$$\begin{pmatrix} x' \\ 1 \end{pmatrix} = \begin{pmatrix} \Lambda & a \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ 1 \end{pmatrix} \quad ; \quad \left[\begin{pmatrix} x \\ 1 \end{pmatrix} \rightarrow \begin{pmatrix} d \rightarrow \text{number of dimensions} \\ 1 \rightarrow 3^{rd} \text{ dimension: } z \text{ is fixed} \end{pmatrix} \right]$$

It implies the idea that “rotating in higher dimensions” can relate to translation in lower dimensions.

for example: The rotation of laser points in $2D$ plane. When projected, it results in an apparent translation along the z axis.

Now let Lorentz transformation be:

$$\Lambda' = \mathbb{I} + W' \tag{4.8}$$

$$\therefore U(\Lambda', a') = U(\mathbb{I} + W', a') = \hat{\mathbb{I}} + i a' \hat{P} + \frac{i}{2} W_{mn} \hat{J}^{mn} \tag{4.9}$$

here, $\frac{1}{2} \rightarrow$ factor here is to avoid double counting

as W_{mn} is antisymmetric matrix

$\hat{P} \rightarrow$ an operator whether it is translational

or not will be given by its “**Algebra**”

$$W_{mn} = -W_{nm} \quad ; \quad \hat{J}^{mn} = -\hat{J}^{nm} \quad ; \quad a' = \text{a number}$$

Let’s get back to Wigner once more. Using equation (4.8) in $\Lambda \Lambda' \Lambda^{-1}$ we get,

$$\Lambda \Lambda' \Lambda^{-1} = \Lambda (\mathbb{I} + W') \Lambda^{-1} = \mathbb{I} + \Lambda W' \Lambda^{-1} \tag{4.10}$$

$$\text{and, } \Lambda a' + (\mathbb{I} - \Lambda \Lambda' \Lambda^{-1}) a = \Lambda a' - \Lambda W' \Lambda^{-1} a \tag{4.11}$$

Recall equation (4.6):

$$U(\Lambda, a) U(\Lambda', a') U^\dagger(\Lambda, a) = U(\Lambda \Lambda' \Lambda^{-1}, \Lambda a' + (\mathbb{I} - \Lambda \Lambda' \Lambda^{-1}) a) \tag{4.6}$$

Now combining equation (4.6), (4.9), (4.10), (4.11) we will get:

$$\begin{aligned} \therefore U(\Lambda, a) U(\Lambda', a') U^\dagger(\Lambda, a) &= U(\Lambda, a) \left[\hat{\mathbb{I}} + i a' \hat{P} + \frac{i}{2} W_{mn} \hat{J}^{mn} \right] U^\dagger(\Lambda, a) \\ \Rightarrow U(\Lambda \Lambda' \Lambda^{-1}, \Lambda a' + (\mathbb{I} - \Lambda \Lambda' \Lambda^{-1}) a) &= U(\Lambda, a) \left[\mathbb{I} + i a'_n \hat{P}^n + \frac{1}{2} W_{mn} \hat{J}^{mn} \right] U^\dagger(\Lambda, a) \\ \Rightarrow U(\mathbb{I} + \Lambda W \Lambda^{-1}, \Lambda a - \Lambda W \Lambda^{-1} a) &= U(\Lambda, a) \left[\mathbb{I} + i a'_n \hat{P}^n + \frac{1}{2} W_{mn} \hat{J}^{mn} \right] U^\dagger(\Lambda, a) \\ \Rightarrow \mathbb{I} + i a'_n U(\Lambda, a) \hat{P}^n U^\dagger(\Lambda, a) + \frac{i}{2} W_{mn} U(\Lambda, a) \hat{J}^{mn} U^\dagger(\Lambda, a) \\ &= \mathbb{I} + i(\Lambda a' - \Lambda W \Lambda^{-1})_n \hat{P}^n + \frac{i}{2} (\Lambda W \Lambda^{-1})_{pq} \hat{J}^{pq} \end{aligned} \tag{4.12}$$

$$\begin{aligned} \Rightarrow i a'_n U(\Lambda, a) \hat{P}^n U^\dagger(\Lambda, a) + \frac{i}{2} W_{mn} U(\Lambda, a) \hat{J}^{mn} U^\dagger(\Lambda, a) \\ = i(\Lambda a')_n \hat{P}^n + \frac{i}{2} (\Lambda W \Lambda^{-1})_{pq} \hat{J}^{pq} - i(\Lambda W \Lambda^{-1})_n \hat{P}^n \end{aligned} \tag{4.13}$$

Now, matching the equations containing a' on both side of the equation (4.13), yields,

$$i a'_n U(\Lambda, a) \hat{P}^n U^\dagger(\Lambda, a) = i(\Lambda a')_m \hat{P}^m = i \Lambda_m^s a'_s \hat{P}^m = i \Lambda_m^n a'_n \hat{P}^m$$

$$\boxed{\therefore U(\Lambda, a) \hat{P}^n U^\dagger(\Lambda, a) = \Lambda_m^n \hat{P}^m}$$

Thus we can see that, $\hat{P}$ transforms like a vector under the group. It is a constant matrix. It doesn't depend on the coordinates or the values of a' .

Assignment-1:

How does $\hat{J}^{rs}$ transform under Poincaré transformation? (It's a semi direct product)

Note:

Direct Product and Semi-direct product

In group theory, a direct product of two groups means that the groups combine in a way where elements of one group do not affect the operation of the other. That is:

$$(a_1, b_1)(a_2, b_2) = (a_1 a_2, b_1 b_2)$$

Here, two group elements become separated completely.

Poincaré group combines Lorentz transformations and spacetime translations as:

$$(\Lambda', a')(\Lambda, a) = (\Lambda' \Lambda, \Lambda' a + a')$$

The second translation is transformed by Λ' , meaning the two parts interact non-trivially. So, this isn't a direct product but a **semi-direct** one. Lorentz part Λ' transforms the translation a , indicating that the two subgroups aren't independent.

Transformation of $\hat{J}^{pq}$ under Lorentz transformation

Assume $a = 0 \equiv$ No translation under Lorentz transformation. From the equation (4.12),

$$\begin{aligned} & \mathbb{I} + i a'_n U(\Lambda, a) \hat{P}^n U^\dagger(\Lambda, a) + \frac{i}{2} W'_{mn} U(\Lambda, a) \hat{J}^{mn} U^\dagger(\Lambda, a) \\ &= \mathbb{I} + i(\Lambda a' - \Lambda W' \Lambda^{-1})_n \hat{P}^n + \frac{i}{2} (\Lambda W' \Lambda^{-1})_{pq} \hat{J}^{pq} \end{aligned}$$

Putting $a = 0$ and comparing the 2^{nd} term on both sides,

$$\begin{aligned} & W'_{pq} U \hat{J}^{pq} U^\dagger = (\Lambda W' \Lambda^{-1})_{ab} \hat{J}^{ab} \\ \Rightarrow & W'_{pq} U \hat{J}^{pq} U^\dagger = \Lambda_a^m W'_{mn} (\Lambda^{-1})^n_b \hat{J}^{ab} \\ & \text{By setting } m = p, n = q ; \text{ we get,} \\ \Rightarrow & W'_{pq} U \hat{J}^{pq} U^\dagger = \Lambda_a^p W'_{pq} (\Lambda^{-1})^q_b \hat{J}^{ab} \\ \Rightarrow & U \hat{J}^{pq} U^\dagger = \Lambda_a^p (\Lambda^{-1})^q_b \hat{J}^{ab} \end{aligned} \tag{4.14}$$

Now, recall the Lorentz transformation,

$$\begin{aligned}\Lambda^\top \eta \Lambda &= \eta \\ \Rightarrow \Lambda^\top \eta &= \eta \Lambda^{-1} \\ \therefore \eta \Lambda^\top \eta &= \Lambda^{-1}\end{aligned}\tag{4.15}$$

Putting this value of Λ^{-1} in equation (4.14) we get,

$$\begin{aligned}U \hat{J}^{pq} U^\dagger &= \Lambda_a{}^p (\eta \Lambda^\top \eta)^q{}_b \hat{J}^{ab} \\ &= \Lambda_a{}^p \eta^q{}_r \Lambda^r{}_s \eta^s{}_b \hat{J}^{ab} \\ &= \Lambda_a{}^p \Lambda^q{}_s \hat{J}^{as} \quad ; [\text{setting, } s=b] \\ &= \Lambda_a{}^p \Lambda^q{}_b \hat{J}^{ab}\end{aligned}$$

$$\boxed{\therefore U \hat{J}^{pq} U^\dagger = \Lambda_a{}^p \Lambda^q{}_b \hat{J}^{ab}}$$

Now substituting, $U = e^{(\frac{i}{2} W_{cd} \hat{J}^{cd})}$ and $\Lambda^p{}_a = \delta^p{}_a + W^p{}_a$ Then,

$$\begin{aligned}e^{(\frac{i}{2} W_{cd} \hat{J}^{cd})} \hat{J}^e e^{(-\frac{i}{2} W_{cd} \hat{J}^{cd})} &= (\delta_a{}^p + W^p{}_a) (\delta^q{}_b + W^q{}_b) \hat{J}^{ab} \quad [\text{BCH Relation}] \\ \Rightarrow \hat{J}^{pq} + \frac{i}{2} W_{cd} [\hat{J}^{cd}, \hat{J}^{pq}] &= (\delta_a{}^p \delta^q{}_b + \delta_a{}^p W^q{}_b + W^p{}_a \delta^q{}_b + W^p{}_a W^q{}_b) \hat{J}^{ab} \\ \Rightarrow \hat{J}^{pq} + \frac{i}{2} W_{cd} [\hat{J}^{cd}, \hat{J}^{pq}] &= \hat{J}^{pq} + W^q{}_b \hat{J}^{pb} + W^p{}_a \hat{J}^{aq} \\ \Rightarrow \frac{i}{2} W_{cd} [\hat{J}^{cd}, \hat{J}^{pq}] &= W_{mb} \eta^{mq} \hat{J}^{pb} + W_{na} \eta^{np} \hat{J}^{aq} \\ \Rightarrow \frac{i}{2} W_{cd} [\hat{J}^{cd}, \hat{J}^{pq}] &= W_{cd} \eta^{cq} \hat{J}^{pd} + W_{cd} \eta^{cp} \hat{J}^{dq} \quad [b=d, m=c, a=d, n=c] \\ \Rightarrow \frac{i}{2} W_{cd} [\hat{J}^{cd}, \hat{J}^{pq}] &= W_{cd} \times \frac{1}{2} (\eta^{cq} \hat{J}^{pd} - \eta^{dq} \hat{J}^{pc}) + W_{cd} \times \frac{1}{2} (\eta^{cp} \hat{J}^{dq} - \eta^{dp} \hat{J}^{cq}) \\ &\quad \left[\text{Here we used, } AB = \frac{1}{2}(AB + BA) + \frac{1}{2}(AB - BA) \right] \\ \text{Omitting } W_{cd} \text{ on both sides,} \\ \Rightarrow [\hat{J}^{cd}, \hat{J}^{pq}] &= -i (\eta^{cq} \hat{J}^{pd} - \eta^{dq} \hat{J}^{pc} + \eta^{cp} \hat{J}^{dq} - \eta^{dp} \hat{J}^{cq}) \\ \therefore [\hat{J}^{cd}, \hat{J}^{pq}] &= -i (\eta^{cq} \hat{J}^{pd} - \eta^{dq} \hat{J}^{pc} + \eta^{pc} \hat{J}^{dq} - \eta^{pd} \hat{J}^{cq})\end{aligned}$$

Summary

Here we have successfully found the algebra of angular momentum without the relation $\hat{J} = \vec{r} \times \vec{p}$, just simply using the algebra of the Poincaré group.

In addition, we know in 3D, $\hat{J}^{mn} = i \epsilon^{amn} \hat{J}_a$

Assignment-2:

- Recover 3D rotation algebra from angular momentum.

(Class-5) Lecture 2: Sat, May 24, 2025

A Glimpse of GR via The Classical Field Theory**5.1 Road to GR**

[General relativity is a classical field theory because no quantization is necessarily involved here]

Now, in freely falling coordinates (FFC), the line element is given by:

$$ds^2 = \eta_{ab} d\xi^a d\xi^b \quad (5.1)$$

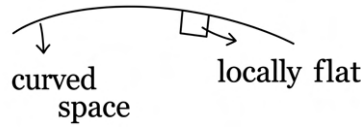
Notation: η_{ab} : Flat Minkowski metric

ξ^a : Freely falling coordinates (FFC)

x^a : General coordinates

Note:

- In Special Relativity (SR), there is no representation of gravity.
- Only in curved space, there is an existence of gravity ; hence, acceleration arises.



However, the FFC are locally defined as:

$$\begin{aligned} \xi^a &= \xi^a(x) \\ \Rightarrow d\xi^a &= \frac{\partial \xi^a}{\partial x^m} dx^m \\ \Rightarrow d\xi^a &\equiv M^a_m dx^m \quad (\text{linear relation}) \end{aligned} \quad (5.2)$$

Substituting (5.2) into (5.1), we get:

$$ds^2 = \eta_{ab} \frac{\partial \xi^a}{\partial x^m} \frac{\partial \xi^b}{\partial x^n} dx^m dx^n \quad (5.3)$$

$$\text{here, } \left[\eta_{ab} \frac{\partial \xi^a}{\partial x^m} \frac{\partial \xi^b}{\partial x^n} \right] = \text{symmetric, depends on } x$$

Let us define this term as:

$$G_{mn}(x) \equiv \eta_{ab} \frac{\partial \xi^a}{\partial x^m} \frac{\partial \xi^b}{\partial x^n}$$

Then:

$$ds^2 = G_{mn}(x) dx^m dx^n$$

$G_{mn}(x)$ is called the metric which describes distance in a curved spacetime and has the following properties:

1. **Symmetry:** $G_{mn} = G_{nm}$
2. **Non-degeneracy:** If $\det(G_{mn}) \neq 0$, then G_{mn} is invertible.

Such a manifold is known as a *Riemannian manifold* (or pseudo-Riemannian in Lorentzian spacetime).

Action: Recall the relativistic action:

$$s = m \int \sqrt{\eta_{ab} \dot{\xi}^a \dot{\xi}^b} d\tau \quad (\text{non-polynomial Lagrangian})$$

$$s' = \frac{1}{2} \int \left(\frac{1}{e} \dot{\xi}^a \dot{\xi}^b \eta_{ab} - em^2 \right) d\tau$$

- First term in s' : kinetic term.
- Second term in s' : Lagrange multiplier (enforces mass-shell condition).

When $m = 0$, both s and s' are same, they yield the same dynamics.

Note:

For flat spacetime in Cartesian coordinates, the trajectory of a free particle is:

$$x^i = x_0^i + \left(\frac{p^i}{m} \right) t$$

This equation -

- is valid both in Cartesian coordinates and flat spacetime.
- all quantities here carry upper indices (contravariant).

Conjugate Momenta:

$$p_i = \frac{\partial \mathcal{L}}{\partial \dot{q}^i}$$

- p_i and q^i have different index positions because they live in dual spaces.
- The momentum p_i lives in the **cotangent space** (dual vector space) to the configuration space.

Consider, for example, a scalar function:

$$\phi(x) = \phi'(\xi)$$

where x could be any coordinate system (e.g., spherical, Cartesian, etc.). Then,

$$\frac{\partial \phi'(\xi)}{\partial \xi^a} = \frac{\partial \phi(x)}{\partial x^m} \cdot \frac{\partial x^m}{\partial \xi^a} \quad (5.4)$$

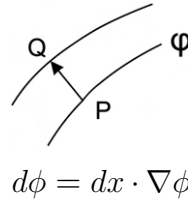
Recall equation (5.3):

$$ds^2 = \eta_{ab} \frac{\partial \xi^a(x)}{\partial x^m} \frac{\partial \xi^b(x)}{\partial x^n} dx^m dx^n = G_{mn}(x) dx^m dx^n$$

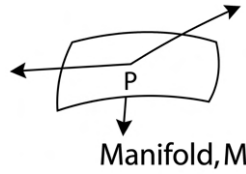
These equations (5.3) and (5.4) are inverse to each other — they reflect the transformation between coordinates and dual coordinates.

Note:**Phase Space**

- The cotangent space of a manifold is called the **phase space**.
- The collection of all x and p coordinates also forms the phase space.
- The collection of all initial conditions again constitutes a phase space.

Differential and Gradient

- The gradient $\nabla\phi$ lives in the dual space.
- If $v \rightarrow \mathbb{R}$, and $\langle w, v \rangle \in \mathbb{R}$ then w and v are not in the same space.
- w is the dual vector space of v .

Tangent Bundle

A **tangent bundle** is the collection of all tangent spaces for all points on a manifold:

- Let $T_p(M)$ be the tangent space at point p on manifold M .
- Then, $\cup_p T_p(M) \equiv T(M)$ is the tangent bundle of M .

Summary: The differential and the gradient are **dual** to each other.

5.2 Particle trajectory in a curved spacetime

From the action:

$$S[x] = \frac{1}{2} \int d\tau \, \xi^a \xi^b \eta_{ab} = \frac{1}{2} \int d\tau \, G_{mn}(x) \dot{x}^m \dot{x}^n \quad ; \quad [\dot{x}^m \dot{x}^n = \text{squared 4-velocity term}]$$

Now, from the action principle (i.e., $\delta S = 0$), we can derive the equation of motion of the particle.

$$\delta S = \frac{1}{2} \int \left[\delta G_{mn}(x) \frac{dx^m}{d\tau} \frac{dx^n}{d\tau} + 2G_{mn}(x) \frac{d}{d\tau}(\delta x^m) \frac{dx^n}{d\tau} \right] d\tau$$

Here Second term is symmetric in m, n , thus the term doubles. Then applying integration by parts $\left[u = \frac{d}{d\tau}(\delta x^m) , v = G_{mn}(x)\dot{x}^n = G_{mn}(x) \frac{dx^n}{d\tau} \right]$,

$$\begin{aligned}
\delta S &= \frac{1}{2} \int \left[\left\{ \frac{\partial G_{mn}}{\partial x^p} \dot{x}^m \dot{x}^n \delta x^p \right\} + 2 G_{mn} \frac{dx^n}{d\tau} \delta x^m \right] - 2 \frac{d}{d\tau} \left(G_{mn} \frac{dx^n}{d\tau} \right) \delta x^m d\tau \\
&= \frac{1}{2} \int \left[\frac{\partial G_{mn}}{\partial x^p} \dot{x}^m \dot{x}^n \delta x^p - 2 \left(G_{mn} \ddot{x}^n + \frac{\partial G_{mn}}{\partial x^p} \dot{x}^p \dot{x}^n \right) \delta x^m \right] d\tau \\
&= \int \left[\frac{1}{2} \partial_p G_{mn} \dot{x}^m \dot{x}^n \delta x^p - G_{mn} \ddot{x}^n \delta x^m - \partial_p G_{mn} \dot{x}^p \dot{x}^n \delta x^m \right] d\tau \\
&= - \int \left[G_{mn} \ddot{x}^n + \left(\partial_p G_{mn} \dot{x}^p \dot{x}^n - \frac{1}{2} \partial_m G_{np} \dot{x}^n \dot{x}^p \right) \right] \delta x^m d\tau \\
&= - \int \left[G_{mn} \ddot{x}^n + \frac{1}{2} (\partial_p G_{mn} + \partial_n G_{mp} - \partial_m G_{np}) \dot{x}^n \dot{x}^p \right] \delta x^m d\tau
\end{aligned}$$

Here, in the first line, the term $2 G_{mn} \frac{dx^n}{d\tau} \delta x^m \Big| = 0$ at the boundary. δx^μ is arbitrary, therefore to satisfy the action principle, the integrand must vanish which gives:

$$\begin{aligned}
G_{mn} \ddot{x}^n + \frac{1}{2} (\partial_p G_{mn} + \partial_n G_{mp} - \partial_m G_{np}) \dot{x}^n \dot{x}^p &= 0 \\
\Rightarrow \ddot{x}^n + \frac{1}{2} G^{nm} (\partial_p G_{mn} + \partial_n G_{mp} - \partial_m G_{np}) \dot{x}^n \dot{x}^p &= 0
\end{aligned}$$

In the second term,

$$\frac{1}{2} G^{nm} (\partial_p G_{mn} + \partial_n G_{mp} - \partial_m G_{np}) = \Gamma^m_{np} \rightarrow \text{Christoffel Symbol}$$

$$\boxed{\therefore \ddot{x}^n + \Gamma^m_{np} \dot{x}^n \dot{x}^p = 0}$$

This is the **geodesic equation**.

Therefore, the acceleration is proportional to the square of velocity, $\dot{x}^2$.

But in *electromagnetism*:

$$\ddot{x}^m = F^m_n \dot{x}^n \quad (\text{Lorentz force equation})$$

So here, acceleration is proportional to a 4-vector.

Exercise

$$S = \frac{1}{2} \int m \dot{\xi}^2 d\tau + q \int A_b(\xi) \dot{\xi}^b d\tau$$

[This latter interaction term is geometrical as well]

- Show that we can recover the Lorentz force from here.

5.3 Schur's Lemma

Schur's Lemma involves two such equations. Before understanding the lemma, we need to understand some definitions first.

Representation

Let:

$$AB = C \quad \Rightarrow \quad R(A)R(B) = R(C)$$

Here, R is a matrix-valued map that forms an algebra. This is called a **representation**.

Invariant Subspaces

It begins with the question: are there subspaces on which the representation maps onto itself? The answer is **null-space** $R(0) \rightarrow 0$.

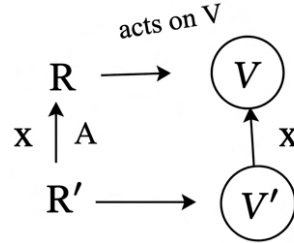
- In the case of rotation, the invariant subspace is the **axis** of rotation.
- Except for the entire space and the trivial zero vector, any other invariant subspace (if exists) implies the representation is **reducible**.

Irreducible Representations (Irreps)

If no invariant subspace exists, it is called an **irreducible representation**. For irreducible representations:

- No non-trivial invariant subspaces exist.
- Any operator commuting with all elements of the representation is proportional to the identity.

Statement of Schur's Lemma



Let V and V' be vector spaces, and let R and R' be two representations of a group G . That is, to each group element $g \in G$, we assign a matrix (or linear transformation) $R(g)$ acting on V , and $R'(g)$ acting on V' , such that:

$$R(g_1 g_2) = R(g_1) R(g_2), \quad R'(g_1 g_2) = R'(g_1) R'(g_2) \quad \forall g_1, g_2 \in G$$

Suppose there exists a linear map (or matrix) $X : V \rightarrow V'$ such that for all $g \in G$,

$$X R(g) = R'(g) X$$

$$\text{Then } X = \lambda \mathbb{I}$$

I.e., it's proportional to an identity matrix.

For example, if

$$[J_1, J_2] = iJ_3$$

$$\text{and } [J^2, J_i] = 0$$

$$\text{Then } J^2 \propto \mathbb{I}$$

Intertwiner

Let R, R' be two irreps and

$$RX = XR'$$

To make this possible, there exists two possibilities for X .

1. X is invertible.

2. Then:

$$X^{-1}RX = R'$$

That is, X *intertwines* the two representations—it commutes with the group action. If X commutes with all the matrices of an irrep, then $X = \lambda \mathbb{I}$, i.e., it's proportional to the identity matrix.

Note: Commutator is a derivative

Anything that follows the product rule, is called a **derivation**.

$$\frac{d}{dx}(uv) = \frac{du}{dx} v + u \frac{dv}{dx}$$

$$[A, uv] = [A, u]v + u[A, v]$$

Therefore, **Commutation** is a **derivation**.

Exercises: Check the commutations:

$$[J, p] \sim p$$

$$[p, p] \sim 0$$

$$[J, J] \sim J$$

$$[J, p^2] \sim 0$$

$$[J, p_a p^a] = [J, p_a] p^a + p_a [J, p^a] \sim 0$$

Notes regarding the exercises:

- We can do QFT just because of p^2 .
- No quantum version of Lorentz group exists.
- If $[A, \mathcal{V}] = 0$, then $\mathcal{V}$ can only be constant.

(Class-6) Lecture 3: Sat, May 31, 2025

Lie Derivative**Some Clarifications from Previous Lecture**

- The Lorentz group doesn't have any invariant subalgebra.

Invariant Subgroup: A subgroup $N \subset G$ is **invariant** if -

$$g n g^{-1} \in N \quad \forall g \in G$$

That means, conjugating any element of N by any element of G keeps it inside N . Therefore, for any group element g , if

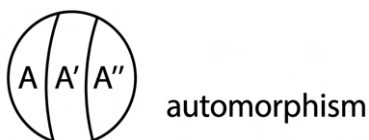
$$\begin{aligned} g g^{-1} &= g'' \quad (\text{conjugating each like similarity transformation}) \\ \Rightarrow h g h^{-1} \cdot h g h^{-1} &= h g'' h^{-1} \quad (\text{where } h g h^{-1} = g, h \in G) \\ \Rightarrow g g^{-1} &= g g^{-1} \quad \therefore h e h^{-1} = e \rightarrow \text{unaffected} \end{aligned}$$

Invariant

Under conjugation, things get mapped, but some things remain unchanged — that is called **invariant**.

Automorphism

An **automorphism** is a structure-preserving map from a mathematical object to itself.

**Invariant Subalgebra**

A subalgebra $h \subset g$ is said to be **invariant** if:

$$[X, H] \in h \quad \forall X \in g, H \in h$$

That is, the Lie bracket of any element of the full algebra with any element of the subalgebra lies inside the subalgebra.

Therefore, if $h = e^A$ where $A \in \text{Lie algebra}$ and $h \in \text{Lie group}$, then:

$$\begin{aligned} h g h^{-1} &= g' \\ \Rightarrow e^A e^B e^{-A} &\sim e^{B+[A,B]} = e^{B'} \quad (\because [A, B] \sim B') \end{aligned}$$

(g doesn't have to be g' , it just has to map into the subalgebra.)

Because boosts and rotations are mixed under commutators, they don't form invariant subspaces.

Notes

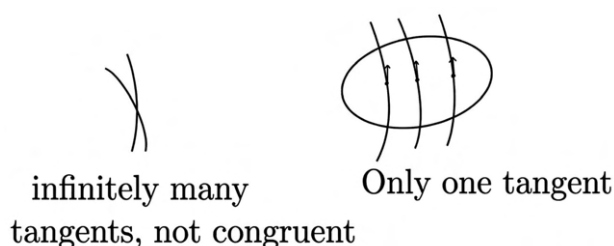
- The Lorentz group also has no unitary representation because it is **non-compact**.
- The Poincare group has unitary representation because it is **compact**.
- Every finite group has a unitary representation.
- **Peter-Weyl theorem** states that “compact groups have the same structure as finite groups.”
- Hence, compact groups too have a unitary representation.
- On the other hand, boost transformations form an infinite set of elements.
- Boosts are parameterized by the *rapidity* (or velocity), which is a continuous real parameter.
- Boosts are generated by the non-compact part of the Lorentz algebra $SO(1,3)$.
- Since the group is continuous and non-compact, it has infinitely many elements—like translations or rotations. Hence, **no unitary representation**.

6.1 Lie Derivative

6.1.1 Congruence

A congruence is a family of smooth curves such that through every point in a region (or patch) of spacetime, exactly one curve passes.

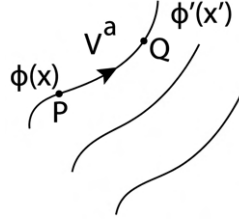
- Congruence implies a well-defined vector field and vice versa.
- *Diffeomorphism*: A map from spacetime into itself in a continuous manner, again in the same space.



Defining Lie Derivative

We have learned about congruence because —“The Lie derivative measures how a tensor field changes along a flow, and a congruence provides exactly that flow.”

- Lie derivative only needs congruence/vector field.
- Covariant derivative needs dot product (metric structure).



Let point P be mapped into Q due to congruence. Let the coordinate transformation be:

$$x^a \rightarrow x'^a = x^a + t v^a,$$

- It is a flow where t is a flow parameter (an infinitesimal step)
- v^a is the vector field generating the flow.

Therefore, the Lie derivative of a scalar field ϕ is defined as:

$$\mathcal{L}_v \phi = \lim_{t \rightarrow 0} \frac{\phi_{-t}(x') - \phi(x)}{t}$$

This describes the evolution of the field along the flow lines defined by the congruence.

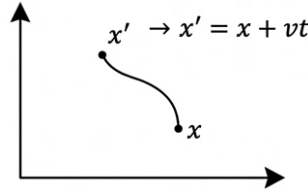
Note: Why $\phi_{-t}(x')$?

The evolution of the system itself. Negative time means going back to its own position.

$$\boxed{\mathfrak{D}} \xrightarrow{t} \boxed{\mathfrak{C}}$$

$\varphi(x) \qquad \varphi'(x')$

6.2 Lie Derivative for a Scalar Field



Here, only coordinates change, the field doesn't change. Therefore, for a scalar field, it is simply, $\phi'(x') = \phi(x)$, no involvement of direction. We want to calculate the field at the initial position, i.e.,

$$\begin{aligned} \phi'(x) &= \phi'(x' - vt) \\ &= \phi'(x') - t v^a \partial'_a \phi'(x') + \dots \quad [\text{Expanding in Taylor series}] \\ &= \phi(x) - t v^a \partial'_a \phi(x) \quad [\text{Only keeping the first-order term in } t] \end{aligned}$$

Therefore, the Lie derivative:

$$\begin{aligned} \mathcal{L}_v \phi &= \lim_{t \rightarrow 0} \frac{\phi'(x) - \phi(x)}{t} \\ &= \lim_{t \rightarrow 0} \frac{-t v^a \partial'_a \phi(x)}{t} \\ &= -v^a \partial_a \phi(x) \\ &= -(\vec{v} \cdot \nabla) \phi(x) \\ &= -\hat{v} \phi(x) \end{aligned}$$

This is nothing but the directional derivative in $3D$.

Hence, the Lie derivative of a scalar field is just the directional derivative of ϕ along the vector v^a . And it's a linear combination of partial derivatives weighted by the components of the vector field v^a .

6.3 Lie Derivative of a Vector Field

Similarly, $V(x)$ is a vector field defining the flow, and $W(x)$ is another vector field (the one we want to differentiate). Now, the lie derivative:

$$\mathcal{L}_V W^a = \lim_{t \rightarrow 0} \frac{W'^b(x) - W^a(x)}{t}$$

(where W'^b is the new vector field at position Q , while W^a is at position P)

This tells us how the vector field W changes when dragged along the flow generated by V .

Therefore,

$$\begin{aligned} x &\rightarrow x' + Vt \\ W^a(x) &\rightarrow W'^b(x') = \frac{\partial x'^a}{\partial x^m} W^m(x) \\ \Rightarrow W'^b(x') &= \frac{\partial}{\partial x^m} (x^a + V^a t) W^m(x) \\ \Rightarrow W'^b(x + Vt) &= (\delta_m^a + t \partial_m V^a) W^m(x) \\ \Rightarrow W'^b(x) + t V^b \partial'_b W'^b(x) &= \delta_m^a W^m(x) + t W^m(x) \partial_m V^a \\ \Rightarrow W'^b(x) - W^a(x) &= t (W^m \partial_m V^a - V^b \partial'_b W'^b) \\ \Rightarrow \frac{W'^b(x) - W^a(x)}{t} &= W^m \partial_m V^a - V^b \partial'_b W'^b \end{aligned}$$

$$\boxed{\therefore \mathcal{L}_V W^a = W^m \partial_m V^a - V^b \partial'_b W'^b = W^m \partial_m V^a - V^b \partial_b W^a}$$

This is the Lie derivative of a vector field. Let's simplify it,

$$\begin{aligned}
 (\mathcal{L}_{\hat{V}} W^a)(\phi) &= (W^m \partial_m V^a - V^b \partial_b W^a) \phi = (W^b \partial_b V^a - V^b \partial_b W^a) \phi \quad [m = b] \\
 &= (W^b \partial_b V^c - V^b \partial_b W^c) \partial_c \phi \\
 &= (W^b \partial_b V^c \partial_c \phi + W^b V^c \partial_b \partial_c \phi) - (V^b \partial_b W^c \partial_c \phi + W^b V^c \partial_b \partial_c \phi) \\
 &= W^b \partial_b (V^c \partial_c \phi) - V^b \partial_b (W^c \partial_c \phi) \\
 &= (\hat{W} \hat{V} - \hat{V} \hat{W}) \phi(x) \\
 &= [\hat{W}, \hat{V}] \phi(x) = -[\hat{V}, \hat{W}] \phi(x)
 \end{aligned}$$

$$\boxed{\therefore \mathcal{L}_{\hat{V}} W^a = -[\hat{V}, \hat{W}]^a}$$

Hence, two vector fields are mapped to another vector field. $[\hat{V}, \hat{W}]$ creates a vector field out of the two vector fields $\hat{V}$ and $\hat{W}$.

Note:

Recall the Jacobi identity:

$$[[A, B], C] + [[B, C], A] + [[C, A], B] = 0$$

The Jacobi identity is true for all algebra.

Exercise: Show that-

$$[\mathcal{L}_{\hat{V}}, \mathcal{L}_{\hat{W}}] = \mathcal{L}_{\hat{V}} \mathcal{L}_{\hat{W}} - \mathcal{L}_{\hat{W}} \mathcal{L}_{\hat{V}} = \mathcal{L}_{[\hat{V}, \hat{W}]}$$

SO far we've seen that a Lie derivative maps -

- Scalar to scalar
 - Vector to vector
 - Tensor to tensor.
- All of the same type.

Covector

A **covector** is a type of linear map that acts on vectors and returns real or complex numbers.

- A vector is an element of the space V .
- A covector is an element of the dual space V^* .
- Dot product of a vector and covector produces a scalar.

$$(\text{Vector})^b \cdot (\text{Covector})_b = \text{Scalar}$$

6.4 Lie Derivative of a Covector

$$\mathcal{L}_{\hat{V}}(W^a U_a) = -V^c \partial_c (W^a U_a)$$

$$\Rightarrow (\mathcal{L}_{\hat{V}} W^a) U_a + W^a (\mathcal{L}_{\hat{V}} U_a) = -\{ V^c (\partial_c W^a) U_a + V^c W^a (\partial_c U_a) \}$$

$$\Rightarrow (W^c \partial_c V^a - V^c \partial_c W^a) U_a + W^a (\mathcal{L}_{\hat{V}} U_a) = -V^c (\partial_c W^a) U_a - V^c W^a (\partial_c U_a)$$

$$\Rightarrow W^c \partial_c V^a U_a + W^a (\mathcal{L}_{\hat{V}} U_a) = -V^c W^a \partial_c U_a$$

$$\Rightarrow W^a (\mathcal{L}_{\hat{V}} U_a) = -(W^c \partial_c V^a U_a + V^c W^a \partial_c U_a)$$

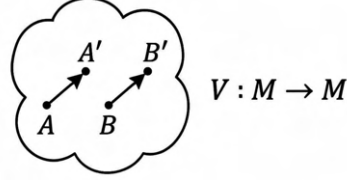
$$\Rightarrow W^a (\mathcal{L}_{\hat{V}} U_a) = -(W^a \partial_a V^c U_c + V^c W^a \partial_c U_a) = -W^a (\partial_a V^c U_c + V^c \partial_c U_a)$$

$$\boxed{\therefore \mathcal{L}_{\hat{V}} U_a = -(V^c \partial_c U_a + \partial_a V^c U_c)}$$

Exercise: Find the Lie derivative of a tensor field

$$\mathcal{L}_{\hat{V}} \tilde{T}_b^a$$

(Class-7) Lecture 4: Wed, June 18, 2025

Covariant Derivative**Deriving Lie Derivative from the Vector Field**

- Vector field maps from manifold to manifold.
- All Lie groups are manifolds, but all manifolds are not Lie groups.
- Lie derivative can be defined on any manifold.
- In a manifold, there is no composition rule.

Previous class discussion

- Adjoint map and Lie derivative are not same.
- Adjoint definition :

$$\langle v, Au \rangle = \langle A^*v, u \rangle$$

Whereas, Lie derivative:

$$\mathcal{L}_v u = -[u, v] = [v, u]$$

- Lie derivative and Lie algebra are not same.

$$f = f(x, y) \quad ; \quad g = g(x, y)$$

$$f = f(x, y) \quad ; \quad g = g(x, y)$$

$$v = f \frac{\partial}{\partial x} + g \frac{\partial}{\partial y}$$

$$[v, v'] = \frac{\partial f}{\partial x} \frac{\partial g}{\partial y} - \frac{\partial g}{\partial x} \frac{\partial f}{\partial y} \quad (7.1)$$

This (7.1) is not constant for Lie derivative. However,

$$[T^a, T^b] = f^{ab}{}_c T^c \quad (7.2)$$

The term $f^{ab}{}_c$ in (7.2) is **constant** in Lie algebra or Lie group.

Now, ξ^a = local cartesian coordinate. We know that,

$$ds^2 = (d\xi^a)^2 = \delta_{ab} d\xi^a d\xi^b \quad (7.3)$$

But we also know that,

$$\begin{aligned} d\xi^a &= \frac{\partial \xi^a}{\partial x^m} dx^m \\ \Rightarrow d\xi^a &= M^a_m dx^m \\ \therefore dx^m &= (M^{-1})^m_a d\xi^a \end{aligned} \quad (7.4)$$

Notes :

- A vector has to transform according to (7.4).
- If M^{-1} doesn't exist, we can only go to one way, known as singular. Example - the origin.

Using (7.4) to (7.3) we get

$$\begin{aligned} ds^2 &= \delta_{ab} \frac{\partial \xi^a}{\partial x^m} \frac{\partial \xi^b}{\partial x^n} dx^m dx^n \\ \therefore ds^2 &= G_{mn} dx^m dx^n \quad ; \quad G_{mn} = G_{nm} \quad (\text{symmetric}) \end{aligned}$$

If such symmetric G_{mn} exists which is also invertible, the manifold is called **Riemannian manifold**.

Since ds^2 is a scalar, it can also be given as:

$$\langle v, u \rangle \equiv g_{ab} V^a V^b$$

7.1 Killing Vector

[It is connected to symmetry hence related to a conserved quantity via Noether's theorem.]

Definition: A vector field X on M is called a Killing vector field if the Lie derivative of the metric with respect to X is zero:

$$\mathcal{L}_X g_{ab} = 0$$

This means that the metric doesn't change along the flow of X , i.e., distances and angles are preserved under the transformation generated by X .

Cyclic Coordinates

A cyclic coordinate is a generalized coordinate q_i that doesn't appear explicitly in the Lagrangian $\mathcal{L}(q, \dot{q}, t)$.

$$\frac{\partial \mathcal{L}}{\partial q_i} = 0$$

Which implies q_i is cyclic.

Also,

$$[\mathcal{L}_u, \mathcal{L}_v] = \mathcal{L}_{[u,v]}$$

Two Killing vectors also constitute another Killing vector.

7.2 Covariant Derivative

The **covariant derivative** is a generalization of the ordinary derivative to curved spaces (or to situations where vectors live in different tangent spaces at each point). To define it, we impose two basic requirements:

1. It must satisfy the **Leibniz rule** (like an ordinary derivative):

$$D(ab) = (Da)b + a(Db), \quad a, b \in A,$$

where A is the algebra of functions.

2. It should reduce to the usual derivative on scalars and vanish on constants:

$$D(c) = 0, \quad c \in \mathbb{R}.$$

To illustrate, $D(ca) = (Dc)a + c(da) = c(da)$
 $\therefore (Dc)$ needs to be zero.

Metric Compatibility

We also demand that the derivative preserves the metric:

$$Dg_{ab} = 0 \quad \Rightarrow \quad \nabla_m g_{ab} = 0.$$

Scalars

For a scalar field ϕ , the covariant derivative reduces to the ordinary derivative:

$$\nabla_a \phi = \partial_a \phi.$$

So scalars are straightforward, ∇_a hits on a scalar and produces a vector.

- $\nabla \rightarrow$ generic case
- $\partial \rightarrow$ flat space only

Vectors

For a vector, the situation is more subtle. In general,

$$\nabla_a v^b \neq \partial_a v^b.$$

To see why, consider a change of basis:

$$v^b \rightarrow v'^c = M^c_b v^b.$$

Differentiating:

$$\partial'_d v'^c = \partial'_d (M^c_b v^b) = M^c_b \partial'_d v^b + (\partial'_d M^c_b) v^b.$$

Here, the extra term

$$(\partial'_d M^c_b) v^b$$

is a problem — it mixes derivatives of the transformation matrix itself. The extra additive term spoils the transformation law of $\partial'_d v'^c$. It keeps the traces of previous map. In other words, the ordinary derivative of a vector does **not** transform like a vector.

Solution: Introduce a Connection

Therefore, to define a consistent derivative of tensors, we need to cancel this unwanted term. To cancel that “bad” term, one might try to demand

$$\partial'_d M^c_b = 0,$$

But this is not true in general coordinates. Instead, we add a correction term (the connection Γ) to the derivative:

$$\nabla_a v^b = \partial_a v^b + \Gamma_a^b{}_c v^c.$$

This ensures that the transformation law works out correctly. More generally, we write

$$\nabla_a = \partial_a + A_a,$$

Where A_a is the connection (called “connexion” in older books). Hence,

$$\nabla'_a(MU) = M \nabla_a U \quad \text{and} \quad \nabla'_a \rightarrow M^{-1} \nabla_a M.$$

Then the covariant derivative of a vector is:

$$\nabla_a v^b = \partial_a v^b + \Gamma_a^b{}_c v^c,$$

where $\Gamma_a^b{}_c$ are the connection coefficients (Christoffel symbols in Riemannian geometry).

Similarly, for a covector:

$$\nabla_a v_b = \partial_a v_b - \Gamma_{ab}^c v_c.$$

Summary

$$\begin{aligned} \nabla_a \phi &= \partial_a \phi && \text{(scalar)} \\ \nabla_a v^b &= \partial_a v^b + \Gamma_a^b{}_c v^c && \text{(vector)} \\ \nabla_a v_b &= \partial_a v_b - \Gamma_{ab}^c v_c && \text{(covector)} \end{aligned}$$

Now let,

$$\begin{aligned} U^a V_a &= \phi \\ \Rightarrow \nabla_c(U^a V_a) &= \partial_c \phi \\ \Rightarrow (\nabla_c U^a) V_a + U^a (\nabla_c V_a) &= \partial_c \phi \end{aligned}$$

Then,

$$\begin{aligned} \nabla_a(V_b W^c) &= (\nabla_a V_b) W^c + V_b (\nabla_a W^c) \\ &= (\partial_a V_b) W^c - (\Gamma_{ab}^d V_d) W^c - V_b (\partial_a W^c) - V_b (\Gamma_a^b{}_c W^c) \\ &= \partial_a(V_b W^c) - \Gamma_{ab}^d V_d W^c - \Gamma_a^b{}_c W^c V_b \\ \therefore \nabla_a g_{bc} &= \partial_a g_{bc} - \Gamma_{ab}^d g_{dc} - \Gamma_{ac}^d g_{bd} \end{aligned}$$

Previously, we know that:

$$\begin{aligned} \mathcal{L}_V W_a &= -(V^c \partial_c W_a + W_c \partial_a V^c) \\ \therefore \mathcal{L}_V U^a &= V^c \partial_c U^a - U^c \partial_c V^a \end{aligned}$$

HW

- Calculate $\mathcal{L}_v g_{ab}$ **Hint** : Let choose,

$$g_{ab} = P_a Q_b$$

$$\mathcal{L}_v(P_a Q_b) = (\mathcal{L}_v P_a) Q_b + P_a (\mathcal{L}_v Q_b)$$

$$\mathcal{L}_v g_{ab} = V^c \partial_c g_{ab} + g_{ac} \partial_b V^c + g_{cb} \partial_a V^c$$

- Verify if the partial derivative is switched with the covariant derivative ($\partial \rightarrow \nabla$) the Lie derivative won't change.

Here one way is given for the second task,

$$\begin{aligned} \mathcal{L}'_V U^a &= V^c \nabla_c U^a - U^c \nabla_c V^a \\ &= V^c (\partial_c U^a + \Gamma_c^a{}_b U^b) - U^c (\partial_c V^a + \Gamma_c^a{}_b V^b) \\ &= (V^c \partial_c U^a - U^c \partial_c V^a) + \Gamma_c^a{}_b U^b V^c - \Gamma_c^a{}_b U^c V^b \\ &= \mathcal{L}_V U^a + \Gamma_c^a{}_b U^b V^c - \Gamma_b^a{}_c U^b V^c \\ &= \mathcal{L}_V U^a \quad [\Gamma_c^a{}_b = \Gamma_b^a{}_c] \quad (\text{Torsion-free condition}) \\ \therefore \mathcal{L}'_V U^a &= \mathcal{L}_V U^a \end{aligned}$$

One covariant derivative and another covariant derivative is related with their connection:

$$\begin{aligned} [\nabla_a, \nabla_b] \phi &= 0 \\ [\partial_a, \partial_b] \phi &= 0 \end{aligned}$$

Also,

$$\begin{aligned} \nabla_c (g_{ab} U^b) &= g_{ab} (\nabla_c U^b) + (\nabla_c g_{ab}) U^b \\ &= g_{ab} \nabla_c U^b \end{aligned}$$

This suggests that the latter term must be zero,

$$\boxed{\nabla_c g_{ab} = 0}$$

This is known as the **Metricity condition**.

(Class-8) Lecture 5: Sat, June 21, 2025

Classical Field Theory

Recall the Line element:

$$ds^2 = g_{ab} dx^a dx^b$$

Notes

In GR, one takes $g_{ab} = g_{ab}(x)$ itself as a dynamical field: $g_{ab}(x) = g_{ba}(x)$
 Also, in the **Finsler metric** g not only depends on x but also the derivative,

$$g_{ab} = g_{ab}(x, \partial x)$$

It is used in transportation.

Now,

$$g'_{ab}(x') = g_{mn} \frac{\partial x^m}{\partial x'^a} \frac{\partial x^n}{\partial x'^b}$$

$$G \rightarrow G' = M^T G M \quad ; \quad M = \frac{\partial x}{\partial x'}$$

From multivariable calculus,

$$\begin{aligned} d^n x &\rightarrow d^n x' = \left| \frac{\partial x'}{\partial x} \right| d^n x \\ \Rightarrow d^n x' &= \det(M^{-1}) d^n x = \frac{1}{\det(M)} d^n x \\ \Rightarrow \frac{\partial x'}{\partial x} &= \det \left(\frac{\partial x'^m}{\partial x^n} \right) \quad (\text{Jacobian}) \end{aligned}$$

Now,

$$\begin{aligned} \det(G') &= \det(M^T G M) = \det(M^T) \det(G) \det(M) \\ &= [\det(M)]^2 \det(G) \\ \therefore \sqrt{\det(G')} &= |\det(M)| \sqrt{\det(G)} \\ \Rightarrow \sqrt{g'} d^n x' &= \sqrt{g} d^n x \\ \therefore \sqrt{g} d^n x &\text{ is an invariant volume element} \end{aligned}$$

8.1 DeWitt Notation

$$a \cdot b = a_\mu b^\mu$$

Quantum Mechanics (QM) says, in the Schrödinger picture:

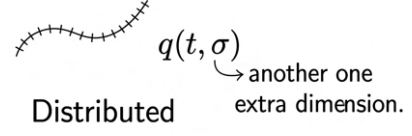
$$\begin{aligned} \psi(x) &= \langle x | \psi \rangle \quad \rightarrow \quad x \text{ component of } \psi \\ \Rightarrow \langle \phi | \psi \rangle &= \int \langle \phi | x \rangle \langle x | \psi \rangle d\mu(x) \\ &= \int \phi^*(x) \psi(x) d\mu(x) = \phi_i^* \psi_i \quad \rightarrow \quad \text{DeWitt} \end{aligned}$$

8.2 Point vs Field

localized at only one point



Point



Distributed

Field

- **Point:** $q(t)$, in only one point.
- **Field:** $\phi(x) = \phi(t, x^i)$

Now action,

$$S = \int \mathcal{L} dt = \int \left[\int d^3x \mathcal{L}[\phi] \right] dt$$

Tenet:

- Action is invariant under a set of transformations.
- $\delta S = 0 \rightarrow$ equation of motion
- $\mathcal{L}[\phi]$ will be a polynomial of ϕ and its first-order derivative

Classification of symmetries

There are several ways to classify symmetries. Symmetries can be discrete or continuous. They can also be global or local. For elementary particles, we can define two general classes of symmetries:

- **Spacetime symmetries:** These symmetries correspond to transformations on a field theory acting explicitly on the spacetime coordinates,

$$x^\mu \mapsto x'^\mu(x^\nu), \quad \mu, \nu = 0, 1, 2, 3.$$

Examples are rotations, translations and, more generally, Lorentz- and Poincaré transformations defining the global symmetries of special relativity, as well as general coordinate transformations that are the local symmetries defining general relativity.

- **Internal symmetries:** These are symmetries that correspond to transformations of the different fields in a field theory,

$$\Phi^a(x) \mapsto M^a_b \Phi^b(x).$$

Roman indices a, b label the corresponding fields. If M^a_b is constant then the symmetry is a global symmetry; in case of spacetime-dependent $M^a_b(x)$ the symmetry is called a local symmetry.

Now,

$$\begin{aligned} t &\rightarrow -t && \text{(time reversal)} \\ x &\rightarrow -x && \text{(space inversion)} \end{aligned}$$

These are discrete symmetries — **no conserved quantity**. However if,

$$x'^{\mu} = (\Lambda w x)^{\mu} + a w^{\mu} \rightarrow \text{continuous symmetry}$$

According to Noether's theorem, there's a conserved quantity for **continuous symmetry**.

Notes

Coleman–Mandula Theorem (No-go Theorem):

A sarcastic way to put it is - Internal symmetry cannot be married off to space-time symmetry within the framework of Lie group theory. **Example:** Generators:

$$M = (J + P) + T$$

Can't write like this so that:

$$[J, T] \neq 0$$

and

$$JT^+ |-\frac{1}{2}\rangle = J | \frac{1}{2}\rangle$$

Example:

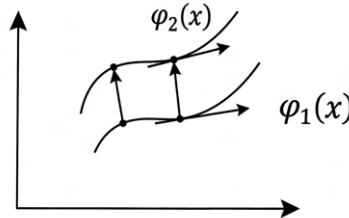
$$\begin{aligned} S[\phi] &= \alpha \phi_i \phi_i + \beta \partial \phi_i \partial \phi_i + \gamma_i \phi_i \\ &= \int d^4x [\alpha \phi(x) \phi(x) + \beta \partial_{\mu} \phi(x) \partial^{\mu} \phi(x) + \gamma_i(x) \phi_i(x)] \end{aligned}$$

[looking at this, it is too cumbersome without the DeWitt notation]

Now taking variation,

$$\delta S = 2\alpha \phi_i \delta \phi_i + 2\beta \partial \phi_i \delta(\partial \phi_i) + \gamma_i \delta \phi_i$$

The variation is assumed to be between two differentiable configurations.



$$\begin{aligned} \delta\phi(x) &= \phi_2(x) - \phi_1(x) \\ \Rightarrow \partial [\delta\phi(x)] &= \partial(\phi_2(x) - \phi_1(x)) \\ &= \partial\phi_2(x) - \partial\phi_1(x) \\ \therefore \partial(\delta\phi) &= \delta(\partial\phi) \end{aligned}$$

So, **variation and differentiation commute**. Hence,

$$\begin{aligned}
 \delta S &= 2\alpha \phi_i \delta \phi_i + 2\beta \partial \phi_i \partial (\delta \phi_i) + \gamma_i \delta \phi_i \\
 &= (2\alpha \phi_i + \gamma_i) \delta \phi_i - 2\beta \partial^2 \phi_i \delta \phi_i + 2\beta \partial \phi_i \delta \phi_i \Big|_{\text{boundary}} \\
 &= \delta \phi_i (2\alpha \phi_i + \gamma_i - 2\beta \partial^2 \phi_i)
 \end{aligned}$$

Since $\delta \phi_i$ is arbitrary,

$$\begin{aligned}
 2\alpha \phi_i + \gamma_i - 2\beta \partial^2 \phi_i &= 0 \\
 \Rightarrow (2\beta \partial^2 - 2\alpha) \phi_i &= \gamma_i
 \end{aligned}$$

- If $\alpha = \alpha(x)$:

$$(2\beta \partial^2 - 2\alpha) \phi_i = \gamma_i \quad (\text{No problem})$$

- But if $\beta = \beta(x)$:

$$2\partial (\beta \partial \phi_i) - 2\alpha_i \phi_i = \gamma_i$$

To match with the Klein-Gordon equation,

$$(\partial^2 + m^2) \phi = 0$$

We need

$$\begin{aligned}
 2\beta &= +1 \\
 2\alpha &= -m^2 \\
 \gamma &= 0
 \end{aligned}$$

Exercise: Do the similar procedure for this action -

$$S[\phi] = \alpha \phi_i \phi_i + \beta \partial \phi_i \partial \phi_i + \gamma_i \phi_i + K_i \partial \phi_i$$

Note: The last term is important for π -meson.

Note

A huge problem in Quantum Mechanics is linear independence which results in the potential $V(x)$. $V(x)$ is not being a spacetime invariant. A fundamental theory should not have the potential $V(x)$.

Klein-Gordon Action:

$$\begin{aligned}
 S &= \frac{1}{2} (\partial \phi)^2 - \frac{1}{2} m^2 \phi^2 + J \phi \\
 &= \int d^4x \left[\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 + J(x) \phi(x) \right]
 \end{aligned}$$

8.3 Electromagnetism

Equation of Motion (EOM):

$$\partial_\mu F^{\mu\nu} = j^\nu \quad \rightarrow \quad 4 \text{ equations}$$

Where,

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$$

$$\vec{B} = \vec{\nabla} \times \vec{A}$$

$$\vec{E} = -\vec{\nabla} A_0 - \frac{\partial \vec{A}}{\partial t}$$

Thus we get two sets of equations from the equation of motion:

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\varepsilon_0} \quad (\text{M1})$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{j} + \frac{\partial \vec{E}}{\partial t} \quad (\text{M3})$$

The remaining two are found from:

$$\partial_{[\mu} F_{\nu\lambda]} = \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} + \partial_\lambda F_{\mu\nu} = 0 \quad [\text{Bianchi identity}]$$

$$\vec{\nabla} \cdot \vec{B} = 0 \quad (\text{M2})$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad (\text{M4})$$

Notes

- There are total 8 Maxwell's (scaler) equations — 2 scalar divergence equations, 6 scalar components from the 2 vector curl equations.
- The vector equations evolve with time; their validity is confirmed from the previous two equations.
- Maxwell's EOM are equivalent to Hamilton's EOM, not the Lagrange's.

Now for the action,

$$S = \int \mathcal{L}(\phi, \partial\phi) \sqrt{-g} d^4x$$

The determinant of g is negative so we put a minus sign inside to make the quantity under the square root positive. However we omit it in flat space:

$$\begin{aligned} \delta S &= \int \left[\frac{\partial \mathcal{L}}{\partial \phi} \delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta(\partial_\mu \phi) \right] d^4x \\ &= \int \left[\frac{\partial \mathcal{L}}{\partial \phi} \delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\mu(\delta\phi) \right] d^4x \\ &= \int \left[\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) \right] \delta\phi d^4x + \int d^3x \eta^\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta\phi \\ &= 0 \quad \left(\int d^3x \eta^\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta\phi = 0 \rightarrow \text{Dirichlet condition} \right) \end{aligned}$$

Verifying Dirichlet condition

Let's verify this,

$$\begin{aligned}\nabla \cdot (\psi \nabla \phi) &= \nabla \psi \cdot \nabla \phi + \psi \nabla^2 \phi \\ \Rightarrow \int \nabla \phi \cdot \nabla \psi \, d^3x &= - \int \psi \nabla^2 \phi \, d^3x + \int \nabla \cdot (\psi \nabla \phi) \, d^3x \\ &= - \int \psi \nabla^2 \phi \, d^3x + \int \psi (n \cdot \nabla) \phi \, dS\end{aligned}$$

Post-class discussions

Time-like Space:

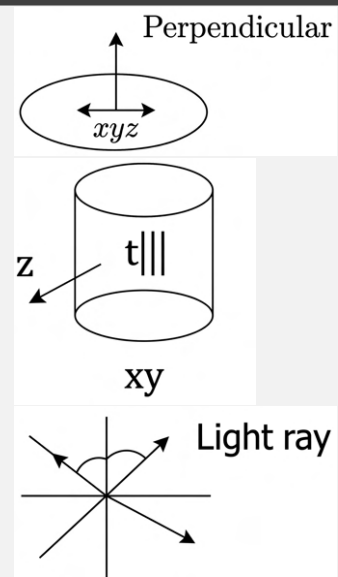
- Perpendicular is in a time-like direction.

Space-like Space:

- Perpendicular is in z-direction.

Null-like Space:

- Null object's perpendicular is also null.
- E.g., light ray.



(Class-9) Lecture 6: Sat, June 28, 2025

Gauge Theory**9.1 Gauge Field****9.1.1 Free Field**Maxwell field $\equiv$ EM fieldElectromagnetism is **not** a theory of **field strength tensor**, but a **theory of potential**.

Now,

$$\mathcal{L} = -\frac{1}{4}F^{ab}F_{ab} \quad ; \quad F_{ab} = \partial_{[a}A_{b]} = \frac{1}{2}(\partial_a A_b - \partial_b A_a)$$

Note about the factor- $\frac{1}{2}$

- If we put $\frac{1}{4}$ in the Lagrangian, the $\frac{1}{2}$ is necessary in F_{ab} , otherwise we can omit it.
- ' $\frac{1}{2}$ ' was also used in *Penrose's* book

To find the equation of motion from the Lagrangian, we vary A ,

$$\begin{aligned}
 \delta\mathcal{L} &= -\frac{1}{4}F^{ab} \delta F_{ab} - \frac{1}{4}\delta F^{ab} F_{ab} \\
 &= -\frac{1}{2}F_{ab} \delta F^{ab} \\
 &= -\frac{1}{2}F_{ab} \delta(\partial^a A^b - \partial^b A^a) \\
 &= -F_{ab} \delta(\partial^a A^b) \\
 \therefore \delta\mathcal{L} &= -F_{ab} \partial^a(\delta A^b) = 0 \quad [\text{however, } \delta A^b \neq 0] \\
 \therefore \partial^a F_{ab} &= 0 \quad (\text{Free Maxwell Equation})
 \end{aligned} \tag{9.1}$$

Derivation of (9.1)

$$F_{ab} (\partial^a C^b - \partial^b C^a) = F_{ab} \partial^a C^b - F_{ab} \partial^b C^a \tag{1}$$

$$= F_{ab} \partial^a C^b - F_{ba} \partial^a C^b \tag{2}$$

$$= F_{ab} \partial^a C^b + F_{ab} \partial^a C^b \quad [\because F_{ab} = -F_{ba}] \tag{3}$$

$$\therefore F_{ab} (\partial^a C^b - \partial^b C^a) = 2F_{ab} \partial^a C^b = 2F_{ab} \partial^a A^b \quad [C \rightarrow A] \tag{4}$$

9.1.2 In Presence of a Source

$$\begin{aligned}
 \partial^a F_{ab} &= J_b \\
 \Rightarrow \partial^a F_{ab} - J_b &= 0
 \end{aligned}$$

and variation:

$$F_{ab} \partial^a(\delta A^b) - J_b \delta A^b = 0$$

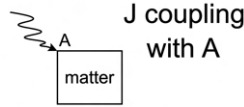
Thus in presence of a source the Lagrangian becomes,

$$\mathcal{L} = -\frac{1}{4}F_{ab}F^{ab} - J_b A^b$$

$$J_b A^b \rightarrow \text{Interaction term}$$

Notes

- Everything else other than the gauge field is matter.
- In the theory of gravitation, electromagnetism will act as a matter.



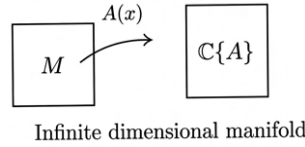
Therefore,

$$J^b = -\frac{\delta \mathcal{L}_{\text{matter}}}{\delta A^b(x)}$$

Question: When can a current couple to a photon?

Answer: Let's do a gauge transformation.

$$A_a \rightarrow A'_a = A_a + \partial_a F \quad (9.2)$$



Notes

- It's like a translation in internal space.
- Translation of a configuration space.
- Transformation is thought as a translation in extra dimension.

Now, we want the action (Lagrangian) to be invariant under gauge transformation (9.2).

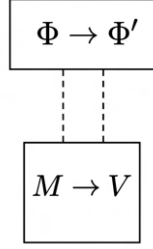
$$\begin{aligned} -J_b A^b &\rightarrow -J_b (A^b + \partial^b F) \\ &= -J_b A^b - J_b \partial^b F \\ &= -J_b A^b + (\partial^b J_b) F \quad [\text{integration by parts}] \end{aligned}$$

- F must vanish on boundaries
- F is not arbitrary.
- Therefore the current must be conserved.

$$\therefore \partial^b J_b = 0 \quad ; \quad \text{for all } J$$

- Classical EOM $\neq$ Whole Configuration Space.
- Here, we extremize action.

9.2 Covariant Derivative



$$\phi \rightarrow \phi' = M\phi \quad ; \quad M = M(x)$$

$$\partial_\mu \phi \rightarrow \partial_\mu \phi' = \partial_\mu (M\phi) = M\partial_\mu \phi + (\partial_\mu M)\phi$$

The latter term $(\partial_\mu M)\phi$ denotes there is another space dependency on ϕ

M is a matrix valued. Now we have to add terms whose linear addition removes this garbage (extra term). Therefore,

$$D_\mu \phi \rightarrow MD_\mu \phi \equiv D'_\mu \phi' \equiv D'_\mu (M\phi)$$

Derivative is not hermitian, it is anti-hermitian. Hence we write,

$$D_\mu \phi = \partial_\mu \phi + iA_\mu \phi \quad \text{instead of} \quad \partial_\mu \phi + A_\mu \phi.$$

Also D_μ is an associative operation.

$$\begin{aligned} D'_\mu &= M D_\mu M^{-1} \\ \Rightarrow (\partial_\mu + iA'_\mu) &= M(\partial_\mu + iA_\mu)M^{-1} \\ \Rightarrow \partial_\mu + iA'_\mu &= \partial_\mu + M(\partial_\mu M^{-1}) + iMA_\mu M^{-1} \quad [\because M\partial_\mu M^{-1} = \partial_\mu] \\ \Rightarrow iA'_\mu &= M(\partial_\mu M^{-1}) + iMA_\mu M^{-1} \\ \Rightarrow iA'_\mu &= M\partial_\mu M^{-1} + iMA_\mu M^{-1} \\ \therefore A'_\mu &= MA_\mu M^{-1} - iM\partial_\mu M^{-1} \end{aligned}$$

Findings:

- D is matrix valued.
- D_μ transforms covariantly $\rightarrow$ Tensor
- A_μ doesn't transform covariantly $\rightarrow$ Not a tensor

Now,

$$\begin{aligned} \tilde{F}_{\mu\nu} &= \frac{1}{i}[D_\mu, D_\nu] \\ &= \frac{1}{i}[\partial_\mu \mathbb{I} + iA_\mu, \partial_\nu \mathbb{I} + iA_\nu] \\ &= \frac{1}{i}\{i[\mathbb{I}\partial_\mu, A_\nu] + i[A_\mu, \mathbb{I}\partial_\nu] + i^2[A_\mu, A_\nu]\} \quad (\because [\partial_\mu, \partial_\nu] = 0) \\ \therefore \tilde{F}_{\mu\nu} &= \partial_\mu A_\nu - \partial_\nu A_\mu + i[A_\mu, A_\nu] \\ \therefore [D_\mu, D_\nu] &\text{ is a curvature} \end{aligned}$$

Now,

$$\begin{aligned}
 \tilde{F}_{\mu\nu} &= M F_{\mu\nu} M^{-1} \\
 &= -\frac{1}{i} [D_\mu, D_\nu] \\
 &= -\frac{1}{i} [M D_\mu M^{-1}, M D_\nu M^{-1}] \\
 &= -\frac{1}{i} M [D_\mu, D_\nu] M^{-1} \\
 \therefore \tilde{F}_{\mu\nu} &= M \tilde{F}_{\mu\nu} M^{-1}
 \end{aligned}$$

Now, for a transformation $\phi \rightarrow \phi' = M\phi$

$$A_\mu \rightarrow M A_\mu M^{-1} - i M \partial_\mu M^{-1} \quad (9.3a)$$

$$\bar{A}_\mu \rightarrow M \bar{A}_\mu M^{-1} - i M \partial_\mu M^{-1} \quad (9.3b)$$

Subtracting (9.3b) to (9.3a) we get

$$(\bar{A}_\mu - A_\mu) \rightarrow M (\bar{A}_\mu - A_\mu) M^{-1}$$

Therefore, the variation $(\bar{A}_\mu - A_\mu)$ of a connection acts as a tensor, not A_μ itself.

Extra information

- a) **Affine space:** the origin is not known.
- b) If A is a matrix, we can define some norm out of it:
 - $\det A \rightarrow$ not additive
 - $\text{Tr } A \rightarrow$ additive

Now,

$$\begin{aligned}
 S_{YM} &= -\frac{1}{4} \int d^D x \text{Tr} (\tilde{F}_{\mu\nu} \tilde{F}^{\mu\nu}) \\
 \delta S_{YM} &= -\frac{1}{2} \int d^D x \text{Tr} (\tilde{F}^{\mu\nu} \delta \tilde{F}_{\mu\nu}) = 0
 \end{aligned} \quad (9.4)$$

Here,

$$\begin{aligned}
 \delta \tilde{F}_{\mu\nu} &= \frac{1}{i} \delta [D_\mu, D_\nu] \\
 &= \frac{1}{i} \delta (D_\mu D_\nu - D_\nu D_\mu) \\
 &= \frac{1}{i} \{ (\delta D_\mu) D_\nu + D_\mu (\delta D_\nu) - (\delta D_\nu) D_\mu - D_\nu (\delta D_\mu) \} \\
 &= -\frac{1}{i} \{ [\delta D_\mu, D_\nu] + [D_\mu, \delta D_\nu] \} \\
 &= [\delta A_\mu, D_\nu] + [D_\mu, \delta A_\nu] \\
 \therefore \delta \tilde{F}_{\mu\nu} &= [D_\mu, \delta A_\nu] - [D_\nu, \delta A_\mu]
 \end{aligned} \quad (9.5)$$

Recall for electromagnetism,

$$\delta F_{\mu\nu} = \partial_\mu (\delta A_\nu) - \partial_\nu (\delta A_\mu) = [\partial_\mu, \delta A_\nu] - [\partial_\nu, \delta A_\mu]$$

Using (9.5) in (9.4) we get,

$$\begin{aligned}
 \delta S_{YM} &= -\frac{1}{2} \int d^D x \operatorname{Tr} \left(\tilde{F}^{\mu\nu} \left\{ [D_\mu, \delta A_\nu] - [D_\nu, \delta A_\mu] \right\} \right) \\
 &= -\frac{1}{2} \int d^D x \operatorname{Tr} \left(\tilde{F}^{\mu\nu} [D_\mu, \delta A_\nu] - \tilde{F}^{\mu\nu} [D_\nu, \delta A_\mu] \right) \\
 &= - \int d^D x \operatorname{Tr} \left(\tilde{F}^{\mu\nu} [D_\mu, \delta A_\nu] \right) \\
 \therefore \delta S_{YM} &= \int d^D x \operatorname{Tr} \left([D_\mu, \tilde{F}^{\mu\nu}] \delta A_\nu \right) = 0
 \end{aligned}$$

$$\therefore [D_\mu, \tilde{F}^{\mu\nu}] = 0 \quad (\text{Here, } \rightarrow A^1 + A^2 \rightarrow A^3 \text{ highly nonlinear})$$

Derivation of the trace term under the integral sign,

$$\operatorname{Tr}(A[X, B]) = -\operatorname{Tr}([X, A]B)$$

$$\begin{aligned}
 \operatorname{Tr}(A[X, B]) &= \operatorname{Tr}(AXB - ABX) \\
 &= \operatorname{Tr}(XAB) + \operatorname{Tr}(AXB) \\
 &= -\operatorname{Tr}(\{XA - AX\}B)
 \end{aligned}$$

$$\therefore \operatorname{Tr}(A[X, B]) = -\operatorname{Tr}([X, A]B)$$

$$\therefore \operatorname{Tr}(\tilde{F}_{\mu\nu}[D_\mu, \delta A_\nu]) = -\operatorname{Tr}([D_\mu, \tilde{F}_{\mu\nu}]\delta A_\nu)$$

(Class-10) Lecture 7: Sat, July 5, 2025

Path Integral Formalism**10.1 Quantization****What is Quantization?**

In quantum mechanics, canonical quantization is a recipe that takes us from the Hamiltonian formalism of classical dynamics to the quantum theory. The recipe tells us to take the generalized coordinates q_a and their conjugate momenta p^a and promote them to operators.

Example: In the harmonic oscillator, we quantize by defining:

- Ladder operators: $a, a^\dagger$
- Number states: $|n\rangle$

Then, we can calculate overlaps like:

$$\langle m | e^{i\hat{H}t/\hbar} | n \rangle$$

Overlaps

In quantum mechanics, states are represented as vectors in a Hilbert space. The overlap between two quantum states $|\psi\rangle$ and $|\phi\rangle$ is their inner product:

$$\langle \phi | \psi \rangle \quad (\text{Complex number})$$

This is only possible because we quantized—i.e., built a Hilbert space of states and defined how operators act.

No Hilbert Space = No quantization.

Why Quantization?

- Matches with experimental results.
- The ultimate goal is to calculate **overlaps**.

Note

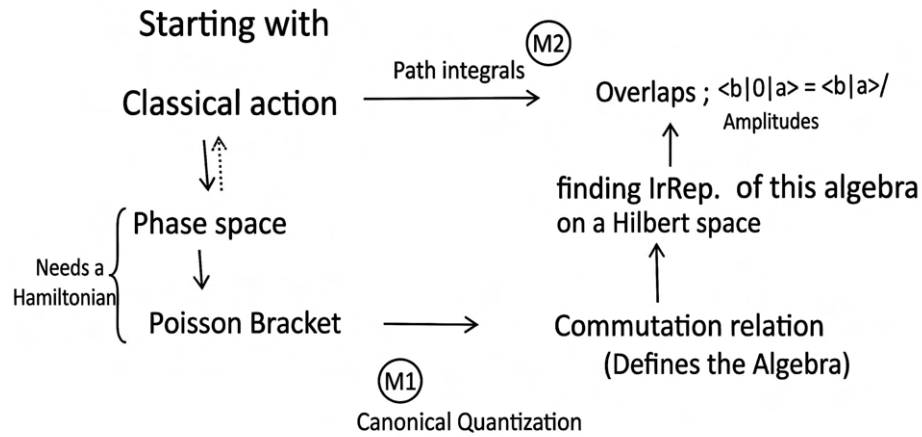
Why in quantum mechanics we take $[a, a^\dagger] = 1$ but why not $a^\dagger |0\rangle = 0$?

Because if $a^\dagger |0\rangle = 0$, then:

$$(x + ip)\psi = 0 \quad \Rightarrow \quad \left(x - \frac{d}{dx}\right)\psi = 0$$

If $\psi \sim Ae^{+x^2/2}$, it won't be normalizable.

10.2 Quantization Methods



GNS Construction

GNS Construction : Algebra $\rightarrow$ Hilbert space

Regular Representation of a Group (Algebra):

$$gg' = g'g$$

Think of a vector:

$$R(g) \begin{pmatrix} g_1 \\ g_2 \\ \vdots \\ g_n \end{pmatrix} = \begin{pmatrix} g'_1 \\ g'_2 \\ \vdots \\ g'_n \end{pmatrix}$$

- Regular representation is not necessarily an irreducible representation (irrep).

Adjoint Representation

$$[T(g), g] = g''$$

- Adjoint Rep. is an irrep.

Also,

Algebra $\rightarrow$ existence of a cyclic vector $\equiv$ choice of a vacuum state

Back to Hamiltonian Formulation

The Hamiltonian depends on a choice of coordinates. It's not a covariant concept.

1. We find H from conjugate momenta.
2. Then need Lagrangian $\mathcal{L}$
3. Then need **dot** i.e., time derivatives
4. Which is frame dependent
5. Proper time needs to be defined

Hence, not good from the vantage point of General Relativity.

Haag's Theorem

Statement: The interaction picture does not rigorously exist in interacting relativistic quantum field theory.

10.3 Pictures in Quantum Mechanics

Pictures	State	Operator
Schrödinger	✓	×
Heisenberg	×	✓

QFT $\rightarrow$ Infinite Degrees of Freedom

Schrödinger Picture: Needs a state vector which is a **functional** of the degrees of freedom (DOFs).

In one particle,

- $\hat{x} \rightarrow$ can be independent of time
- $\psi \rightarrow$ can be time dependent

$\therefore \hat{x} |\psi\rangle \rightarrow$ eigenvalue can be dependent on time.

In Quantum Field Theory,

$$\psi(x) \rightarrow \psi[\phi]$$

This is because x is just a **label**, so relabeling does not change the wavefunction.

Also for one particle,

$$\psi \sim e^{-x^2/2}$$

But for many particles,

$$\psi \sim e^{-x_1^2/2} \cdot e^{-x_2^2/2} \dots \sim \prod_i e^{-x_i^2/2}$$



For Hamiltonian:

$$H = \sum_i p_i^2 + V = - \sum_i \left(\frac{\partial^2}{\partial x_i^2} \right) + V$$

As we can see there are infinite derivatives $\left(\frac{\partial}{\partial x} \right)$ and we need to find their eigenvalues. This is why the Heisenberg picture is often preferred where, **states are independent of time**.

Notice, in Quantum Mechanics:

$$[x_i, p_j] = i\hbar\delta_{ij}$$

- Eigenvalues are unbounded - $1 \leq i, j \leq \mathbb{N}$ - Degrees of freedom are **finite**

In Quantum Field Theory:

$$[\phi(x), \pi(y)] = i\delta^3(x-y)\hat{\mathbb{I}}$$

- Degrees of freedom are **infinite** - Representations of the commutation relations are **not unitarily equivalent** **Another way to realize this:**

$$[x_i, p_j] = i\hbar\delta_{ij}$$

The trace of the right-hand side is non-zero:

$$\text{Tr}(\mathbb{I}) \neq 0$$

But the trace of the left-hand side:

$$\text{Tr}(xp - px) = \begin{cases} 0 & (\text{finite d.o.f.}) \\ \neq 0 & (\text{infinite d.o.f.}) \end{cases}$$

Free Particle Wavefunction

$$\begin{aligned} \psi(x, t) &= e^{i(kx - \omega t)} \\ &= e^{\frac{i}{\hbar}(px - Et)} \\ &= e^{\frac{i}{\hbar} \int (p \frac{dx}{dt} - E) dt} \\ &= e^{\frac{i}{\hbar} \int \mathcal{L} dt} \\ &= e^{\frac{i}{\hbar} S} \end{aligned}$$

Also,

$$\psi(x, t) = \langle x | \psi, t \rangle_S = \langle x, t | \psi \rangle_H$$

We are interested in this "kernel":

$$\langle x', t' | x, t \rangle$$

In path integrals, the kernel often refers to the propagation amplitude between states. For example, in quantum mechanics we write:

$$K(x_f, t_f; x_i, t_i) = \langle x_f, t_f | x_i, t_i \rangle = \int \mathcal{D}x(t) e^{\frac{i}{\hbar} S[x]},$$

which is called the *kernel of propagation*.

Why This Kernel?

We assume we are dealing with linear equations:

$$\hat{L}\phi(x) = J(x) \quad ; \quad \text{Effect} \rightarrow \text{Cause} \quad (10.1)$$

This is a local equation. Recall Ohm's law:

$$\begin{aligned} J &= \sigma E \\ \Rightarrow J(x) &= \sigma(x) E(x) \quad (\text{Local, like Maxwell's equations}) \\ \text{But, } V &= IR \quad (\text{Bilocal: Involves two points}) \end{aligned}$$

Inversion of (10.1) yields:

$$\phi(x) = \int G(x, y) J(y) dy$$

Where:

- $G(x, y)$: Green's function (kernel).
- $\hat{L} G(x, y) = \delta(x, y)$; here, $\delta(x, y)$ is a localized delta function, not $\delta(x - y)$, since the system lacks translation symmetry.

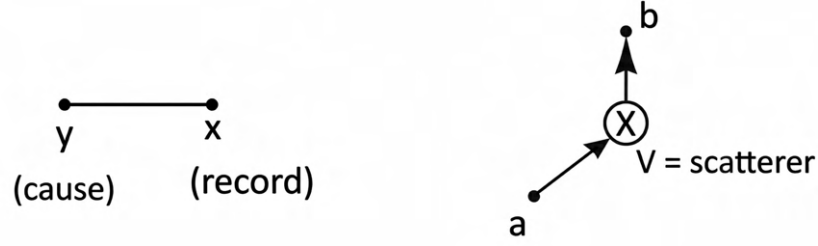
Checking (10.1):

$$\begin{aligned} \hat{L}\phi(x) &= \int \hat{L} G(x, y) J(y) dy \\ &= \int \delta(x, y) J(y) dy \\ &= J(x) \\ \therefore \phi(x) &= \int G(x, y) J(y) dy \end{aligned}$$

Important Notes:

- $\delta(x, y) \rightarrow$ Localized cause
- $G(x, y)$ is not necessarily symmetric
- Operator $\hat{L}$ possesses translation symmetry

Effect at point x caused by a source of unit strength localized at point y .

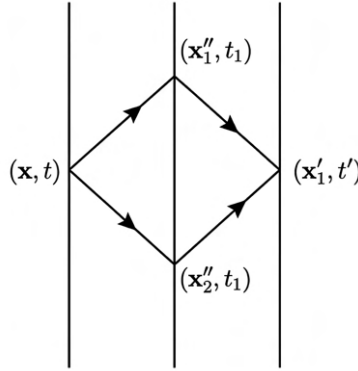


That is:

$$\langle b|V|a\rangle$$

From right to left.

10.4 Double Slit



$$\begin{aligned}\langle x', t'|x, t\rangle &= \sum_i \langle x', t'|x''_i, t_2\rangle \langle x''_i, t_2|x, t\rangle \\ &= \int dx'' \langle x', t'|x''\rangle \langle x''|x, t\rangle\end{aligned}$$

Completeness Theorem:

$$\int dx'' |x''\rangle \langle x''| = \mathbb{I} \quad \rightarrow \quad \text{must be a complete set.}$$

All calculations hereafter are to be done on a discrete space-time and we will take the continuum limit.

Hence,

$$\langle x', t'|x, t\rangle = \int \prod_{i=1}^N dx_i \langle x', t'|x_{i-1}\rangle \langle x_{i-1}|x_{i-2}\rangle \langle x_{i-2}|\dots\rangle \langle x_1|x, t\rangle$$

Now, we need to measure:

$$\langle x', t + dt|x, t\rangle$$

Recall,

$$\langle x, t|\psi\rangle_H = \langle x|\psi, t\rangle_S$$

Schrödinger equation:

$$\begin{aligned}
 i \frac{\partial}{\partial t} |\psi(t)\rangle &= \hat{H} |\psi\rangle \\
 \Rightarrow |\psi(t)\rangle &= e^{-i\hat{H}t} |\psi(0)\rangle = e^{-i\hat{H}t} |\psi\rangle_H \\
 \therefore \langle x|\psi, t\rangle_S &= \langle x|e^{-i\hat{H}t} |\psi\rangle_H
 \end{aligned}$$

$$\boxed{\therefore \langle x', t + dt|x, t\rangle = \langle x', \Delta t|x, 0\rangle = {}_H\langle x'|e^{-i\hat{H}\Delta t}|x\rangle_H}$$

$$\left[\text{here, } {}_H\langle x'|e^{-i\hat{H}\Delta t}|x\rangle_H \rightarrow \text{measure at the same time} \right]$$

Note as $\Delta t \rightarrow 0$,

$$\langle x', \Delta t|x, 0\rangle = \delta(x - x') \quad (\text{compare at the same time})$$

Hence,

$$\begin{aligned}
 \langle x'|e^{-i\hat{H}\Delta t}|x\rangle &= \int dp \langle x'|e^{-i\hat{H}\Delta t}|p\rangle \langle p|x\rangle \\
 &= \int dp dp' \langle x'|p'\rangle \langle p'|e^{-i\hat{H}\Delta t}|p\rangle \langle p|x\rangle \\
 &= \int dp dp' \frac{1}{2\pi} e^{ip'x'} \langle p'|e^{-i\hat{H}\Delta t}|p\rangle e^{-ipx} \\
 &\quad \left[\because \langle x|p\rangle = \frac{1}{\sqrt{2\pi}} e^{ipx}; \hbar = 1 \right] \\
 &= \frac{1}{2\pi} \int dp dp' e^{i(p'x' - px)} \langle p'|e^{-i\hat{H}\Delta t}|p\rangle
 \end{aligned}$$

Now, for a free particle,

$$\hat{H} = \frac{\hat{p}^2}{2m} \quad [V(x) \text{ not discussed here}]$$

Hence,

$$\begin{aligned}
 \langle p'|e^{-i\hat{H}\Delta t}|p\rangle &= \langle p'|e^{-i\frac{\hat{p}^2}{2m}\Delta t}|p\rangle \\
 &= e^{-i\frac{p^2}{2m}\Delta t} \langle p'|p\rangle \\
 &= e^{-i\frac{p^2}{2m}\Delta t} \delta(p - p')
 \end{aligned}$$

Therefore,

$$\begin{aligned}
 \langle x'|e^{-i\hat{H}\Delta t}|x\rangle &= \frac{1}{2\pi} \int dp dp' e^{i(p'x' - px)} e^{-i\frac{p^2}{2m}\Delta t} \delta(p - p') \\
 &= \frac{1}{2\pi} \int dp e^{i(px' - px)} e^{-i\frac{p^2}{2m}\Delta t}
 \end{aligned}$$

Using the Gaussian integral identity:

$$\int_{-\infty}^{\infty} dz e^{-az^2 + bz} = \sqrt{\frac{\pi}{a}} e^{\frac{b^2}{4a}}$$

We get,

$$\begin{aligned}
&= \frac{1}{2\pi} \int dp e^{-\left(\frac{i\Delta t}{2m}\right)^2 p^2} e^{ip[x'-x]} \\
&= \frac{1}{2\pi} \left(\frac{\pi}{\frac{i\Delta t}{2m}} \right)^{1/2} e^{-\frac{(x'-x)^2}{4i\Delta t/(2m)}} \\
&= \left(\frac{m}{2\pi i\Delta t} \right)^{1/2} e^{i\frac{m}{2} \left(\frac{x'-x}{\Delta t} \right)^2 \Delta t} \\
&= \left(\frac{m}{2\pi i\Delta t} \right)^{1/2} e^{i\frac{m}{2} v^2 \Delta t} \\
&= A e^{i\mathcal{L}\Delta t} \quad ; \quad \left[A = \left(\frac{m}{2\pi i\Delta t} \right)^{1/2} \right]
\end{aligned}$$

So,

$$\langle x', t' | x, t \rangle = A \int dx' dx'' \prod_i e^{\frac{im}{2} \left[\frac{x_{i+1} - x_i}{\Delta t} \right]^2 \Delta t} = \int \mathcal{D}x e^{iS[x]}$$

Where $\mathcal{D}x = A \prod dx_i$ is an infinite-dimensional factor.

(Class-11) Lecture 8 : Sat, July 12, 2025

Green's Function in QFT**11.1 Time-Ordered Products**

In QFT, the time-ordered product of operators is an operation that arranges a sequence of field operators such that they are ordered with **decreasing time arguments** (latest time first).

Why do we need time-ordered products?**Issue: 1**

We need the time-ordered product in QFT because, in the interaction picture, it ensures the correct chronological ordering of operators (as quantum operators do not commute) in *Dyson's formula*, which is essential for computing the perturbative expansion of the S-matrix while preserving causality.

For scattering, which is also an operator:

$$\hat{S} = \lim_{\substack{t_i \rightarrow -\infty \\ t_f \rightarrow +\infty}} \mathcal{U}(t_f, t_i) = \hat{T} \exp \left(-\frac{i}{\hbar} \int_{t_i}^{t_f} H_{\text{int}}(t) dt \right)$$

Here, $\hat{T} \rightarrow$ amounts to time ordering operator

For additive systems:

$$\hat{T}[A + B] = \hat{T}[A] + \hat{T}[B]$$

Say, if we are dealing with an otherwise free scalar field interacting with a **classical source** $J(x)$, which can be thought of as disturbance from outside, then $H_{\text{int}}(t)$ becomes:

$$H_{\text{int}}(t) = \int d^3x J(x) \hat{\phi}(x)$$

$J(x)$ is not an operator, it's just a classical source. It has no fluctuation in time.

$$\hat{S} = \hat{T} \left[\exp \left(-i \int J(x) \hat{\phi}(x) d^4x \right) \right] \quad ; \quad [\hbar = 1 \text{ hereafter}]$$

Expanding yields:

$$\begin{aligned} \hat{S} &= \hat{T} \left[\sum_{n=0}^{\infty} \frac{(-i)^n}{n!} \prod_{i=1}^n \int d^4x_i [J(x_1)J(x_2) \cdots J(x_n) \hat{\phi}(x_1) \hat{\phi}(x_2) \cdots \hat{\phi}(x_n)] \right] \\ &= \sum_{n=0}^{\infty} \frac{(-i)^n}{n!} \prod_{i=1}^n \int d^4x_i [J(x_1)J(x_2) \cdots J(x_n) \hat{T} \hat{\phi}(x_1) \hat{\phi}(x_2) \cdots \hat{\phi}(x_n)] \end{aligned}$$

Here, $\hat{T}$ will act only on the field operators. $J(x_i)$ is not an operator at all. Hence,

$$\hat{T}[\phi(x)] = \phi(x) \quad ; \quad \text{one field only, no time-ordering needed.}$$

$$\hat{T}[\phi(x_1)\phi(x_2)] = \theta(t_1 - t_2)\phi(x_1)\phi(x_2) + \theta(t_2 - t_1)\phi(x_2)\phi(x_1)$$

- For bosonic field, there will be a “+ve” sign in between the two terms.
- For fermionic field, there will be a “-ve” sign in between.

- $x_i = (t_i, \vec{x}_i)$ - For each field, a classical source is needed to generate it.
- $\theta(\text{argument}) = \theta(x)$ - This is called the **Heaviside Step Function**, a piecewise-defined function used to “turn on” expressions depending on the sign of its argument.

$$\theta(x) = \begin{cases} 1 & x > 0 \\ 0 & x < 0 \end{cases} \quad \left[\text{sometimes } \theta(0) = \frac{1}{2} \text{ is also used} \right]$$

•

$$\theta(t_1 - t_2) = \begin{cases} 1 & ; \text{ if } t_1 \text{ is later in time than } t_2 \\ 0 & ; \text{ if } t_1 \text{ is earlier} \end{cases}$$

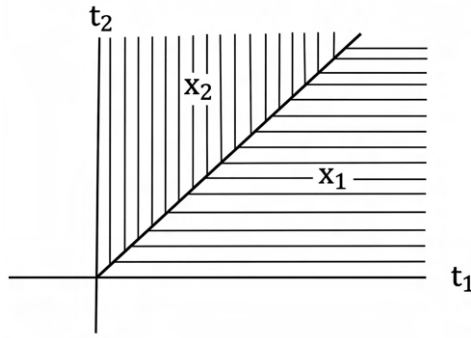
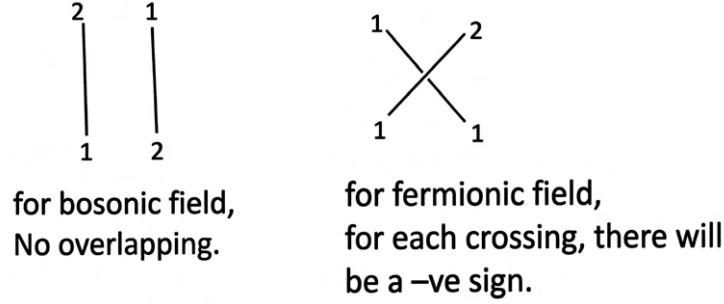


Figure Explanation

The diagram is a spacetime plot in the $t_1 - t_2$ plane. Each point corresponds to a pair of times (t_1, t_2) .

- **Diagonal line:** $t_1 = t_2$
→ This is the boundary where both operators are evaluated at the same time.
- **Region above the diagonal:** $t_2 > t_1$
→ Labeled as x_2 : corresponds to the $\theta(t_2 - t_1)$ term.
→ The time-ordered product becomes: $\phi(x_2)\phi(x_1)$
- **Region below the diagonal:** $t_1 > t_2$
→ Labeled as x_1 : corresponds to the $\theta(t_1 - t_2)$ term.
→ The time-ordered product becomes: $\phi(x_1)\phi(x_2)$

Pictorial View



Similarly, for three fields $\phi(x_1), \phi(x_2), \phi(x_3) \equiv \phi_1, \phi_2, \phi_3$, there are $3! = 6$ permutations; so 6 terms will appear.

$$\begin{aligned} \hat{T}[\phi(x_1)\phi(x_2)\phi(x_3)] = & \theta(t_1 - t_2)\theta(t_2 - t_3)\phi_1\phi_2\phi_3 + \theta(t_1 - t_3)\theta(t_3 - t_2)\phi_1\phi_3\phi_2 \\ & + \theta(t_2 - t_1)\theta(t_1 - t_3)\phi_2\phi_1\phi_3 + \theta(t_3 - t_2)\theta(t_2 - t_1)\phi_3\phi_2\phi_1 \\ & + \theta(t_3 - t_1)\theta(t_1 - t_2)\phi_3\phi_1\phi_2 + \theta(t_2 - t_3)\theta(t_3 - t_1)\phi_2\phi_3\phi_1 \end{aligned}$$

No need to give the hat sign again on operators ; as $\hat{T}$ always acts on operators.

11.2 Green's Function

Definition: The vacuum expectation value of a time-ordered product of field operators is called a *Green's function* (or *correlation function*) in quantum field theory.

Therefore, for a scalar field $\phi(x)$, the 2-point Green function is:

$$G(x_1, x_2) = \langle 0 | \hat{T}[\phi(x_1)\phi(x_2)] | 0 \rangle$$

This is also called the *Feynman propagator*.

Why do we need time-ordered products?

Issue: 2

For example, consider the Klein-Gordon (KG) field with no self-interaction. The Lagrangian is:

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi_i\partial^\mu\phi_i - \frac{1}{2}m^2\phi_i^2$$

From this, the equation of motion will be:

$$(\square + m^2)\phi_i = 0 \quad \Rightarrow \quad (\partial^2 + m^2)\phi_i = 0$$

Reminder - DeWitt notation

$$\phi_i^2 = \phi_i \phi_i = \int \phi(x)\phi(x) dx$$

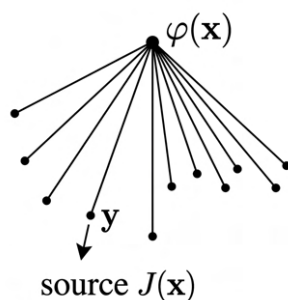
Now, if you need to observe something, you need interaction, which will change the energy or momentum of the system. Hence, in the presence of interaction, the equation of motion will become:

$$(\square + m^2)\phi_i = J_i$$

And the solution would be:

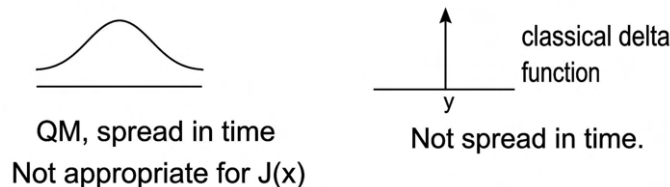
$$\phi_i = \phi_i^0 + \int G(x, y) J(y) d^4 y$$

Where ϕ_i^0 is the solution in the absence of J . It is like the background/noise without the source. This equation is valid for linear superposition.



Interpretation

Here, $J(y)$ should be a classical source. When we have no idea of the input, but we want to calibrate the interacting system, then we need a source that is well defined and has no fluctuation, if it exists at all.



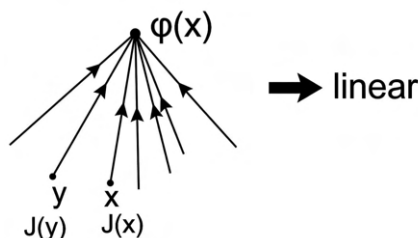
Question

How do we identify a QFT, and how do we say that we know a QFT?

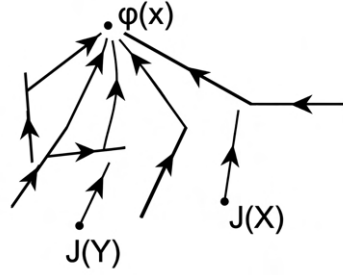
Answer

QFT is identified by all its Green's functions. Thus by knowing all the Green's functions, we can say that we know the theory is a QFT.

In interacting theory, it suffices to know the 2-point linear function $J(x)$ to get the Green's function.



If it's not linear



Non-linear

then the solution will be,

$$(\square + m^2)\phi_i = J_i - \lambda\phi_i^2 = J''$$

and

$$\phi_i = \phi_i^0 + \int G(x, y) [J(y) - \lambda\phi_i^2] d^4y$$

Therefore, in terms of Green's function, the equation of motion of the KG field will be -

$$(\square + m^2)G(x, y) = i\delta^4(x - y)$$

where,

$$G(x - y) = \langle 0 | T[\phi(x)\phi(y)] | 0 \rangle \quad ; \quad | 0 \rangle = \text{vacuum state}$$

In other words, $G_F(x, y)$ is the solution to the inhomogeneous Klein-Gordon equation. It's the function that gives the response of the scalar field to a point-like source at y .

Example: Harmonic Oscillator

We know that the differential equation for a driven harmonic oscillator,

$$\left(\frac{d^2}{dt^2} + \omega_0^2 \right) x(t) = \frac{f(t)}{m}$$

Now we define the Green's function $G(t - t')$ as the solution to:

$$\left(\frac{d^2}{dt^2} + \omega_0^2 \right) G(t - t') = \delta(t - t')$$

which means the system's response to a point impulse at time t' .

And given:

$$\delta(t) = \frac{1}{2\pi} \int_{\mathbb{R} \rightarrow (-\infty, +\infty)} d\omega e^{-i\omega t}$$

Setting $t' = 0$, then:

$$G(t) = \frac{1}{2\pi} \int_{\mathbb{R}} d\omega G(\omega) e^{-i\omega t}$$

Where $G(\omega) \rightarrow 0$ as $\omega \rightarrow \pm\infty$, and ω is the driving frequency.

Now, applying the operator $\left(\frac{d^2}{dt^2} + \omega_0^2\right)$ to both sides:

$$\begin{aligned} \left(\frac{d^2}{dt^2} + \omega_0^2\right) G(t) &= \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} G(\omega) \left(\frac{d^2}{dt^2} + \omega_0^2\right) e^{-i\omega t} = \delta(t) \\ &\Rightarrow \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} (-\omega^2 + \omega_0^2) G(\omega) e^{-i\omega t} = \frac{1}{2\pi} \int d\omega e^{-i\omega t} \\ &\Rightarrow (\omega_0^2 - \omega^2) G(\omega) = 1 \\ &\Rightarrow G(\omega) = \frac{1}{\omega_0^2 - \omega^2} \\ \therefore G(t) &= \frac{1}{2\pi} \int_{\mathbb{R}} \left(\frac{1}{\omega_0^2 - \omega^2}\right) e^{-i\omega t} d\omega \end{aligned}$$

Exercise:

- Find the solution of the $G(t)$.

11.3 Quantization

First, the unequal-time commutation relation between $\hat{x}$ and $\hat{p}$ for a quantum harmonic oscillator will be as follows:

In the Heisenberg picture, the operators evolve with time, not the states. Therefore, the time evolution of an operator $\hat{\mathcal{O}}$ is given by:

$$\hat{\mathcal{O}}(t) = e^{i\hat{H}t/\hbar} \hat{\mathcal{O}}(0) e^{-i\hat{H}t/\hbar}.$$

Now, for a harmonic oscillator with Hamiltonian:

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2$$

By solving the Heisenberg equations we get:

$$\begin{aligned} x(t) &= x(0) \cos(\omega t) + \frac{p(0)}{m\omega} \sin(\omega t) \\ \hat{p}(t) &= \hat{p}(0) \cos(\omega t) - m\omega \hat{x}(0) \sin(\omega t) \end{aligned}$$

Now,

$$\begin{aligned} [x(t), p(t')] &= \left[x(0) \cos(\omega t) + \frac{p(0)}{m\omega} \sin(\omega t), p(0) \cos(\omega t') - m\omega x(0) \sin(\omega t') \right] \\ &= \cos(\omega t) \cos(\omega t') [x(0), p(0)] + \frac{\sin(\omega t) \cos(\omega t')}{m\omega} [p(0), p(0)] \\ &\quad - m\omega \cos(\omega t) \sin(\omega t') [x(0), x(0)] - \frac{\sin(\omega t) m\omega \sin(\omega t')}{m\omega} [p(0), x(0)] \\ &= \cos(\omega t) \cos(\omega t') (i\hbar) + \sin(\omega t) \sin(\omega t') (i\hbar) \quad (\because [x, p] = i\hbar; [p, x] = -i\hbar) \\ &= i\hbar [\cos(\omega t) \cos(\omega t') + \sin(\omega t) \sin(\omega t')] = i\hbar \cos(\omega(t - t')) \\ \therefore [x(t), p(t')] &= i\hbar \cos(\omega(t - t')) \end{aligned}$$

If $t' = t$, then $\cos \omega(t - t') = 1$.

$$\therefore [x(t), p(t)] = i\hbar \quad ; \quad \text{the usual result}$$

Hence, for a particle:

$$[x_i(t), p_j(t)] = i\hbar \delta_{ij}$$

For a field:

$$\begin{aligned} [\phi(\mathbf{x}), \pi(\mathbf{y})] &= i\hbar \delta^3(\mathbf{x} - \mathbf{y}) \quad \text{and,} \\ [\phi(\mathbf{x}), \phi(\mathbf{y})] &= [\pi(\mathbf{x}), \pi(\mathbf{y})] = 0 \end{aligned}$$

Recall:

$$\langle 0 | T [\phi(x) \phi(y)] | 0 \rangle = G(x - y)$$

Now:

$$\begin{aligned} &(\partial_x^2 + m^2) \langle 0 | T [\phi(x) \phi(y)] | 0 \rangle \\ \Rightarrow &\left(\frac{\partial^2}{\partial t_x^2} - \nabla_x^2 + m^2 \right) \langle 0 | \theta(t_x - t_y) \phi_x \phi_y + \theta(t_y - t_x) \phi_y \phi_x | 0 \rangle \end{aligned}$$

Breaking down the differentiation part: it is symmetric in $x \leftrightarrow y$

$$(-\nabla_x^2 + m^2) [\theta(t_x - t_y) \phi(x) \phi(y)] = \theta(t_x - t_y) [(-\nabla_x^2 + m^2) \phi_1] \phi_2$$

and replacing $\pi = \dot{\phi}$ and $\phi(x_1) = \phi_1$, $\phi(x_2) = \phi_2$, $t_x = t_1$, $t_y = t_2$

$$\begin{aligned} \frac{\partial^2}{\partial t^2} [\theta(t_1 - t_2) \phi_1 \phi_2] &= \partial_{t_1} [\delta(t_1 - t_2) \phi_1 + \theta(t_1 - t_2) \dot{\phi}_1] \phi_2 \\ &= [\delta'(t_1 - t_2) \phi_1 + \delta(t_1 - t_2) \dot{\phi}_1 + \theta(t_1 - t_2) \ddot{\phi}_1 + \delta(t_1 - t_2) \dot{\phi}_1] \phi_2 \\ &= \delta'(t_1 - t_2) \phi_1 \phi_2 + 2\delta(t_1 - t_2) \dot{\phi}_1 \phi_2 + \theta(t_1 - t_2) \ddot{\phi}_1 \phi_2 \end{aligned}$$

Where we have used :

1. The derivative of step function is the delta function:

$$\frac{d\theta}{dt} = \delta(t)$$

- 2.

$$\frac{d^2}{dx^2}(uv) = \frac{d^2u}{dx^2} + 2\frac{du}{dx}\frac{dv}{dx} + \frac{d^2v}{dx^2}$$

Now, putting in the KG equation yields,

Lesson:

Path integrals naturally calculate the time-ordered correlators.

(Class-12) Lecture 9 : Sat, July 16, 2025

Schwinger-Dyson Equation

Path integral begins by defining a partition function and postulated as:

$$Z[g] = \int_{\text{all config}} \mathcal{D}\phi e^{iS[\phi, g]}$$

Here, $\mathcal{D}\phi \equiv$ measure over paths = spacetime history
 g = a parameter, can also be a set g as well.

12.1 Lemma

$$\int_{\text{all config}} \mathcal{D}F(x) \frac{\delta G[F(x)]}{\delta F} = 0$$

It follows from Newton's fundamental theorem of calculus:

$$\int_{x=a}^{x=b} \frac{dF}{dx} dx = F(b) - F(a)$$

Now,

$$\begin{aligned} \delta G[F(x)] &= G[F + \delta F] - G[F] \quad ; \quad \text{There will be NO variation in } x \text{ here.} \\ \Rightarrow \frac{\delta G[F(x)]}{\delta F} &= \lim_{\delta F \rightarrow 0} \frac{G[F + \delta F] - G[F]}{\delta F(x)} \end{aligned}$$

Hence,

$$G[F + f] - G[F] = 0$$

Because it is just translation in function.

Therefore, the lemma follows: the space function to be considered is closed.

Hence the result :

$$\begin{aligned} \int \mathcal{D}\phi \frac{\delta \left[e^{\frac{i}{\hbar} S[\phi]} \right]}{\delta \phi} &= 0 \\ \Rightarrow i \int \mathcal{D}\phi e^{iS[\phi]} \frac{\delta S}{\delta \phi} &= 0 \end{aligned}$$

Under the assumption that this integral is dominated by a single configuration and the fact $e^{iS[\phi]} \neq 0$, hence

$$\frac{\delta S}{\delta \phi} = 0 \quad : \text{ classical EOM.}$$

Now, if there is a classical source $J(x)$

$$\begin{aligned} Z[J] &= \mathcal{N} \int \mathcal{D}\phi e^{i(S[\phi] + J \cdot \phi)} = e^{iW[J]} \quad \text{and} \\ J \cdot \phi &= \int d^4x J(x) \phi(x) \end{aligned}$$

The choice of the normalization constant $\mathcal{N}$ will be fixed by demanding $Z[0] = 1$ and W can be real or complex.

Now, if we want to have expectation value of ϕ :

$$\langle \phi(J) \rangle = \int \mathcal{D}\phi \phi e^{i(S[\phi] + J \cdot \phi)} \Big|_{J=0}$$

Now, replacing $\phi \rightarrow \frac{1}{i} \frac{\delta}{\delta J}$ and $\frac{i}{\hbar} \longleftrightarrow 1$, we get:

$$\begin{aligned} \langle \phi(J) \rangle &= \frac{1}{i} \int \mathcal{D}\phi \frac{\delta}{\delta J(x)} e^{i(S[\phi] + J \cdot \phi)} \\ &= \frac{1}{i} \frac{\delta}{\delta J(x)} \int \mathcal{D}\phi e^{i(S[\phi] + J \cdot \phi)} = \frac{1}{i} \frac{\delta}{\delta J} [e^{iW[J]}] \end{aligned}$$

Where J is the conjugate or the Fourier transform of ϕ .

Now, from the previous lemma:

$$\begin{aligned} &\int \mathcal{D}\phi \frac{\delta S}{\delta \phi} e^{i(S[\phi] + J \cdot \phi)} = 0 \\ \Rightarrow &\frac{\delta S}{\delta \phi} \Big|_{\phi \rightarrow \frac{1}{i} \frac{\delta}{\delta J}} \int \mathcal{D}\phi e^{i(S[\phi] + J \cdot \phi)} = 0 \\ \Rightarrow &\frac{\delta S}{\delta \phi} \Big|_{\phi \rightarrow \frac{1}{i} \frac{\delta}{\delta J}} e^{iW[J]} \mathbb{I} = 0 \end{aligned}$$

Now lets look at,

$$\begin{aligned} \partial_x [e^{i\theta(x)} \psi(x)] &= (\partial_x e^{i\theta(x)}) \psi(x) + e^{i\theta(x)} \partial_x \psi(x) \\ &= e^{i\theta(x)} [\partial_x(x) + i(\partial_x \theta(x))] \psi(x) \end{aligned}$$

Applying this in the previous eqn we get,

$$\begin{aligned} &e^{iW[J]} \left[\frac{\delta S}{\delta \phi} \Big|_{\phi \rightarrow \phi_c + \frac{1}{i} \frac{\delta}{\delta J}} \right] \mathbb{I} = 0 \\ \therefore &\left[\frac{\delta S}{\delta \phi} \Big|_{\phi \rightarrow \phi_c + \frac{1}{i} \frac{\delta}{\delta J}} \right] \mathbb{I} = 0 \quad [\because e^{iW[J]} \neq 0] \end{aligned}$$

Not same as classical one. It's the quantum mechanically corrected classical equation. This is called the famous **Schwinger–Dyson equation**.

Example

12.2 Anharmonic oscillator

In DeWitt notation

$$S = \frac{1}{2} \partial \phi \partial \phi - \frac{1}{2} m^2 \phi^2 - \frac{1}{3!} g \phi^3$$

(potential)

Now, we start with the Lagrangian density:

$$\mathcal{L} = \frac{1}{2} (\partial_\mu \phi) (\partial^\mu \phi) - \frac{1}{2} m^2 \phi^2 - \frac{g}{3!} \phi^3.$$

The action is

$$S[\phi] = \int d^4x \mathcal{L}.$$

Therefore, variation of the action is:

$$\delta S = \int d^4x \left[(\partial_\mu \phi)(\partial^\mu \delta\phi) - m^2 \phi \delta\phi - \frac{g}{2} \phi^2 \delta\phi \right].$$

Applying integration by parts to the first part of the kinetic term:

$$= - \int d^4x (\partial_\mu \partial^\mu \phi) \delta\phi + \text{boundary term}.$$

The boundary term vanishes, so

$$\delta S = \int d^4x \left[-(\partial_\mu \partial^\mu \phi) \delta\phi - m^2 \phi \delta\phi - \frac{g}{2} \phi^2 \delta\phi \right].$$

Then factoring out $\delta\phi$:

$$\delta S = \int d^4x \delta\phi(x) \left(-\partial^2 \phi(x) - m^2 \phi(x) - \frac{g}{2} \phi^2(x) \right).$$

Thus

$$\frac{\delta S}{\delta\phi(x)} = -\partial^2 \phi(x) - m^2 \phi(x) - \frac{g}{2} \phi^2(x).$$

The Euler–Lagrange equation reads

$$\frac{\delta S}{\delta\phi(x)} = 0 \quad \Rightarrow \quad (-\partial^2 - m^2)\phi(x) - \frac{g}{2}\phi^2(x) = 0.$$

Final Result

$$\boxed{(-\partial^2 - m^2)\phi(x) - \frac{g}{2}\phi^2(x) = 0}$$

Hence, the correct classical equation of motion for the cubic anharmonic oscillator in QFT is

$$\frac{\delta S}{\delta\phi} = (\partial^2 + m^2)\phi + \frac{g\phi^2}{2} = 0$$

Now, the Schwinger–Dyson equation for this system is obtained by shifting ϕ ,

$$\phi : \phi \rightarrow \phi + \frac{\hbar}{i} \frac{\delta}{\delta J}$$

That is,

$$\begin{aligned} \phi \mathbb{I} &\rightarrow \phi + \frac{\hbar}{i} \frac{\delta}{\delta J} \mathbb{I} = \phi \\ \phi \mathbb{I} &\rightarrow \left(\phi + \frac{\hbar}{i} \frac{\delta}{\delta J} \right) \phi = \phi^2 + \frac{\hbar}{i} \frac{\delta \phi}{\delta J} \end{aligned}$$

Therefore, the equation of motion turns out to be

$$\left(\square + m^2 \right) \phi + \frac{g\phi^2}{2} + \frac{\hbar}{2i} \frac{\delta \phi}{\delta J} = 0$$

(Class-13) Lecture 10 : Sat, Aug 2, 2025

Green's Function (n-point)**13.1 ϕ^4 Theory**

Recall from the previous class:

$$\left(\frac{\delta S}{\delta \phi} + J \right) \mathbb{I} \Big|_{\phi \rightarrow \phi_c + \frac{\hbar}{i} \frac{\delta}{\delta J}} = 0$$

The Lagrangian is

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)^2 - \frac{1}{2}m^2 \phi^2 - \frac{g}{4!} \phi^4$$

The equation of motion will be:

$$\left[\left((\square + m^2) + \frac{g}{3!} \phi^3 \right) \phi \right] \mathbb{I} \Big|_{\phi \rightarrow \phi_c + \frac{\hbar}{i} \frac{\delta}{\delta J}} = 0$$

This can also be written as $(\phi \mathbb{I})$.

Now, in particular, a shift is observed:

$$\begin{aligned} \frac{g}{3!} \phi^3 \mathbb{I} &= \frac{g}{3!} \left(\phi_c + \frac{\hbar}{i} \frac{\delta}{\delta J} \right) \left(\phi_c + \frac{\hbar}{i} \frac{\delta}{\delta J} \right) \phi_c \\ &= \frac{g}{3!} \left(\phi_c + \frac{\hbar}{i} \frac{\delta}{\delta J} \right) \left(\phi_c^2 + \frac{\hbar}{i} \frac{\delta \phi_c}{\delta J} \right) \\ &= \frac{g}{3!} \left[\phi_c^3 + \underbrace{3 \frac{\hbar}{i} \phi_c \frac{\delta \phi_c}{\delta J} + \left(\frac{\hbar}{i} \right)^2 \frac{\delta^2 \phi_c}{\delta J^2}}_{\text{Quantum correction}} \right] \end{aligned}$$

Why is the $\frac{\delta \phi_c}{\delta J} = G_{ii} = G$ is the Green's function?

Because we know,

$$\hat{A} \phi = J \quad ; \quad \phi = \phi + GJ \quad \therefore \frac{\delta \phi}{\delta J} = G$$

A quick rundown to Statistical Mechanics

Partition function:

$$Z = \sum_{\text{configs}} e^{-\beta H} \longrightarrow \int d(\text{phase space}) e^{-\beta H(p,q,\dots)} = e^{-\beta F}$$

Here, F = free energy. There are two kinds of free energy: **Helmholtz** and **Gibbs**. Also for this system, the Internal energy U is the Hamiltonian.

$$\underbrace{\boxed{S} \boxed{S} \boxed{S} \boxed{S}}_{\lambda} \dots U(S,V,N)$$

$$U(\lambda S, \lambda V, \lambda N) = \lambda U(S, V, N)$$

Euler equation for a homogeneous function:

$$S \frac{\partial U}{\partial S} + V \frac{\partial U}{\partial V} + N \frac{\partial U}{\partial N} = U$$

$$\therefore ST - PV + \mu N = U \quad \left[\frac{\partial U}{\partial S} = T, \quad \frac{\partial U}{\partial V} = -P \right]$$

This is called the Gibbs–Duhem equation. Here, the S, V, N are pure extensive variables. After applying the Legendre transformation, we can get:

$$\begin{array}{ccc} U(S,V) & \longrightarrow & A(T,V) \quad \text{Helmholtz} \\ \downarrow & & \downarrow \\ H(S,P) & \longrightarrow & G(T,P) \quad \text{Gibbs} \\ \text{Enthalpy (mixed)} & & \end{array}$$

- Enthalpy $H(S, P)$ and Helmholtz energy $A(T, V)$ both have extensive and intensive variables.
- Gibbs free energy $G(T, P)$ contains only intensive variables.

What is the Landau theory of phase transition?

→ The system can be considered as a field. Here, ϕ is called the *order parameter* because it behaves differently in different phases. Also the Hamiltonian is not unique.

$$\uparrow \otimes \rightleftharpoons \downarrow \otimes \rightarrow \Phi_{ab}$$

The relativistic superconductor is similar with QCD. Microscopic d.o.f = quark, gluon, but we see meson.

Back to lecture

From the quantum field

$$\phi \rightarrow \phi_c + \frac{\hbar}{i} \frac{\delta}{\delta J}, \quad \phi_c \equiv \text{Landau order parameter}$$

$$H = H_0 + J\phi$$

Remember, E, H are the field J , and D, B are the induction (not field).

Then,

$$Z = e^{-\beta F} = e^{\beta A} \Rightarrow Z[J] = \int \mathcal{D}\phi e^{i(S+J\cdot\phi)} = e^{iW[J]}$$

Note: All of these are Schwingerian ways of doing things.

13.2 Free theory

Free theory means there is no interaction present. The action is written as in the presence of a classical source:

$$S = \frac{1}{2} \phi^T A \phi + J^T \phi, \quad A = -(\square + m^2)$$

$$S = \frac{1}{2} (\partial\phi)^2 - \frac{1}{2} m^2 \phi^2$$

$$\therefore Z[J] = \int \underline{\mathcal{D}\phi} e^{\frac{1}{2} \phi^T A \phi + J^T \phi} \quad [\mathcal{D}\phi \equiv \prod d\phi]$$

A = symmetric matrix with *positive* eigenvalues.

$$\Lambda = \text{diag}(\lambda_1, \lambda_2, \lambda_3, \dots), \quad x^T A x = y^T \Lambda y, \quad y = R x, \quad R^T A R = \Lambda, \quad \text{Jacobian} = \det R$$

Hence,

$$\therefore Z[J] = \int \prod_{i=1}^N dx_i e^{-x^T A x + b^T x}$$

Measure is rotation invariant.

Now, recall the integration for a multilinear function:

$$\int \prod_i dy_i e^{-\lambda_i y_i^2 + (b'^T y)} \quad \text{where } b' = R^T b$$

$$= \prod_{i=1}^N \left(\frac{\pi}{\lambda_i} \right)^{1/2} \exp \left[\frac{1}{4} b'^T A^{-1} b' \right]$$

$$= \frac{\pi^{N/2}}{\sqrt{\det A}} \exp \left[\frac{1}{4} b'^T A^{-1} b' \right]$$

$$b^T b = b'^T R^T R b' = b'^T b'$$

This is for a real scalar field. If it was for complex scalar field, we would have $2\sqrt{\det A}$ terms for both real and imaginary parts as:

$$\sqrt{\det A} \times \sqrt{\det A} = \det A$$

Now,

$$Z[J] = \prod_{i=1}^{\infty} \frac{\sqrt{\pi}}{\sqrt{\det A}} e^{\left[\frac{1}{4} i J^T \left(\frac{2}{\det A}\right) i J\right]} = \frac{N}{\sqrt{\det A}} e^{-\frac{1}{2} J^T A^{-1} J}$$

Note: Everything we are writing are in DeWitt notation:

$$A \cdot B = \int d^4x A^m(x) B^n(x)$$

Aside, if A has a zero eigenvalue:

$$S = \frac{1}{2} \sum_{i=1}^N x^T A x + b^T x = \frac{1}{2} \sum_{i=2}^N \lambda_i y_i^2 + \dots$$

$$\int_{-\infty}^{\infty} \prod_{i=2}^N dy_i e^{(\dots)}$$

y_1 can be separated out so it won't blow up — $\int_{-\infty}^{\infty} dy_1 = \infty$, otherwise $y_1 \rightarrow 0$ zero mode. If **NO** zero mode, we can expand each convergent term.

n -point correlator:

$$\langle \phi_1 \phi_2 \dots \phi_N \rangle = \frac{1}{i^N} \frac{1}{Z[J]} \frac{\delta^n}{\delta J_1 \delta J_2 \dots \delta J_N} \Big|_{J=0} \equiv \langle 0 | T[\hat{\phi}_1 \hat{\phi}_2 \dots \hat{\phi}_N] | 0 \rangle$$

To normalize $Z[J]$, we need $\langle \mathbb{I} \rangle = 1$ and we will demand $Z[0] = 1$. We also use the assumption: $\hat{A}$ is independent of J .

$$Z[J] = e^{-\frac{i}{2} J_x (A^{-1})_{xy} J_y} \quad ; \quad J_x (A^{-1})_{xy} J_y = \int d^D x d^D y J(x) (A^{-1})_{xy} J(y)$$

For free theory, it cannot make phase transition itself.

$$\langle \phi_i \rangle = \frac{1}{i} \frac{\delta Z_0[J]}{\delta J} \Big|_{J=0} = 0$$

and

$$\begin{aligned} \langle \phi_a \phi_b \rangle &= - \frac{\delta^2 Z_0}{\delta J_a \delta J_b} \Big|_{J=0} \\ &= - \frac{\delta}{\delta J_b} \left[(-i) (A^{-1})_{xa} e^{-\frac{i}{2} J_m (A^{-1})_{mn} J_n} \right] \Big|_{J=0} \\ &= -i (A^{-1})_{ba} = -i (A^{-1})_{ab} \end{aligned}$$

Here, $\bar{q}_b M_{ab} q_a + J_{ab} (i q_b \bar{q}_a)$.

Again by the same argument:

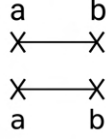
$$\langle \phi_1 \phi_2 \phi_3 \rangle = 0 \quad \text{and} \quad \langle \phi_1 \phi_2 \phi_3 \dots \phi_{n+1} \rangle = 0 \quad (\text{in } \phi^4)$$

This is not true in ϕ^3 theory because it is neither even nor odd.

- Fourier of even = even
- Fourier of odd = odd
- What you choose as order parameter will depend on your reality.

$$\langle \phi_a \phi_b \rangle = -i(A^{-1})_{ab} \quad ; \quad \langle \phi_a \phi_b \phi_c \rangle = 0$$

Pictorially,

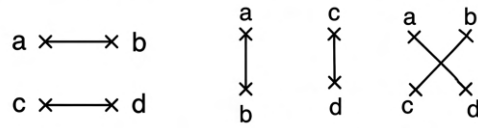


The '×' represents the source.

Now, we will calculate $\langle \phi_a \phi_b \phi_c \phi_d \rangle$. Recall,

$$\begin{aligned} Z[J] &= e^{-\frac{i}{2} J_x (A^{-1})_{xy} J_y} \\ &= \frac{\delta^2 Z_0}{\delta J_a \delta J_b} \Big|_{J=0} \quad [\text{Say } A^{-1} = V] \\ &= \frac{\delta}{\delta J_b} \left((-i) J_x V_{xa} e^{\frac{i}{2} J_m V_{mn} J_n} \right) \Big|_{J=0} \\ &= -i \left[V_{ab} e^{-\frac{i}{2} J_m V_{mn} J_n} - i V_{mb} J_m e^{-\frac{i}{2} J_m V_{mn} J_n} \times J_x V_{xa} \right] \\ &= -i [V_{ab} - (V_{mb} J_m)(V_{na} J_n)] e^{-\frac{i}{2} J_m V_{mn} J_n} \end{aligned}$$

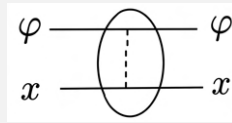
$$\begin{aligned} \langle \phi_a \phi_b \phi_c \phi_d \rangle &= \left[(V_{cd} - V_{cm} V_{dn} J_m J_n) \times e^{-\frac{i}{2} J_\alpha V_{\alpha\beta} J_\beta} \right] \\ &= \left[V_{cd} \frac{\delta^2}{\delta J_a \delta J_b} \left(e^{-\frac{i}{2} J_\alpha V_{\alpha\beta} J_\beta} \right) - V_{cm} V_{dn} \frac{\delta^2}{\delta J_a \delta J_b} \left(J_m J_n e^{-\frac{i}{2} J_\alpha V_{\alpha\beta} J_\beta} \right) \right] \Big|_{J=0} \end{aligned}$$



Exercise : Do this calculation for 6-point function and calculate $\langle \phi_a \phi_b \phi_c \phi_d \phi_e \phi_f \rangle$

Post class discussions:

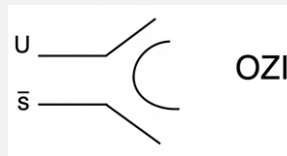
Extras: New particle finding



$$n \rightarrow p + e^- + \bar{\nu}_e \quad (\text{semi-leptonic})$$

$$K \rightarrow \pi + \pi + \pi \quad (\text{no leptonic})$$

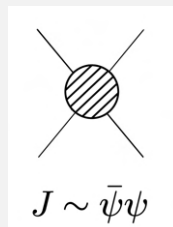
$$\mu^- \rightarrow e^- + \nu_\mu + \bar{\nu}_e \quad (\text{leptonic})$$



Here,

$$J_A J_B = J_A A + J_B A + A^n$$

$$(J_A + J_B)^n = J_A^n + J_B^n + 2J_A J_B$$



(Class-14) Lecture 11 : Sat, Aug 9, 2025

Feynman Diagram

In free field theory, we have

$$S = \frac{1}{2} \phi^T A \phi + J^T \phi \quad , \quad \text{condition: } A^T = A$$

Exercise: Consider a field theory for two fields ϕ, χ and whose action is given by

$$S = \frac{1}{2} \phi^T A \phi + \frac{1}{2} \chi^T A \chi + \chi^T B \phi - \phi^T B \chi \quad , \quad [B = -B]$$

Find the propagators for this field.

What to think of QFT?

$$\begin{array}{ccc}
 S & \xrightarrow{\quad} & \Gamma[\varphi_c] \\
 \text{EOM} \downarrow & Z[\mathbf{J}] = \int \mathcal{D}\varphi e^{i(S[\varphi] + \mathbf{J} \cdot \varphi)} & \downarrow \\
 \frac{\delta S}{\delta \varphi} = -J & \xrightarrow{\text{Quantum correction}} & \left. \frac{\delta \Gamma}{\delta \varphi_c} \right|_{\varphi = \varphi_c + \frac{\hbar}{i} \frac{\delta}{\delta J}} = J \mathbb{I}
 \end{array}$$

Therefore, we can also write:

$$\frac{\delta \Gamma}{\delta \phi_c} = -J$$

The goal of QFT is to find $\Gamma[\phi_c]$

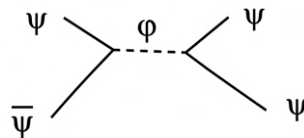
We assume “action” S by considering symmetry. There is no experimental basis beneath this. It can be assumed to be anything by considering the symmetry and it is not unique as well. Also, what we call ϕ is not necessarily unique too.

14.1 Introduction to Feynman Diagram

Yukawa Lagrangian:

$$\mathcal{L} = \bar{\psi}(i\not{\partial} - m)\psi + g\phi\bar{\psi}\psi + \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2$$

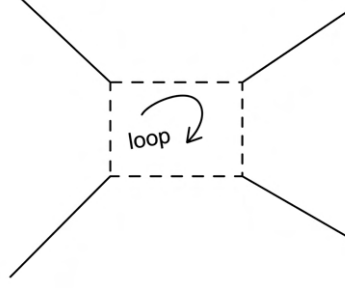
Here, 2 incoming particles + interaction + 2 outgoing particles. Hence, Yukawa interaction term $g\phi\bar{\psi}\psi$ and corresponding diagram:



Tree Diagram

Note: It's not a quantum theory yet.

14.2 Box Diagram

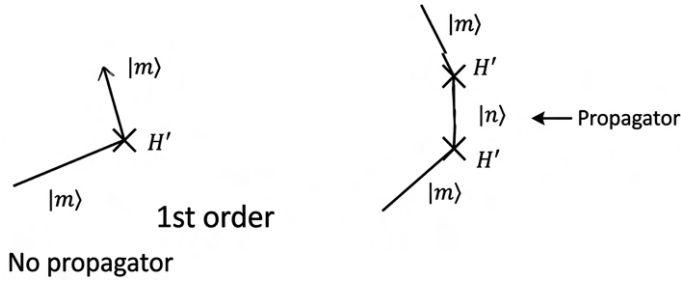


Here is one loop inside.

Now, from perturbation theory, we know:

- First order: $\Delta E_m^{(1)} = \langle m | H' | m \rangle$
- Second order: $\Delta E_m^{(2)} = \sum_{n \neq m} \frac{\langle m | H' | n \rangle \langle n | H' | m \rangle}{E_m - E_n}$.

Pictorial view:



If $E^{(1)}$ is overestimated then $E^{(2)}$ shifts it down as:

1. The numerator is clearly positive.
2. The denominator is clearly negative.

The shift in ground state is always negative.

Now, from the eigenvalue equation:

$$\begin{aligned} \hat{H}\psi &= E\psi \quad , \quad E = H \\ \Rightarrow (\hat{H} - H) |\psi\rangle &= 0 \\ \Rightarrow (\hat{H} - H_0) |\psi\rangle &= V |\psi\rangle \quad , \quad \hat{H} = H_0 + V \end{aligned}$$

Its $(\hat{H} - H_0)$ inverse is the propagator: $A^{-1} \equiv \Delta$

From the last class, we get:

$$Z_0[J] = Z_0[0] e^{-\frac{i}{2} J^T \Delta J}.$$

And the n -point Green function:

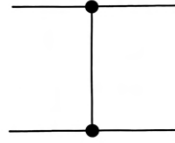
$$G_N = \frac{1}{i^N} \frac{1}{Z[0]} \frac{\delta^n Z[J]}{\delta J_1 \delta J_2 \cdots \delta J_n} \bigg|_{J=0}.$$

Then, the 4-point Green function can be written as a pictorial view as:

$$G_4 = \begin{array}{c} 1 \text{---} A_1 \text{---} 2 \\ 3 \text{---} A_2 \text{---} 4 \end{array} + \begin{array}{c} 1 \\ | \\ 3 \end{array} \begin{array}{c} 2 \\ | \\ 4 \end{array} + \begin{array}{c} 1 \quad 2 \\ \diagdown \quad \diagup \\ 3 \quad 4 \end{array} \quad J_i = J(x_i)$$

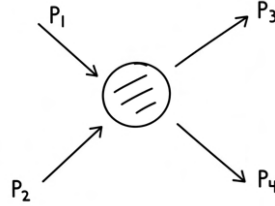
Here, A_1 and A_2 are the amplitudes.

These diagrams are called the *disconnected diagrams* because there is no connection or vertex between them.



An example of a connected diagram

14.3 Mandelstam variables



Here, $\vec{p}_1, \vec{p}_2, \vec{p}_3, \vec{p}_4$ are the 3-vectors and hence, total number of components are $3 \times 4 = 12$

Analogy

The number of covariant Maxwell's equations = 2. There are two constraints:

$$\nabla \cdot \mathbf{B} = 0 \quad , \quad \nabla \cdot \mathbf{E} = \rho,$$

As both $\mathbf{B}$ and $\mathbf{E}$ are 3-vectors, the number of independent variables are: $3 - 1 = 2$

Hence, total number of independent variables are: $2 + 2 = 4$

Also, in 2×2 scattering matrix, there are 2 independent variables.

In the center-of-mass frame, a particular Poincaré transformation has 10 independent parameters. Hence, we will eliminate 10 parameters, and the remaining independent parameters will be: $12 - 10 = 2$.

Now, this various relation between incoming and outgoing momenta can be considered

as:

$$\begin{aligned}(p_1 + p_2)^2 &= s = (E_1 + E_2)^2 - (\mathbf{p}_1 + \mathbf{p}_2)^2 = m_1^2 + m_2^2 - 2(E_1 E_2 + \mathbf{p}_1 \cdot \mathbf{p}_2) \\(p_1 - p_3)^2 &= t = (E_1 - E_3)^2 - (\mathbf{p}_1 - \mathbf{p}_3)^2 = m_1^2 + m_3^2 - 2(E_1 E_3 - \mathbf{p}_1 \cdot \mathbf{p}_3) \\(p_1 - p_4)^2 &= u = (E_1 - E_4)^2 - (\mathbf{p}_1 - \mathbf{p}_4)^2 = m_1^2 + m_4^2 - 2(E_1 E_4 - \mathbf{p}_1 \cdot \mathbf{p}_4)\end{aligned}$$

From the mass-shell (on-shell) condition $p^2 = m^2 \Rightarrow E^2 - \mathbf{p}^2 = m^2$, by calculation we can also find:

$$s + t + u = m_1^2 + m_2^2 + m_3^2 + m_4^2$$

Proof: We consider the $2 \rightarrow 2$ scattering process

$$p_1 + p_2 \rightarrow p_3 + p_4,$$

with on-shell conditions $p_i^2 = m_i^2$ for $i = 1, \dots, 4$. Expanding each:

$$\begin{aligned}s &= (p_1 + p_2)^2 = p_1^2 + p_2^2 + 2p_1 \cdot p_2 = m_1^2 + m_2^2 + 2p_1 \cdot p_2, \\t &= (p_1 - p_3)^2 = p_1^2 + p_3^2 - 2p_1 \cdot p_3 = m_1^2 + m_3^2 - 2p_1 \cdot p_3, \\u &= (p_1 - p_4)^2 = p_1^2 + p_4^2 - 2p_1 \cdot p_4 = m_1^2 + m_4^2 - 2p_1 \cdot p_4.\end{aligned}$$

Adding them:

$$s + t + u = 3m_1^2 + m_2^2 + m_3^2 + m_4^2 + 2p_1 \cdot p_2 - 2p_1 \cdot p_3 - 2p_1 \cdot p_4.$$

Now use momentum conservation $p_1 + p_2 = p_3 + p_4$, so that

$$p_2 = p_3 + p_4 - p_1.$$

Hence

$$\begin{aligned}2p_1 \cdot p_2 - 2p_1 \cdot p_3 - 2p_1 \cdot p_4 &= 2p_1 \cdot (p_3 + p_4 - p_1) - 2p_1 \cdot p_3 - 2p_1 \cdot p_4 \\&= -2p_1^2 \\&= -2m_1^2.\end{aligned}$$

Therefore,

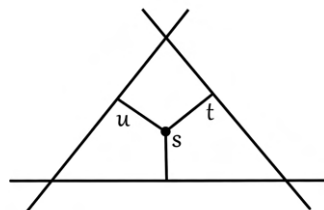
$$s + t + u = m_1^2 + m_2^2 + m_3^2 + m_4^2,$$

which is the required identity.

In the pion case, all masses are equal, hence:

$$s + t + u = 4m_\pi^2$$

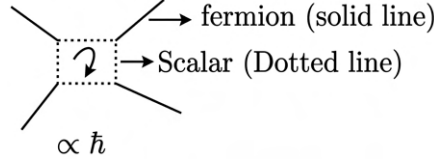
Geometry: In equilateral triangle,



$$s + u + v = \text{sum of the perpendicular distance} = \text{sum of total height area}$$

In free theory, only propagators exist. There is nothing beyond the propagator. Also there can also be products of propagators as well.

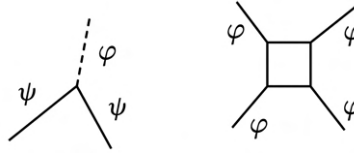
A case for interacting theory



The order of the diagram depends on the order of $\hbar$.

The Lagrangian,

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)^2 - \frac{1}{2}m^2 \phi^2 + g \bar{\psi} \psi \phi + \lambda \phi^4$$



The last term $\lambda \phi^4$ comes from quantum (non-renormalizable) corrections. There is no classical analog of it. Some key points to remember:

- If the theories don't match the parameters, they are not renormalizable.
- Every fundamental theory should be renormalizable.
- From the partition function,

$$Z[J] = \int \mathcal{D}\phi e^{\frac{i}{\hbar}[S_0[\phi] - V(\phi) + J\phi]}$$

Here, $V(\phi)$ is the interaction term, chosen to be polynomial. Because non-polynomial terms can be handled in $(1+1) = 2$ dimensions.

14.4 A Detour of Engineering Dimension in QFT

Dimensions are measured in energy units:

$$[M] = [E] = 1, \quad [L] = [T] = \frac{1}{[M]} = \frac{1}{[E]} = -1$$

The action S is dimensionless as $S \propto \hbar = 1$

Bosonic Field

$$S_{\text{bosonic}} = \int d^D x \mathcal{L}(q, \dots)$$

Here

$$[\mathcal{L}] = D, \quad [d^D x] = -D, \quad [(\partial_\mu \phi^2)] = D \Rightarrow [\partial_\mu \phi] = \frac{D}{2} \quad \therefore [\phi] = \frac{D}{2} - 1$$

Hence,

$$\mathcal{L} = \frac{1}{2} \underbrace{(\partial_\mu \phi)^2}_D - \frac{1}{2} \underbrace{m^2 \phi^2}_{D=2+2\left(\frac{D}{2}-1\right)} \quad (1)$$

$$D, \quad D = 2 + 2\left(\frac{D}{2} - 1\right) \quad (2)$$

Fermions

$$\mathcal{L}_D = \bar{\psi}(i\not{D} - m)\psi$$

Here we have:

$$[\bar{\psi}i\not{D}\psi] = D, \quad [\bar{\psi}\psi] = D - 1 \quad \therefore [\psi] = E^{\frac{D-1}{2}}$$

In the Yukawa term $[G_F(\bar{\psi}\psi)^2] = D = 4$, for $D = 4$:

$$[\bar{\psi}\psi] = D - 1 = 4 - 1 = 3, \quad [(\bar{\psi}\psi)^2] = 3 \times 2 = 6$$

Thus:

$$[G_F] = -6 + 4 = -2 \quad (\text{needed for } D = 4 \text{ renormalizability}).$$

Extra Example

$$\sin(pa) = pa - \frac{(pa)^3}{3!} + \frac{(pa)^5}{5!} + \dots$$

Here p, a are both dimensionless, so it can be expanded into a series.

Back to the Partition Function Again:

$$\begin{aligned} Z[J] &= \int \mathcal{D}\phi \, e^{\frac{i}{\hbar}[S_0[\phi] - V(\phi) + J\phi]} \\ &= e^{-\frac{i}{\hbar}V(\frac{1}{i}\frac{\delta}{\delta J})} \underbrace{\int \mathcal{D}\phi \, e^{\frac{i}{\hbar}[S_0[\phi] + J\phi]}}_{e^{\frac{i}{2}J\Delta J}} \quad [\text{Fourier transformation}] \\ &= e^{-\frac{i}{\hbar}V(\frac{1}{i}\frac{\delta}{\delta J})} \times e^{\frac{i}{2}J\Delta J} \end{aligned}$$

Example

Potential, $V(\phi) = \frac{\lambda}{4!}\phi^4$. Then:

$$\begin{aligned} Z[J] &= e^{-\frac{i}{\hbar}\frac{\lambda}{4!}\frac{\delta^4}{\delta J_i^4}} \left(e^{-\frac{i}{2}J_m\Delta_{mn}J_n} \right) \\ \Rightarrow Z[J] &= \left[1 - \frac{i}{\hbar}\frac{\lambda}{4!}\frac{\delta^4}{\delta J_i^4} + \dots \right] \left(e^{-\frac{i}{2}J_m\Delta_{mn}J_n} \right) \end{aligned}$$

Looking at the Second Term, lets consider:

$$\begin{aligned} &e^{\frac{i}{2}J_m\Delta^{mn}J_n} \\ &= \\ &1 - \frac{i}{2} \underbrace{\frac{m}{J_m} \Delta_{mn} \frac{n}{J_n}}_{\text{}} + \frac{1}{2!} \left(\frac{-i}{2} \right) \underbrace{\frac{m}{J_m} \Delta_{mn} \frac{n}{J_n}}_{\text{}} \underbrace{\frac{p}{J_p} \Delta_{pq} \frac{a}{J_a}}_{\text{}} + \dots \end{aligned}$$

Now, focus on the derivative term:

$$\frac{\delta^4}{\delta J_{j_1} \dots \delta J_{j_4}} \left[\underbrace{(J_m \Delta_{mn} J_n)}_M \underbrace{(J_p \Delta_{pq} J_q)}_M \right]$$

$$\begin{aligned}
&= \frac{\delta^3}{\delta J_i^3} \left[2(J_p \Delta_{pq} J_q) \frac{\partial}{\partial J_i} (J_m \Delta_{mn} J_n) \right] \\
&= \frac{\delta^3}{\delta J_i^3} [4(J_p \Delta_{pq} J_q) \Delta_{in} J_n]
\end{aligned}$$

Now:

$$\begin{aligned}
\frac{\partial}{\partial J_i} (J_m \Delta_{mn} J_n) &= \frac{\partial J_m}{\partial J_i} \Delta_{mn} J_n + J_n \Delta_{mn} \frac{\partial J_i}{\partial J_i} \\
&= \delta_{mi} \Delta_{mn} J_n + J_n \Delta_{mn} \delta_{ni} \\
&= \Delta_{in} J_n + J_n \Delta_{ni} \\
\therefore \frac{\partial}{\partial J_i} (J_m \Delta_{mn} J_n) &= 2\Delta_{im} J_m
\end{aligned}$$

Here, $\Delta_{mn} \equiv \Delta_{nm}$ (symmetric) because A is symmetric ($A^T = A$).

The derivative term $\frac{\partial}{\partial J_i}$ amputates the propagator. That is:

$$\begin{aligned}
\frac{\partial}{\partial J_i} \left(\underbrace{\times \text{---} \times}_{\substack{\text{there are 2} \\ \text{choices to amputate it}}} \right) &= \underbrace{\times \text{---} \bullet}_{\text{(Right)}} + \underbrace{\bullet \text{---} \times}_{\text{(Left)}} = 2 \bullet \text{---} \times \\
\frac{\partial}{\partial J_i} (\text{propagator term}) &\Rightarrow \text{propagator is removed}
\end{aligned}$$

Punchline: Differentiating with respect to J amputates J .

Back to the Previous Calculation

$$\begin{aligned}
&= 4 \frac{\partial^2}{\partial J_i^2} [(J_p \Delta_{pq} J_q) \Delta_{in} \delta_{in} + (\Delta_{in} J_n) \times 2\Delta_{pi} J_p] \\
&= 4 \frac{\partial^2}{\partial J_i^2} [(J_p \Delta_{pq} J_q) \Delta_{ii} + 2\Delta_{pi} \Delta_{ni} J_n J_p] \\
&= 4 \frac{\partial}{\partial J_i} \left[\frac{\partial}{\partial J_i} (J_p \Delta_{pq} J_q) \Delta_{ii} + 2\Delta_{pi} \Delta_{ni} \frac{\partial}{\partial J_i} (J_n J_p) \right] \\
&= 4 \frac{\partial}{\partial J_i} [(2\Delta_{iq} J_q) \Delta_{ii} + 2\Delta_{pi} \Delta_{ni} (J_n \delta_{pi} + J_p \delta_{ni})] \\
&= 4 [2\Delta_{iq} \delta_{iq}] + 4 [2\Delta_{pi} \Delta_{ni}] \frac{\partial}{\partial J_i} (J_n \delta_{pi} + J_p \delta_{ni}) \\
&= 8\Delta_{ii} + 8\Delta_{pi} \Delta_{ni} (\delta_{ni} \delta_{pi} + \delta_{ni} \delta_{pi}) \\
&= 8\Delta_{ii} + 8 \times 2 \Delta_{ni} \Delta_{ni} \\
&= 24\Delta_{ii} \Delta_{ii} = 4! \Delta_{ii} \Delta_{ii}
\end{aligned}$$

The $4!$ arises due to the term $\frac{\lambda}{4!} \phi^4$ where a $4!$ term was present in the denominator.

Pictorial view

1. Δ_{ii} = self-interacting term = generates loop.

$$\frac{\partial^2}{\partial J_i^2} \left[\underbrace{\begin{array}{c} \times \text{---} \times \\ \uparrow \quad \uparrow \end{array}}_{\text{Amputated}} \right] \equiv \text{creation of a loop}$$

2. $\Delta_{ii}\Delta_{ii}$ term corresponds to connecting two vertices, giving additional loop contributions.

$$\frac{\partial^4}{\partial J_i^4} \left(\underbrace{\begin{array}{c} \times \text{---} \times \\ \times \text{---} \times \end{array}}_{\text{Amputated at this point}} \right) \equiv \begin{array}{c} i \text{---} i \\ i \text{---} i \end{array} = A_{ii}A_{ii}$$

Here

$$\lambda \phi^4 \rightarrow \begin{array}{c} \bullet \text{---} \bullet \\ \diagup \quad \diagdown \\ \bullet \text{---} \bullet \end{array} \begin{array}{l} \text{connect it to} \\ \text{any point, 3 choices} \\ \rightarrow \text{only left} = 2 \text{ choices} \end{array} = 4 \times 3 \times 2 \times 1 = 4!$$

One can choose any points = 4 choices

Free theory generates only tree graphs. Sequential diagrams of the partition function due to the potential $(V = \frac{\lambda}{4!}\phi^4)$ will be

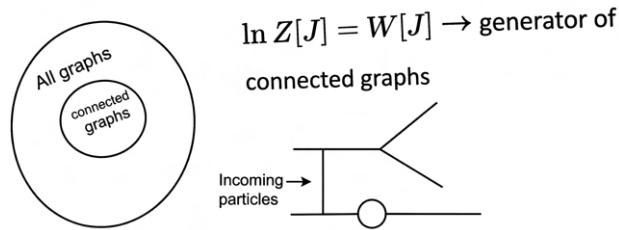
$$\begin{array}{cccccc} \text{---} & + & \text{= } & + & \text{= } & + & \text{= } & + & \text{---} \\ & & \text{8} & + & \text{8} & + & \text{8} & + & \text{---} \\ & & & & \text{8} & + & \text{8} & + & \text{---} \\ & & & & & & \text{8} & + & \text{---} \end{array}$$

Therefore,

$$\Gamma \equiv \bullet + \frac{\hbar}{i} \text{O} + \left(\frac{\hbar}{i} \right)^2 \left[\frac{1}{12} \text{8} + \frac{1}{8} \text{O} + \text{8} + \dots \right]$$

Lessons:

1. Quadratic (free theory) generates only tree graphs.
2. Interaction leads to loops.
3. $Z[J]$ generates all graphs, including disconnected graphs.
4. $W[J]$ generates only connected graphs.



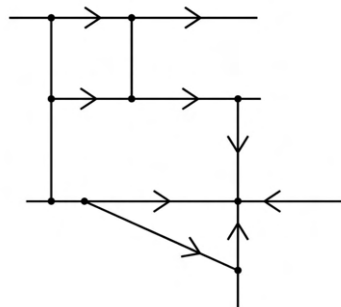
Exercise: For $V = \frac{g\phi^3}{3!}$, Calculate the 2nd order contribution to $Z[J]$ due to this interaction

Connected graph = product of disconnected graphs.



This is the only connected graph in free theory

Post-class discussion:



Free choices of momentum in a loop:

I = No. of internal lines (not function of p)

V = No. of vertices (No. of linear equations)

There is a constant in the diagram (momentum conservation), hence at vertex: $V - 1$.

$\Rightarrow$ Free choices: $I - (V - 1) = I - V + 1$

Trick: $\hbar^n$, where n = number of loops in the diagram.

(Class-15) Lecture 12 : Sat, Aug 23, 2025

Feynman Diagram Continued**15.1 Decomposing Connected Correlators**

From the previous class, we learned that,

$$Z[J] = e^{iW[J]} \quad , \quad W[J] = -i \ln Z[J].$$

Here, $Z[J]$ generates all diagrams, both connected and disconnected. For example:

$$\begin{array}{ccccc}
 \text{---} \bigcirc \text{---} & + & \begin{array}{c} \text{---} \bigcirc \text{---} \\ \text{---} \bigcirc \text{---} \end{array} & + & \begin{array}{c} \diagup \bigcirc \diagdown \\ \diagdown \bigcirc \diagup \end{array} + \dots \\
 \text{connected} & & \text{Disconnected} & & \text{connected} + \dots \\
 W[J] & \text{generates only connected diagrams.} & & &
 \end{array}$$

Remark:

Recall,

$$\langle \phi_1 \phi_2 \dots \phi_n \rangle = \langle 0 | T \{ \phi(x_1) \phi(x_2) \dots \phi(x_n) \} | 0 \rangle = \int \mathcal{D}\phi \, \phi_1 \phi_2 \dots \phi_n e^{iS}$$

And,

$$Z[J] = \langle 0 | T (e^{iJ \cdot \phi}) | 0 \rangle.$$

Then, Green's function for n points is

$$G_n(x_1, \dots, x_n) \equiv G_{1\dots n} = \left. \frac{\delta^n Z[J]}{\delta J_1 \delta J_2 \dots \delta J_n} \right|_{J=0}.$$

And

$$\begin{aligned}
 G_{1\dots n}^c &= \left. \frac{\delta^n W[J]}{\delta J_1 \delta J_2 \dots \delta J_n} \right|_{J=0}, \quad J_i \equiv J(x_i) \quad , \quad (\text{c=connected}) \\
 \Rightarrow G_{1\dots n}^c &= \left. \frac{i(-i) \delta^n (\ln Z[J])}{\delta J_1 \delta J_2 \dots \delta J_n} \right|_{J=0}.
 \end{aligned}$$

If $n = 2$, i.e. the two-point Green function, then

$$\begin{aligned}
 G_{12}^c &= \left. \frac{\delta^2 (\ln Z[J])}{\delta J_1 \delta J_2} \right|_{J=0} \\
 &= \left. \frac{\delta}{\delta J_2} \left(\frac{1}{Z[J]} \frac{\delta Z[J]}{\delta J_1} \right) \right|_{J=0} \\
 &= \left. \left(\frac{1}{Z[J]} \frac{\delta^2 Z[J]}{\delta J_1 \delta J_2} - \frac{1}{Z^2[J]} \frac{\delta Z}{\delta J_1} \frac{\delta Z}{\delta J_2} \right) \right|_{J=0} \\
 &= G_{12} - \left(\frac{1}{Z[J]} \frac{\delta Z}{\delta J_1} \right) \left(\frac{1}{Z[J]} \frac{\delta Z}{\delta J_2} \right) \\
 \therefore G_{12}^c &= G_{12} - G_1^c G_2^c
 \end{aligned}$$

But we need to get back to $\bar{\phi}$ in the argument.

- **Hamiltonian** is defined on the **cotangent** space.
- **Lagrangian** is defined on the **tangent** space.
- Philosophically, $H = \frac{p^2}{2m}$ is more accurate rather than $\frac{1}{2}mv^2$.

15.2 Effective Action

In the path-integral formalism, the generating functional of connected Green's function is $W[J]$. The Legendre transform of $W[J]$ defines the effective action:

$$\Gamma[\phi] = W[J] \pm J \cdot \phi$$

where $J \cdot \phi \equiv \int d^4x J(x)\phi(x)$. In free theory, S and Γ are exactly equal.

From the previous equation:

$$\frac{\delta\Gamma(\phi)}{\delta\bar{\phi}} + J = 0$$

Comparing this with classical EOM:

$$\frac{\delta S}{\delta\phi} + J = 0$$

Thus, quantum corrections are encoded in the *Schwinger–Dyson* equation. SD equation:

$$\left(\frac{\delta S}{\delta\phi} + J \right) \mathbb{I} \Big|_{\phi \rightarrow \phi + \frac{\hbar}{i} \frac{\delta}{\delta J}} = \left(\frac{\delta S}{\delta\phi} \right) \mathbb{I} \Big|_{\phi \rightarrow \phi + \frac{\hbar}{i} \frac{\delta}{\delta J}} = \frac{\delta\Gamma(\phi)}{\delta\phi}.$$

We choose S and observe Γ .

- In QCD, there are different bases: $\delta\Gamma \rightarrow$ Baryons, $\delta S \rightarrow$ quarks, gluons.
- S and Γ can be expanded in terms of polynomial in their respective arguments.

15.2.1 One-Loop Effective Action

We can determine the generic form of the effective action at the one-loop level by solving the Schwinger–Dyson (SD) equation.

We can write Taylor's theorem as:

$$f(x+a) = e^{a \frac{d}{dx}} f(x).$$

Therefore,

$$\frac{\delta\Gamma}{\delta\phi_a} = \frac{\delta S}{\delta\phi_a} \left[\phi + \frac{1}{i} G \frac{\delta}{\delta\phi} \right] = e^{\frac{1}{i} G \frac{\delta}{\delta\phi}} \frac{\delta S}{\delta\phi_a}.$$

Here, $\hat{A}_{ab} = G_{bc} \frac{\delta}{\delta\phi_c}$. Assuming G_{bc} is symmetric, we can write:

$$\hat{A}_{ab} = \frac{1}{2} G_{bc} \frac{\delta}{\delta\phi_c}.$$

Expanding to first order in $\hbar$, the RHS becomes:

$$\frac{\delta\Gamma}{\delta\phi_a} = \frac{\delta S}{\delta\phi_a} + \frac{1}{2i} G_{bc} \frac{\delta^2 S}{\delta\phi_b \delta\phi_c}.$$

But $G_{bc} = \left(\frac{\delta^2 S}{\delta\phi_b \delta\phi_c} \right)^{-1}$, so:

$$\frac{\delta\Gamma}{\delta\phi_a} = \frac{\delta S}{\delta\phi_a} + \frac{1}{2i} \frac{\delta}{\delta\phi_a} \text{Tr} \ln G^{-1}.$$

Thus, the one-loop effective action (up to an additive constant) is:

$$\Gamma = S - \frac{\hbar}{2i} \ln \det G.$$

(Class-16) Lecture 13 : Thrs, Aug 28, 2025

1-Loop Effective Potential**Reference:** Ramond**Digression**

(Historical: Due to Riemann)

16.1 Riemann zeta-function

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$$

Here,

$$\zeta(0) = -\frac{1}{2}, \quad \zeta(-1) = -\frac{1}{12}.$$

Definition

If an operator $\hat{A}$, which is self-adjoint (typically elliptic), then the associated zeta function is

$$\zeta_A(s) \equiv \sum \frac{1}{\lambda_i^s},$$

where $\hat{A}\phi_i = \lambda_i\phi_i$.

16.2 Elliptic Operators

It is a 2nd order differential equation and the sequence of eigenvalues have the same sign. Example: Laplace equation. Here λ can't be complex, but zeta can be complex. Self-adjoint is similar to Hermitian, so if there are infinite eigenvalues, then they are not equal.

All self-adjoint $\Rightarrow$ Hermitian operators, but not all Hermitian operators are self-adjoint.

Target: To calculate $\ln \det A$.The derivative of $\zeta_A(s)$ w.r.t. s is:

$$\begin{aligned} \zeta'_A(s) &= - \sum_n \frac{\ln \lambda_n}{\lambda_n^s} \\ \therefore \zeta'_A(0) &= - \sum_n \ln \lambda_n = - \ln \left(\prod_n \lambda_n \right) = - \ln \det A \end{aligned}$$

If action has an imaginary part, i.e. if e^{-iEt} (E is imaginary), then it means the system is unstable (decay or growth).

Again, we know that

$$W[J] = \frac{1}{i} \ln Z[J]$$

and

$$\begin{aligned}\frac{\delta W}{\delta J_m} &= \phi_m \\ \Rightarrow \frac{\delta \phi_m}{\delta J_n} &= G_{mn} = G_{nm} = \frac{\delta^2 W[J]}{\delta J_m \delta J_n}\end{aligned}$$

Then via Legendre transformation:

$$\begin{aligned}\Gamma[\phi] &= W[J] - J\phi \\ \Rightarrow \frac{\delta \Gamma}{\delta \phi} &= -J \\ \Rightarrow \frac{\delta S}{\delta \phi} \bigg|_{\phi \rightarrow \bar{\phi} + \frac{\hbar}{i} \frac{\delta}{\delta J}} &= \frac{\delta \Gamma}{\delta \bar{\phi}}\end{aligned}$$

$S = \Gamma$, when there is no quantum correction.

Here,

$$\begin{aligned}G_{mn} &= \text{---} \bullet_m \text{---} \bullet_n \text{---} + \\ &+ \text{---} \bigcirc \text{---} + \\ &+ \text{---} \bigcirc \text{---} \bigcirc \text{---} \bigcirc \text{---} \bigcirc \text{---} + \dots\end{aligned}$$

Apart from the first figure, the rest of these are quantum corrections; still, they are junk, i.e., redundant. So we need 1PI (1-particle irreducible) diagram. These are subsets of connected diagrams.

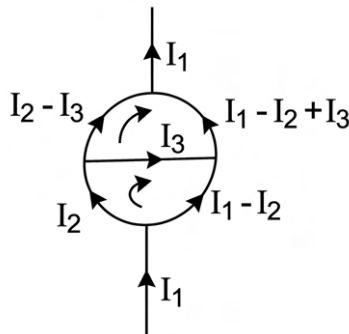
The following is an example of 1PR (1-particle reducible):

$$\text{---} \bigcirc \text{---} \bigcirc \text{---} \equiv \text{1PR}$$

It is 1PR because it can be cut into two pieces and then form 1PI. Whereas 1PI:

$$\text{---} \bigcirc \text{---} = \text{1PI}$$

1PI cannot be cut at the middle to produce another 1PI.



There are 5 internal lines here, among which only two of them are independent (I_2, I_3). So, the number of loops = 2

Now,

$$\frac{\delta^2 W[J]}{\delta J_m \delta J_k} \frac{\delta^2 \Gamma}{\delta \phi_k \delta \phi_n} \Rightarrow \frac{\delta \phi_m}{\delta J_k} \frac{\delta J_k}{\delta \phi_n} = -\delta_{mn} \quad (3)$$

$$\int d^D y \frac{\delta \phi_m(x)}{\delta J(y)} \frac{\delta J(y)}{\delta \phi_n(z)} = \delta^D(x - z)$$

This is an exact relation, not an approximation.

Here,

$$\begin{aligned} \frac{\delta^2 W[J]}{\delta J_k \delta J_n} &= G_{mn} = \text{2-point Green function} \\ \frac{\delta^2 \Gamma}{\delta \phi_k \delta \phi_n} &= \text{Inverse of the 2-point Green function} \end{aligned}$$

Therefore, the 2-point function is the inverse of the quadratic part.

Now, the partition function Z can be defined as:

$$Z = \int \mathcal{D}\phi e^{S_0[x] + Jx} = e^{\frac{i}{2} J A^{-1} J}, \quad W = \frac{1}{2} J A^{-1} J,$$

where

$$S[x] + Jx = \frac{1}{2} x^T A x + Jx.$$

And,

$$\frac{\delta^2 W[J]}{\delta J_m \delta J_n} \frac{\delta^2 J}{\delta \phi_k \delta \phi_n} = G_{mk}^c \Gamma_{kn} = -\delta_{mn}$$

This is how Schwinger's approached to Feynman diagrams.

$$= -\delta_{mn}$$

- Open blobs will be G_{mn}^c .
- Full blobs will be all 1PI's.

Now,

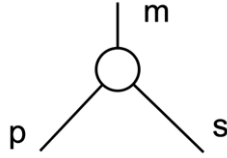
$$\frac{\delta}{\delta J} \left(m \text{---} \bigcirc \text{---} \overset{k}{\text{---}} \bullet \text{---} n \right) = \frac{\delta}{\delta J} (-\delta_{mn}) = 0$$

$$\begin{aligned}
&\Rightarrow \frac{\delta G_{mk}^c}{\delta J_s} \Gamma_{kn} + G_{mk}^c \frac{\delta \Gamma_{kn}}{\delta J_s} = 0 \\
&\Rightarrow G_{smk}^c \Gamma_{kn} = -G_{mk}^c \frac{\delta \bar{\phi}_a}{\delta J_s} \frac{\delta \Gamma_{km}}{\delta \bar{\phi}_a} = -G_{mk}^c G_{sa}^c \Gamma_{akn} \\
&\Rightarrow G_{smk}^c \Gamma_{kn} G_{np}^c = -G_{mk}^c G_{sa}^c \Gamma_{akn} G_{np}^c
\end{aligned}$$

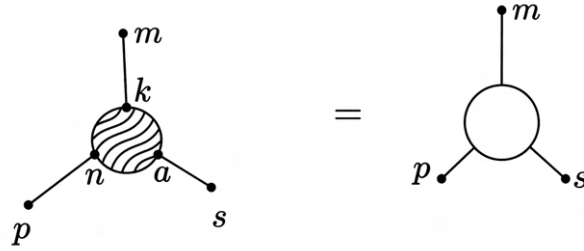
$$\Rightarrow G_{smp}^c = G_{snp}^c = G_{mk}^c G_{pn}^c G_{sa}^c \Gamma_{kma} \quad [\Gamma_{kn} G_{np}^c = -\delta_{kp}]$$

As partial derivatives commute, ordering is not necessary. Thus,

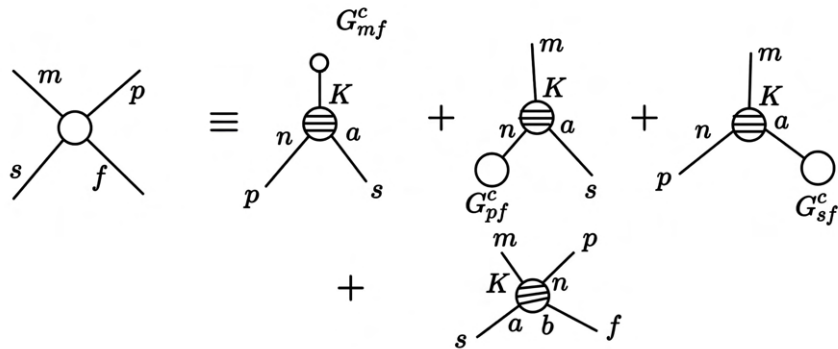
$$G_{smp}^c \equiv \text{3-point function} = \text{all connected}$$



The derivative of Γ :



For 4-points:



Now,

$$\frac{\delta}{\delta J_a} (i - \bigcirc - j) = \begin{array}{c} \bullet a \\ | \\ i - \bigcirc - j \end{array}$$

$$\frac{\delta}{\delta J_n} \left[\begin{array}{c} m - \bigcirc - k - \bigcirc - n \end{array} \right] = \begin{array}{c} f \\ | \\ m - \bigcirc - \text{shaded} - n \end{array} + \frac{\delta}{\delta J_f} \left(\begin{array}{c} m - \bigcirc - k - \text{shaded} - n \end{array} \right)$$

Detour:

$$\frac{\delta}{\delta J_k} (\text{shaded}) = \frac{\delta \varphi_s}{\delta J_k} \frac{\delta}{\delta \varphi_s} (\text{shaded})$$

$$= G_{ks} \text{ } s - \text{shaded} = - \bigcirc - \text{shaded}$$

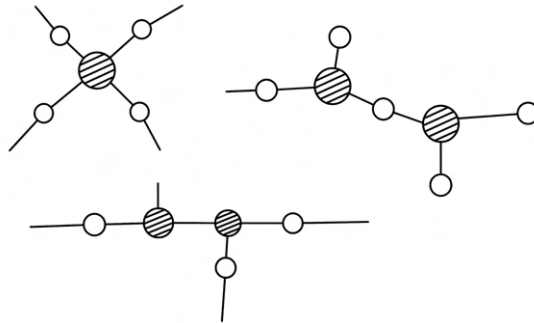
Hence,

$$\left(\frac{\delta}{\delta J_m} \right) [i - \bigcirc - \text{shaded} - j] = \emptyset$$

$$\Rightarrow \begin{array}{c} m \\ | \\ i - \bigcirc - \text{shaded} - j \end{array} = - \begin{array}{c} m \\ | \\ i - \bigcirc - \text{shaded} - j \end{array}$$

$$\Rightarrow \begin{array}{c} m \\ | \\ i - \bigcirc - k \end{array} = - \begin{array}{c} m \\ | \\ i - \bigcirc - \text{shaded} - \bigcirc - k \end{array}$$

For 4-point, the diagrams would be like -



(Class-17) Lecture 14 : Sat, Aug 30, 2025

Dimensional Regularization

The diagram shows two equations. The first equation shows a tadpole diagram (a circle with four external lines) plus a tadpole diagram with a cross through it (representing zero). The second equation shows a tadpole diagram equals a self-energy diagram (a circle with a horizontal line through its center) plus a bubble diagram (two circles connected by a horizontal line). The bubble diagram is underlined.

There will be 3 variations of this (s-channel)

17.1 Zeta function regularization

$$\ln \det A = -\zeta'_A(0)$$

$$\hat{A}\psi_n = \lambda_n \psi_n, \quad A^\dagger = A, \quad \lambda_n > 0$$

$$\zeta_A(s) = \sum_n \frac{1}{\lambda_n^s}$$

17.1.1 Mellin transform

the Mellin transform is a mathematical tool closely related to the Fourier and Laplace transforms, but it works on multiplicative rather than additive scales.

The Mellin transform is like a bridge between integrals, gamma functions, and ζ -functions. That's why it shows up in ζ -function regularization: it connects the heat kernel expansion to the spectral ζ -function. For a function $f(x)$ with $x > 0$, the Mellin transform is defined as

$$\mathcal{M}\{f\}(s) = \int_0^\infty x^{s-1} f(x) dx,$$

where s is a complex variable.

It “weights” the function by powers of x .

Convergence depends on $f(x)$ and the real part of s . If $f(x) = e^{-x}$, then

$$\mathcal{M}\{e^{-x}\}(s) = \int_0^\infty x^{s-1} e^{-x} dx = \Gamma(s).$$

$$\Gamma(s) = \int_0^\infty t^{s-1} e^{-t} dt$$

$$\lambda^{-s} = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-\lambda t} dt$$

$$\zeta_A(s) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} \sum_n e^{-\lambda_n t} dt$$

Define the heat kernel for the operator A :

$$G_A(x, x'; \tau) \equiv \sum_n \psi_n(x) \psi_n^*(x') e^{-\lambda_n \tau}$$

$$\frac{\partial G_A}{\partial \tau} = - \sum_n \lambda_n \psi_n(x) \psi_n^*(x') e^{-\lambda_n \tau} = -A G_A(x, x'; \tau)$$

$$\left(\frac{\partial}{\partial \tau} + A \right) G_A(x, x'; \tau) = 0$$

with boundary condition

$$G_A(x, x'; \tau = 0) = \delta^D(x - x').$$

$$\frac{\partial u}{\partial t} - \alpha \nabla^2 = 0$$

Compare with heat equation:

$$\frac{\partial \phi}{\partial \tau} = A \phi, \quad \hat{A} = -\nabla^2.$$

Recall:

$$G(x, x'; \tau = 0) = \sum_n \psi_n(x) \psi_n^*(x') = \delta(x - x') = \sum_n |x\rangle \langle x|$$

Thus if $x = x'$,

$$\int G_A(x, x; \tau) d^D x = \sum_n e^{-\lambda_n \tau} \int \psi_n(x) \psi_n^*(x) d^D x = \sum_n e^{-\lambda_n \tau}.$$

Hence

$$\zeta_A(s) = \frac{1}{\Gamma(s)} \int_0^\infty \tau^{s-1} \int G_A(x, x; \tau) d^D x d\tau.$$

17.2 Heat kernel

Free scalar field in D dimensions We will work with Euclidean spacetimes. In Minkowski spacetime, the d'Alembertian (box) operator is

$$\square = \partial_\mu \partial^\mu = \frac{\partial^2}{\partial t^2} - \nabla^2,$$

with signature $(+, -, -, -)$.

Under the Wick rotation $t \rightarrow -i\tau$, we have

$$\frac{\partial}{\partial t} = i \frac{\partial}{\partial \tau} \quad \Longrightarrow \quad \frac{\partial^2}{\partial t^2} = -\frac{\partial^2}{\partial \tau^2}.$$

Therefore, the box operator in Euclidean spacetime becomes

$$\square \longrightarrow -\frac{\partial^2}{\partial \tau^2} - \nabla^2 = -\square_E,$$

where

$$\square_E = \frac{\partial^2}{\partial \tau^2} + \nabla^2$$

is the Euclidean Laplacian, which is positive definite.

$$(\square + m^2)\phi = 0 \quad \Rightarrow \quad [-\square_E^2 + m^2]\phi = 0$$

$$\square_E \equiv \partial_\mu \partial^\mu$$

For a translationally invariant system, the field can be expanded in Fourier modes:

$$\phi_n(k, x) = e^{ik \cdot x} \phi_n(\tau), \quad \phi(x, \tau) = \int d^D k \phi(k, \tau) e^{ik \cdot x}.$$

The Euclidean propagator satisfies

$$\frac{\partial}{\partial \tau} G_A(x, x'; \tau) = -A G_A(x, x'; \tau),$$

where $A = -\nabla^2 + m^2$. Expanding in eigenfunctions $A\psi_n = \lambda_n\psi_n$ gives

$$G_A(x, x'; \tau) = \sum_n \psi_n(x) \psi_n^*(x') e^{-\lambda_n \tau}.$$

Fourier transforming gives

$$\frac{\partial \phi_n(k, \tau)}{\partial \tau} = -(k^2 + m^2) \phi_n(k, \tau).$$

Solving this first-order ODE yields

$$\phi_n(k, \tau) = A e^{-(k^2 + m^2)\tau},$$

where $A = \phi(0)$ is a constant determined by the initial condition.

The inverse Fourier transform gives

$$\phi_n(x, \tau) = A \int d^D k e^{-k^2 \tau} e^{ik \cdot x} e^{-m^2 \tau}.$$

Using the standard D-dimensional Gaussian Fourier transform,

$$\int d^D k e^{-\tau k^2} e^{ik \cdot x} = \left(\frac{\pi}{\tau}\right)^{D/2} e^{-x^2/4\tau},$$

we obtain

$$\phi(x, \tau) = A \left(\frac{\pi}{\tau}\right)^{D/2} e^{-x^2/4\tau} e^{-m^2 \tau}.$$

By translation invariance, the heat kernel depends only on $x - x'$:

$$G_A(x, x'; \tau) = G_A(x - x'; \tau) = \int d^D k G_A(k, \tau) e^{ik \cdot (x - x')}.$$

$$\phi(x, \tau) = \int d^D k \phi(k, 0) G_A(k, \tau) e^{ik \cdot x} = \int d^D x' G_A(x - x'; \tau) \phi(x', 0).$$

$$G_A(x - x'; \tau) = A \left(\frac{\pi}{\tau}\right)^{D/2} e^{-(x - x')^2/4\tau} e^{-m^2 \tau}.$$

Initial condition at $\tau = 0$

$$G_A(x - x'; 0) = \delta^D(x - x') = \int d^D k G_A(k, 0) e^{ik \cdot (x - x')}.$$

From the Fourier-space normalization

$$G_A(k; 0) = \frac{1}{(2\pi)^D}.$$

Final heat kernel in position space

$$G_A(x - x'; \tau) = \frac{1}{(4\pi\tau)^{D/2}} e^{-(x - x')^2/4\tau} e^{-m^2 \tau}$$

17.3 What is regularization?

$$\begin{aligned}\ln \det A &= \int dx \langle x | \ln A | x \rangle \\ &= \int d^D k \langle k | \ln A | k \rangle = \int d^D k \ln(k^2 + m^2).\end{aligned}$$

This is divergent, so we have to make some regularization.

Also see

$$\int_0^\infty dx \sin x$$

(no well-defined value as it is oscillating).

We have to take finite test:

$$\int_0^A e^{\lambda x} \sin x dx$$

Hence,

$$\begin{aligned}\zeta_A(s) &= \frac{1}{\Gamma(s)} \int_0^\infty \tau^{s-1} \int G_A(x, x; \tau) d^D x d\tau. \\ \zeta_A(s) &= \frac{1}{\Gamma(s)} \int_0^\infty \frac{\tau^{s-1}}{(4\pi\tau)^{D/2}} e^{-m^2\tau} d\tau d^D x \\ &= \int \frac{d^D x}{\Gamma(s)(4\pi)^{D/2}} \int_0^\infty \tau^{s-\frac{D}{2}-1} e^{-m^2\tau} d\tau \\ &= \int \frac{d^D x}{\Gamma(s)(4\pi)^{D/2}} \frac{\Gamma(s - \frac{D}{2})}{(m^2)^{s-D/2}}\end{aligned}$$

In $D = 4$:

$$\frac{\Gamma(s-2)}{\Gamma(s)} = \frac{\Gamma(s-2)}{(s-1)(s-2)\Gamma(s-2)} = \frac{1}{(s-1)(s-2)}.$$

Then,

$$\zeta_A(s) = \frac{1}{(4\pi)^2 (m^2)^{s-2}} \int \frac{d^4 x}{(s-1)(s-2)}$$

We start from the spectral zeta function:

$$\zeta_A(s) = \frac{V}{(4\pi)^2} \frac{(m^{-2})^{s-2}}{(s-1)(s-2)}.$$

Expand $(m^2)^{s-2}$

$$(m^2)^{s-2} = m^{-4} (m^2)^s = m^{-4} e^{s \ln m^2} \approx m^{-4} (1 + s \ln m^2 + \mathcal{O}(s^2)).$$

Expand $\frac{1}{(s-1)(s-2)}$ via partial fractions

$$\frac{1}{(s-1)(s-2)} = \frac{1}{s-1} - \frac{1}{s-2}.$$

Expand each term around $s = 0$:

$$\frac{1}{s-1} = -1 - s - s^2 - \dots, \quad \frac{1}{s-2} = -\frac{1}{2} - \frac{s}{4} - \dots$$

Subtracting:

$$\frac{1}{s-1} - \frac{1}{s-2} = -\frac{1}{2} - \frac{3}{4}s + \dots$$

Multiply expansions

$$\zeta_A(s) = \frac{V_4}{(4\pi)^2} m^4 (1 + s \ln m^2)^{-1} \left(-\frac{1}{2} - \frac{3}{4}s + \dots \right)$$

$$\zeta_A(s) = \frac{V_4}{(4\pi)^2} m^4 \left[-\frac{1}{2} + s \left(-\frac{3}{4} - \frac{1}{2} \ln m^2 \right) + \dots \right].$$

$$\zeta'_A(0) = \frac{V_4}{(4\pi)^2} m^4 \left(-\frac{3}{4} - \frac{1}{2} \ln m^2 \right)$$

One-loop effective action

$$\Gamma = \frac{1}{2} \ln \det A = -\frac{1}{2} \zeta'_A(0) = \frac{V_4}{2(4\pi)^2} m^4 \left(\frac{3}{4} + \frac{1}{2} \ln m^2 \right)$$

(Class-18 (Online)) Lecture 15 : Fri, Sep 5, 2025

Dimensional Regularization Continued

From the previous class,

$$\zeta_A(s) = \frac{1}{\Gamma(s)} \int \tau^{s-1} d^4x G_A(x, x, \tau)$$

Then,

$$G_A(x, x', \tau) = \sum_k \psi_k(x) \psi_k^*(x') e^{-\lambda_k \tau}$$

Now from the diffusion equation

$$(\square_E + m^2)\phi = 0$$

$$\begin{aligned} \Rightarrow G_A(x, y, \tau) &= \frac{1}{(4\pi\tau)^{D/2}} \exp\left(-\frac{(x-y)^2}{4\tau}\right) e^{-m^2\tau} \\ \Rightarrow G_A(x, x, \tau) &= \frac{1}{(4\pi\tau)^{D/2}} e^{-m^2\tau} \end{aligned}$$

Hence,

$$\begin{aligned} \zeta_A(s) &= \frac{1}{(4\pi)^{D/2}} \int_0^\infty d\tau \tau^{s-D/2-1} e^{-m^2\tau} \\ &= \frac{1}{(4\pi)^{D/2}} \frac{\Gamma(s-D/2)}{(m^2)^{s-D/2}} \end{aligned}$$

m^2 is assumed to be a parameter independent of x .

If $D = 4$,

$$\zeta'_A(s=0) = \frac{1}{(4\pi)^2} \frac{d}{ds} \left(\frac{\Gamma(s-2)}{(m^2)^{s-2}} \right) \Big|_{s=0}$$

Again, we know the identity of the gamma function is

$$\begin{aligned} \frac{\Gamma(n)}{\lambda^n} &= \int_0^\infty d\sigma \sigma^{n-1} e^{-\lambda\sigma} \\ \Rightarrow K'_A(0) &= \frac{1}{(4\pi)^2} \left[\frac{\Gamma(s-1)\Gamma(2)}{\Gamma(s)} m^4 - \Gamma(s-1)(s-2)m^2 \ln m^2 \right]_{s=0} \\ &= m^4 \left[\frac{2\Gamma'(0) - \Gamma(0) + 3 - 2 \ln m^2}{4} \right] \end{aligned}$$

—

$$\begin{aligned} \Gamma^{(1)} &= \text{Tr} \ln G = \int d^4x \langle a | \ln G | a \rangle \\ &= \int d^4x dx' \langle x | \ln G | x' \rangle \\ &= \int d^4k dk' \langle k | \ln G | k' \rangle \end{aligned}$$

$$= \int \frac{d^4 k}{(2\pi)^4} \ln(k^2 + m^2)$$

$$I(D, I\beta) = \int d^D k F(k, I\beta)$$

Regularization is a way of controlling the infinity in your calculations.

$$I(\Lambda, m^2) = \int_0^\Lambda d^D k \ln(k^2 + m^2)$$

$$\Rightarrow I(D, m^2) = \int_0^\infty d^D k \ln(k^2 + m^2)$$

For $D = 1$ this is divergent as $\Lambda \rightarrow \infty$.

$$\Rightarrow \frac{\partial I(D, m^2)}{\partial m^2} = \int_0^\Lambda \frac{d^D k}{k^2 + m^2}$$

For $D = 1$, this is clearly finite as $\Lambda \rightarrow \infty$.

$$= \alpha(m)$$

$$\Rightarrow \frac{\partial I}{\partial(m^2)} = \int \frac{d^D k}{(k^2 + m^2)} \quad [\text{this is logarithmically divergent as } D = 4]$$

So, we can render these integrals finite by differentiating with respect to the coupling parameters.

For 1D system,

$$\frac{\partial I}{\partial(m^2)} = \int_{-\infty}^{\infty} \frac{dk}{k^2 + m^2} = \frac{\pi}{m}$$

$$\Rightarrow I = \pi \int \frac{1}{m} dm$$

$$= \pi \int \frac{1}{\sqrt{u}} du \quad \text{where } m^2 = u \Rightarrow m = \sqrt{u}$$

$$= 2\pi\sqrt{u} = 2\pi m$$

The one integral to bind them all:

$$\int \frac{d^D k}{(k^2 + m^2)^\alpha} = \int d^{D-1} \Omega \int \frac{k^{D-1} dk}{(k^2 + m^2)^\alpha}$$

Solid angle of a $(D - 1)$ dimensional sphere:

$$S^1 : x^2 + y^2 = R^2, \quad S^2 : x^2 + y^2 + z^2 = R^2$$

And,

$$\int_{-\infty}^{\infty} dx e^{-x^2} = \sqrt{\pi}$$

$$\Rightarrow \prod_{i=1}^D \int_{-\infty}^{\infty} dx_i e^{-x_i^2} = \pi^{D/2}$$

$$\Rightarrow \int e^{-r^2} \prod_i dx_i = \pi^{D/2}$$

$$d^D V = \Omega^{D-1} \times R^{D-1}$$

$$\int e^{-r^2} r^{D-1} dr \Omega^{D-1} = \pi^{D/2}$$

$$\text{Let } r^2 = t \Rightarrow \sqrt{t} = r \Rightarrow dr = \frac{1}{2\sqrt{t}} dt$$

$$\Rightarrow \frac{1}{2} \int_0^\infty e^{-t} t^{\frac{D}{2}-1} dt \Omega^{D-1} = \pi^{D/2}$$

$$\therefore \Omega^{D-1} = \frac{2\pi^{D/2}}{\Gamma(D/2)}$$

Then,

$$\int \frac{d^D k}{(k^2 + m^2)^\alpha} = \frac{2\pi^{D/2}}{\Gamma(D/2)} \int_0^\infty \frac{dk k^{D-1}}{(k^2 + m^2)^\alpha}$$

Let $k = mz$, then

$$= \frac{2\pi^{D/2}}{\Gamma(D/2)} \frac{1}{m^{2\alpha-D}} \int_0^\infty \frac{dz z^{D-1}}{(z^2 + 1)^\alpha}$$

$$\text{Let, } u = z^2 \quad dz = \frac{du}{2\sqrt{u}}$$

$$= \frac{2\pi^{D/2}}{\Gamma(D/2)} \frac{1}{(m^2)^{\alpha-D/2}} \cdot \frac{1}{2} \int_0^\infty \frac{u^{\frac{D}{2}-1}}{(u+1)^\alpha} du$$

$$= \frac{\pi^{D/2}}{\Gamma(D/2)} \int_0^\infty \frac{u^{\frac{D}{2}-1}}{(1+u)^\alpha} du$$

$$= \frac{\pi^{D/2}}{\Gamma(D/2)} B\left(\frac{D}{2}, \alpha - \frac{D}{2}\right)$$

$$= \frac{\pi^{D/2}}{\Gamma(D/2)} \frac{\Gamma\left(\frac{D}{2}\right) \Gamma\left(\alpha - \frac{D}{2}\right)}{\Gamma(\alpha)}$$

$$= \frac{\pi^{D/2} \Gamma\left(\alpha - \frac{D}{2}\right)}{\Gamma(\alpha)} \frac{1}{(m^2)^{\alpha-D/2}}$$

This is the master integral in dimensional regularization. It's what people mean by "the one integral to bind them all."

(Class-19 (Online)) Lecture 16 : Wed, Sep 10, 2025

1-Loop Corrected Effective Action**19.1 Alternate explanation of the 1-loop correction**

$$Z[J] = \int \mathcal{D}\phi e^{\frac{i}{\hbar}(S(\phi)+J\phi)} = e^{iW[J]}$$

At $J = 0$,

$$Z[J = 0] = 1, \quad \Rightarrow \quad \Gamma = W - J\bar{\phi}$$

if $J = 0$ then $\Gamma = W$

Then,

$$e^{\frac{i}{\hbar}\Gamma[\bar{\phi}]} = \int \mathcal{D}\phi e^{\frac{i}{\hbar}S(\phi)}$$

Assume we have a background (classical) solution $\bar{\phi}$, around which we expand in a functional Taylor series:

$$S[\phi] = S[\bar{\phi}] + \left. \frac{\delta S}{\delta \phi} \right|_{\bar{\phi}} (\phi - \bar{\phi}) + \frac{1}{2} \left. \frac{\delta^2 S}{\delta \phi^2} \right|_{\bar{\phi}} (\phi - \bar{\phi})^2 + \dots$$

For the classical solution,

$$\left. \frac{\delta S}{\delta \phi} \right|_{\phi=\bar{\phi}} = 0$$

Hence, around this solution,

$$S[\phi] = S[\bar{\phi}] + \frac{1}{2} \left. \frac{\delta^2 S}{\delta \phi \delta \phi} \right|_{\bar{\phi}} (\phi - \bar{\phi})^2 + \dots$$

The 2nd derivative is written as

$$\frac{\delta^2 S}{\delta \phi^2} (\phi - \bar{\phi})^2 = \int d^D x d^D y (\phi - \bar{\phi})_x \left(\frac{\delta^2 S}{\delta \phi^2} \right)_{xy} (\phi - \bar{\phi})_y$$

Therefore,

$$\begin{aligned} e^{\frac{i}{\hbar}\Gamma[\bar{\phi}]} &= e^{\frac{i}{\hbar}S[\bar{\phi}]} \int \mathcal{D}(\delta\phi) e^{\frac{i}{2\hbar}(\delta\phi)^2 \frac{\delta^2 S}{\delta \phi^2}(\delta\phi)} \\ &= e^{\frac{i}{\hbar}S[\bar{\phi}]} \left[\det \left(\frac{\delta^2 S}{\delta \phi^2} \right) \right]^{-\frac{1}{2}} \times \text{const} \\ &= e^{\frac{i}{\hbar}S[\bar{\phi}]} \exp \left[-\frac{1}{2} \ln \det \left(\frac{\delta^2 S}{\delta \phi^2} \right)_{\phi=\bar{\phi}} \right] \end{aligned}$$

Standard integral:

$$\int dx e^{-\frac{1}{2}x^T A x} = \frac{\pi^{N/2}}{\sqrt{\det A}}, \quad A^{-1/2} = (e^{\ln A})^{-1/2}, \quad \ln \det A = \text{Tr} \ln A$$

Taking the logarithm of both sides,

$$\frac{i}{\hbar}\Gamma[\bar{\phi}] = \frac{i}{\hbar}S[\bar{\phi}] - \frac{1}{2} \text{Tr} \ln A$$

$$\Rightarrow \Gamma[\bar{\phi}] = S[\bar{\phi}] + \frac{i\hbar}{2} \text{Tr} \ln \left(\frac{\delta^2 S}{\delta \phi^2} \right)$$

$$\Rightarrow \Gamma[\bar{\phi}] = S[\bar{\phi}] - \frac{i\hbar}{2} \text{Tr} \ln G$$

Example

$$S = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 - V(\phi)$$

$$\frac{\delta^2 S}{\delta \phi(x) \delta \phi(y)} = \left(-(\square + m^2) - V''(\phi) \right) \delta(x - y)$$

Thus,

$$\ln \det \left(\frac{\delta^2 S}{\delta \phi^2} \right) = \text{Tr} \ln \left(-(\square + m^2) - V''(\bar{\phi}) \right) \delta(x - y)$$

Switch to momentum space using:

$$\delta^D(x - y) = \int d^D k e^{ik(x-y)}$$

Hence,

$$\ln \det \left(\frac{\delta^2 S}{\delta \phi^2} \right) = \text{Tr} \ln \left[k^2 - m^2 - V''(\bar{\phi}) \right]_k$$

For a free Lagrangian, $V''(\phi) = 0$:

$$G^{-1} = (p^2 - m^2), \quad G(p) = \frac{1}{p^2 - m^2}$$

Here, mass m is identified as the pole in the Green's function. In the presence of interaction, one sees that the pole is getting shifted from $m^2 \rightarrow m^2 + V''(\bar{\phi})$. This is also known as the **Archimedean bath**.

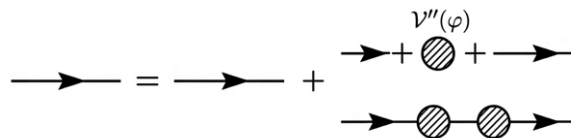
Using the expansion:

$$\frac{1}{A - B} = \frac{1}{A} + \frac{1}{A} B \frac{1}{A} + \frac{1}{A} B \frac{1}{A} B \frac{1}{A} + \dots$$

Therefore,

$$\underbrace{\frac{1}{k^2 - m^2 - V''(\bar{\phi})}}_{\text{2-point function}} = \frac{1}{k^2 - m^2} + \frac{1}{k^2 - m^2} V''(\bar{\phi}) \frac{1}{k^2 - m^2} + \dots$$

This corresponds to the Dyson series.



1-loop effective action

$$\Gamma^n = S + \frac{\hbar}{2i} \bigcirc$$

where,

$$\bigcirc = \bigcirc + \text{diagram with two external lines} + \text{diagram with three external lines} + \text{diagram with four external lines} + \dots$$

This shows that the 1-loop exact effective action sums over all vertex insertions that can be performed.

$$i \int \frac{d^D \bar{k}}{(2\pi)^D} \frac{1}{(\bar{k}^2 + m^2)^\alpha} = \frac{1}{(4\pi)^{D/2}} \frac{\Gamma(\alpha - D/2)}{\Gamma(\alpha) (m^2)^{D/2 - \alpha}}$$

For n such vertex insertions: each minus comes from the propagator in a Euclideanized way,

$$(-1)^n \int \frac{d^D k}{(2\pi)^D} \frac{[V''(\phi)]^n}{(k^2 + m^2)^n}.$$

Using dimensional regularization,

$$\int \frac{d^D k}{(2\pi)^D} \frac{1}{(k^2 + m^2)^n} = \frac{1}{(4\pi)^{D/2}} \frac{\Gamma(n - \frac{D}{2})}{\Gamma(n)} \frac{1}{(m^2)^{n - D/2}}.$$

Thus,

$$= (-1)^n \frac{[V''(\phi)]^n}{(4\pi)^{D/2}} \frac{1}{\Gamma(n)} \frac{\Gamma(n - \frac{D}{2})}{(m^2)^{n - D/2}}.$$

So,

$$\Gamma[\phi] = \frac{\hbar}{2i} \sum_{n=1}^{\infty} \left[-\frac{V''(\phi)}{m^2} \right]^n \frac{\Gamma(n - \frac{D}{2})}{\Gamma(n)} (m^2)^{D/2}.$$

HW: Resume this series (the first terms) to obtain a closed expression.

(Class-20) Lecture 17 : Sat, Sep 13, 2025

Effective Action & Potential Calculation**Reference:** Ramond**20.1 Calculation**

From the last class we had

$$\left(\frac{\partial}{\partial\tau} + \hat{A}\right) H(x, x'; \tau) = 0,$$

where

$$H(x, x'; \tau) = \frac{1}{(4\pi\tau)^{D/2}} \exp\left(-\frac{(x-x')^2}{4\tau} - m^2\tau\right).$$

Now define the zeta function

$$\zeta_A(s) = \frac{1}{\Gamma(s)} \int_0^\infty \tau^{s-1} \int d^D x H(x, x', \tau)$$

We want τ dimensionless. So rescale $\tau \rightarrow \frac{\tau}{m^2}$, then

$$\zeta_A(s) = \frac{V}{(4\pi)^{D/2}\Gamma(s)} \int_0^\infty d\tau \tau^{s-1-D/2} e^{-\frac{\mu^2}{m^2}\tau}.$$

Hence,

$$\zeta_A(s) = \frac{V \Gamma(s - \frac{D}{2})}{(4\pi)^{D/2}\Gamma(s)} \left(\frac{\mu^2}{m^2}\right)^{s-D/2}.$$

We will hereafter continue ourselves with $D = 4$.

$$\begin{aligned} \Rightarrow \zeta_A(s) &= \frac{V}{(4\pi)^2(s-1)(s-2)} \left(\frac{\mu^2}{m^2}\right)^{s-2} \\ \zeta'_A(s=0) &= V \frac{2s-3}{(s-1)(s-2)} \left(\frac{\mu^2}{m^2}\right)^{-2} + \frac{1}{(s-1)(s-2)} \\ &= V \left[\frac{3}{4} \left(\frac{\mu^2}{m^2}\right)^{-2} + \frac{1}{2} \left(\frac{\mu^2}{m^2}\right)^{-2} \ln\left(\frac{\mu^2}{m^2}\right) \right] \\ &= \frac{V}{(4\pi)^2} m^4 \left[\frac{1}{2} \ln\left(\frac{\mu^2}{m^2}\right) + \frac{3}{4} \right] \frac{1}{(\frac{1}{4\pi})^2} \\ &= \frac{Vm^4}{32\pi^2\mu^4} \left[\ln\left(\frac{\mu^2}{m^2}\right) + \frac{3}{2} \right] \end{aligned}$$

Hence, the effective action:

$$\Gamma_1 = -\frac{Vm^4}{64\pi^2\mu^4} \left[\ln\left(\frac{\mu^2}{m^2}\right) + \frac{3}{2} \right]$$

Assignment:

1. Redo today's calculation for $D = 3, 6$
2. Fix the today's μ factor.

If we have a potential $V(\phi) = \frac{\lambda}{4}\phi^4 + \frac{1}{2}m^2\phi^2$, then

$$\frac{\partial^2 V}{\partial \phi^2} = m^2 + 3\lambda\phi^2.$$

Expanding the Action

The action is:

$$S = S[\phi] + \frac{\delta S}{\delta \phi}(\phi - \bar{\phi}) + \frac{1}{2!} \frac{\delta^2 S}{\delta \phi^2}(\phi - \bar{\phi})^2 + \dots$$

We need to focus on the 2nd derivative of the action.

The general framework requires the replacement:

$$m^2 \longrightarrow \frac{d^2 V}{d\phi^2}.$$

Recall, quantum corrections:

$$S_{\text{eff}} = S + \hbar\Gamma_1 + \hbar^2\Gamma_2 + \dots$$

The power of $\hbar$ defines the order of the scattering amplitude:

$$\hbar^{I-V}, \quad \text{1-loop: } L = I - V + 1.$$

Example:

$$V(\phi) = \frac{1}{2}m^2\phi^2 + \frac{\lambda}{4!}\phi^4 \quad \Rightarrow \quad m^2 = \frac{d^2 V}{d\phi^2}.$$

Quantum fluctuations shift the potential. The 1-loop correction is called V_1 .

The general formula in $D = 4$ dimensions is:

$$V_1(\phi) = \frac{\hbar}{64\pi^2} M^4(\phi) \left(\ln \frac{M^2(\phi)}{\mu^2} - \frac{3}{2} \right),$$

where

$M^2(\phi)$ is the field-dependent effective mass, μ is the renormalization scale.

Here,

$$M^2(\phi) = m^2 + \frac{\lambda}{2}\phi^2.$$

Total Effective Potential :

Now we add the quantum correction to the classical part:

$$V_T(\phi) = V(\phi) + V_1(\phi),$$

where

$$V(\phi) = \text{classical potential}, \quad V_1(\phi) = \text{1-loop correction}.$$

Thus,

$$V_T(\phi) = \text{effective potential, which includes quantum effects}.$$

20.2 Effective Mass

The effective mass squared is the curvature of the effective potential at the vacuum:

$$m_{\text{eff}}^2 = \left. \frac{d^2 V_T}{d\phi^2} \right|_{\phi=0}.$$

Explicitly:

$$m_{\text{eff}}^2 = m^2 + \left. \frac{d^2 V_1}{d\phi^2} \right|_{\phi=0}.$$

So the classical mass m^2 is shifted by quantum corrections from V_1 .
Then,

$$V_1 = \frac{\hbar}{64\pi^2} \left[m^2 + \frac{1}{2} \lambda \phi^2 \right]^2 \left[\ln \left(\frac{m^2 + \frac{1}{2} \lambda \phi^2}{\mu^2} \right) - \frac{3}{2} \right].$$

And therefore:

$$V_T = V + V_1.$$

$$\Rightarrow m_{\text{eff}}^2 = \left. \frac{d^2 V_T}{d\phi^2} \right|_{\phi=0} = m^2 + \left. \frac{d^2 V_1}{d\phi^2} \right|_{\phi=0}.$$

Derivative with respect to ϕ

$$\frac{\partial V_1}{\partial \phi} = \frac{\partial}{\partial \phi} \left[\left(m^2 + \frac{\lambda}{2} \phi^2 \right)^2 \ln \left(\frac{m^2 + \frac{\lambda}{2} \phi^2}{\mu^2} \right) - \frac{3}{2} \right].$$

Let $y = m^2 + \frac{\lambda}{2} \phi^2$, then:

$$\frac{\partial y}{\partial \phi} = \lambda \phi,$$

and

$$\frac{\partial V_1}{\partial \phi} = \frac{\partial y}{\partial \phi} \frac{\partial}{\partial y} \left[y^2 \ln \left(\frac{y}{\mu^2} \right) - \frac{3}{2} y^2 \right].$$

$$\frac{\partial}{\partial y} \left[y^2 \ln \left(\frac{y}{\mu^2} \right) \right] = 2y \ln \left(\frac{y}{\mu^2} \right) + y.$$

Thus:

$$\frac{\partial V_1}{\partial \phi} = \lambda \phi \left[2y \ln \left(\frac{y}{\mu^2} \right) - 2y \right].$$

And:

$$V_T' = \lambda \phi \left[2y \ln \left(\frac{y}{\mu^2} \right) - 2y \right] + \lambda \phi \left[2y \ln \left(\frac{y}{\mu^2} \right) + 2y \right].$$

Therefore:

$$V_T' = \lambda \phi 2 \left(m^2 + \frac{\lambda}{2} \phi^2 \right) \ln \left(\frac{m^2 + \frac{\lambda}{2} \phi^2}{\mu^2} \right).$$

20.3 Renormalization Group Idea

Observables change w.r.t. change in μ .

The group of all such changes is called the **Renormalization Group (RG)**.

Alternative way:

$$\Gamma_1 = -\frac{\hbar}{2i} \text{Tr} \ln G_0.$$

$$i\Gamma_1 = \frac{\hbar}{2} \int \frac{d^4 k_E}{(2\pi)^4} \ln(k^2 + m^2).$$

Switch to Euclidean momentum integration:

$$= \frac{\hbar}{8\pi^2} \int_0^\Lambda k^3 dk \ln(k^2 + m^2) = \frac{\hbar}{16\pi^2} \int_0^{\Lambda^2} z \ln(z + m^2) dz,$$

where $z = k^2$.

Differentiating w.r.t. m^2 :

$$\frac{d\Gamma_1}{dm^2} = \frac{\hbar}{16\pi^2} \int_0^{\Lambda^2} \frac{z}{z + m^2} dz = \frac{\hbar}{16\pi^2} \int_0^{\Lambda^2} \left[1 - \frac{m^2}{z + m^2} \right] dz.$$

“DO!!” (perform the integral explicitly)

20.4 Current Algebra

Lagrangian:

$$L = L(\phi, \partial_\mu \phi)$$

Action:

$$S = \int d^4x L$$

Variation:

$$\delta S = \int d^4x \left[\frac{\partial L}{\partial \phi} \delta \phi + \frac{\partial L}{\partial (\partial_\mu \phi)} \delta (\partial_\mu \phi) \right]$$

Euler–Lagrange equation

$$\frac{\partial L}{\partial \phi} = \partial_\mu \left(\frac{\partial L}{\partial (\partial_\mu \phi)} \right)$$

Thus,

$$\delta S = \int d^4x \partial_\mu \left(\frac{\partial L}{\partial (\partial_\mu \phi)} \delta \phi \right)$$

Noether Current

$$j^\mu = \frac{\partial L}{\partial (\partial_\mu \phi)} \delta \phi$$

For symmetries related to Lie groups:

$$\delta \phi = i\epsilon^a T^a \phi$$

General form:

$$J_a^\mu = \frac{\partial L}{\partial (\partial_\mu \phi)} (T^a \phi)$$

Charges:

$$Q^a = \int d^3x \frac{\partial L}{\partial(\partial_\mu \phi)_m} (T_{mn}^a) \phi_n$$

Charge Expression

$$Q^a = \int d^3x \pi_m (T^a)_{mn} \phi_n$$

Observables are bilinears of the fields, not the field itself. These structures also extend to Poincaré symmetry.

Let T^a satisfy a Lie algebra:

$$[T^a, T^b] = i f^{ab}_c T^c$$

π, ϕ satisfy the bosonic equal-time commutation relations (ETCR):

$$[\phi_m(x), \pi_n(y)] = i \delta_{mn} \delta^3(x - y)$$

Check:

$$[Q^a, Q^b] = i f^{abc} Q^c$$

Dirac Lagrangian

$$L = \bar{\psi}(i\not{\partial} - m)\psi$$

Canonical momentum:

$$\frac{\partial L}{\partial(\partial_0 \psi)} = i\psi^\dagger$$

Commutator Identity

$$[A, BC] = [A, B]C + B[A, C]$$

Also:

$$\{A, BC\} = \{A, B\}C - B\{A, C\}$$

Commutation vs Anticommutation

- Fermions (spin 1/2) $\rightarrow$ Anticommutation relations
- Bosons (integer spin) $\rightarrow$ Commutation relations

For bosons:

$$[\phi(x), \phi(y)] = 0$$

For fermions:

$$\{\psi(x), \psi(y)\} = 0$$

$$\{\psi(x), \psi^\dagger(y)\} = 0$$

If $x = y$, then

$$2\psi^2(x) = 0 \quad \Rightarrow \quad \psi^2(x) = 0$$

which is not possible.

20.5 Grassmann Algebra

$$dx_i dx_j = -dx_j dx_i$$

$$dx_i dx_i = 0$$

In Berezin algebra, $\eta_i \eta_j = -\eta_j \eta_i$, $\eta_i^2 = 0$.

General function:

$$f(\eta) = a + a_i \eta_i + a_{ij} \eta_i \eta_j + \dots$$

Maximum number of Taylor coefficients for n Grassmann variables:

$$2^n$$

Differentiation Rules

$$\frac{\partial}{\partial \eta_1} (a_1 \eta_1 + a_{12} \eta_1 \eta_2) = a_1 + a_{12} \eta_2$$

$$\frac{\partial}{\partial \eta_2} (a_2 \eta_2 + a_{12} \eta_1 \eta_2) = a_2 - a_{12} \eta_1$$

Integration Rules

$$\int d\eta \, \eta = 1, \quad \int d\eta = 0$$

Properties

$$\eta^2 = 0, \quad \eta \rightarrow \lambda \eta, \quad d\eta \rightarrow \frac{1}{\lambda} d\eta$$

Integration and differentiation are identical operations:

$$\int d\eta \, \eta = 1$$

Assignment

$$\int d\eta_1 \dots d\eta_N e^{\eta^T A \eta} = \det(A)$$

By induction method:

$$\eta^T = -\eta, \quad A^T = -A$$

(Class-21) Lecture 18 : Sat, Sep 20, 2025

Renormalization of the 2-Point Function in 4D ϕ^3 Theory**21.1 $g\phi^3$ Theory in 4-Dimensions**

From the previous lecture:

$$\int \prod_i d\eta d\bar{\eta} e^{\bar{\eta} A \eta} = \det A, \quad (A^T = -A)$$

Wavefunction Renormalization

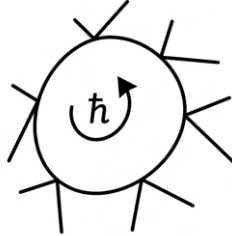
In 1st-order perturbation theory,

$$\begin{aligned} \Delta E_n &= \langle \psi_n | V | \psi_n \rangle \\ |\psi_0\rangle &\longrightarrow |\bar{\psi}_0\rangle = |\psi_0\rangle + \lambda |\psi_1\rangle + \lambda^2 |\psi_2\rangle + \dots \\ &\quad [\text{Unperturbed in the presence of interaction}] \end{aligned}$$

In QFT, there are two expansions:

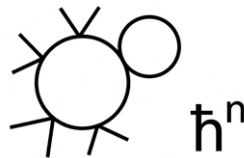
1. $\hbar$ -expansion $\Rightarrow$ WKB approximation,
2. coupling constant λ -expansion in series $\Rightarrow$ Coleman–Weinberg.

Now, the number of loops is equal to the power of $\hbar$.

$$\hbar m^4 \left(\ln \frac{m^2}{\mu^2} - \frac{3}{2} \right) \longrightarrow$$


Zero momentum insertion here.

There exists an infinite order of loops in principle, but in perturbation theory, we consider only up to the 1-loop order.



The mass correction was:

$$m^2 \longrightarrow m^2 + \frac{\lambda}{2} \varphi^2.$$

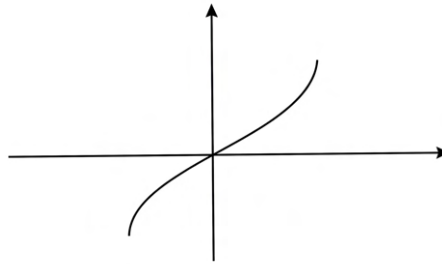
No momentum is involved here. Hence,

$$\hbar \left(m^2 + \frac{\lambda \varphi^2}{2} \right)^2 \ln \left(\frac{m^2 + \frac{\lambda \varphi^2}{2}}{\mu^2} - \frac{3}{2} \right)$$

The effective action expansion is-

$$\begin{aligned} \Gamma &= \Gamma_0 + \Gamma_i \varphi_i + \frac{1}{2} \Gamma_{ij} \varphi_i \varphi_j + \dots \\ \Rightarrow \Gamma &= \circ + \frac{\hbar}{2i} \bigcirc ; [\text{we will only go upto this}] \\ \Rightarrow \Gamma_m(\text{Diff}) &= \text{---} \xrightarrow{p} \text{---} + \frac{\hbar}{2i} \text{---} \bigcirc \text{---} \\ \Rightarrow \Gamma_{mn} &= \text{---} \text{---} \text{---} + \frac{\hbar}{2i} \text{---} \bigcirc \text{---} + \frac{\hbar}{2i} \text{---} \bigcirc \text{---} \\ &\quad \left(\begin{array}{c} \text{linear} \\ \text{unphysical} \\ \text{Tadpole} \end{array} \right) \end{aligned}$$

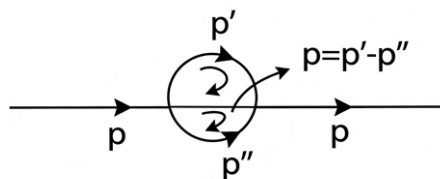
ϕ^3 theory is not practically useful; it is merely a mathematical theory. Because its potential is like this:



Unstable Vacuum

But in $D = 6$, this fits asymptotically, as the same behavior as QCD.

$$\begin{aligned} &\text{Diagram: A horizontal line with momentum } p \text{ entering a loop with momentum } p' \text{ and } p \text{ exiting. A wavy line connects the loop to a vertex.} \\ &\sim \frac{\lambda}{2} \varphi^2 \int \frac{d^4 p'}{p'^2 + m^2} \\ &\quad \underbrace{\hspace{10em}}_{\text{not dependent}} \end{aligned}$$



2-loop **indendepnt** momenta

$$\int \frac{d^4 p' d^4 p''}{(2\pi)^8} \cdot \frac{1}{(p'^2 - m^2)[(p - p' - p'')^2 + m^2][(p - p'')^2 - m^2]}$$

$$\approx F(p^2) \quad ; \quad \text{This integral depends on } p$$

Now,

$$\begin{aligned} \Gamma_{\text{tot, 1-loop}} &= (p^2 + m^2) + \hbar(Ap^2 + B) \\ &= (1 + A\hbar)p^2 + (m^2 + B\hbar) \end{aligned}$$

We have to re-derive in the original S . To do that, we need to replace p by $p \rightarrow (\sqrt{1 + A\hbar})p$.

$$\begin{aligned} S &= \frac{1}{2}\varphi(p^2 + m^2)\varphi + \frac{\lambda\varphi^4}{4!} + \dots \\ \Gamma_{(1\text{-loop})} &= \frac{1}{2}(1 + A\hbar)\varphi(p^2)\varphi + \varphi(m^2 + B\hbar)\varphi \end{aligned}$$

Replace $\varphi' \rightarrow \varphi'\sqrt{1 + A\hbar}$

Coefficient of $\varphi^2 \rightarrow \frac{1}{2}\varphi p^2 \varphi'$

and, $m^2 \rightarrow \frac{m^2 + B\hbar}{1 + A\hbar}$

The number of parameters must match order by order.

φ^4 appears in 2-loop, and φ^3 in 1-loop.

21.2 $g\varphi^3$ Theory

$$\mathcal{L} = \frac{1}{2}\partial_\mu\varphi\partial^\mu\varphi - \frac{1}{2}m^2\varphi^2 + \frac{g}{3!}\varphi^3,$$

$$\Gamma_2 = \frac{(p-m)^2}{2i} + \frac{\hbar}{2i} \text{ (diagram: a circle with two external lines labeled } \mathbf{p} \text{ and } \mathbf{p}' \text{, and a label } \mathbf{p-p}' \text{ below it)}$$

$$\Gamma_{(1\text{-loop})}^{(2\text{-point})} = \int \frac{d^D p'}{(2\pi)^D} \frac{1}{(p'^2 - m^2)[(p - p')^2 - m^2]}$$

It is logarithmically divergent in $D = 4$. So, we need **regularization + renormalization**.

21.3 Feynman–Schwinger Identity

$$\frac{1}{ab} = \int_0^1 \frac{dx}{[ax + b(1-x)]^2} = \int_0^1 \frac{dx}{[b + (a-b)x]^2}$$

The advantage is that it combines two propagators into one propagator in the denominator. The Schwingerian way of doing this is,

$$\frac{1}{a} = \int_0^\infty d\lambda e^{-\lambda a}$$

Now,

$$\begin{aligned}
\frac{1}{(p^2 - m^2)[(p'^2 - m^2)]} &= \int_0^1 \frac{dx}{[(p^2 - m^2)x + (1 - x)(p'^2 - m^2)]^2} \quad [p'' = p - p'] \\
&= \int_0^1 \frac{dx}{[(p'^2 - m^2) + (p^2 - p'^2)x]^2} \\
&= \int_0^1 \frac{dx}{[(p'^2 - m^2) + x(2p \cdot p' - p'^2)]^2} \quad [p^2 - (p - p')^2 = 2p \cdot p' - p'^2] \\
&= \int_0^1 \frac{dx}{[(p - p')^2 - m^2 + x(2p \cdot p' - p'^2)]^2} \\
&= \int_0^1 \frac{dx}{[p'^2(1 - x) - 2pp'(1 - x) + (p^2 - m^2)]^2}
\end{aligned}$$

Recall,

$$\frac{1}{(2\pi)^D} \int \frac{d^D p}{(p^2 + m^2)^\alpha} = \frac{1}{(4\pi)^{D/2}} \frac{\Gamma(\alpha - D/2)}{\Gamma(\alpha)(m^2)^{\alpha - D/2}}.$$

If we set $p \rightarrow p + \bar{p}$, we can absorb divergences.

$$\begin{aligned}
\Gamma_{(1\text{-loop})}^{(2\text{-point})} &= \frac{1}{(2\pi)^D} \int_0^1 dx \int \frac{d^D p'}{[p'^2(1 - x) - 2pp'(1 - x) + (p^2 - m^2)]^2} \\
&= \frac{d^D p}{[p^2 + 2p\bar{p} + m^2]^\alpha} \\
&= \frac{1}{(4\pi)^{D/2}} \frac{\Gamma(\alpha - \frac{D}{2})}{\Gamma(\alpha)(m^2 - \bar{p}^2)^{\alpha - D/2}}
\end{aligned}$$

Only **non-Abelian gauge theory** is asymptotically free.

(Class-22 (Online)) Lecture 19 : Wed, Sep 24, 2025

Gauge Hamiltonian

Gauge Hamiltonian is simply the Hamiltonian formulation of a gauge theory — a theory whose fields have redundant degrees of freedom due to gauge symmetry (like electromagnetism or Yang–Mills).

The Yang–Mills action (in the normalization used in the notes) is

$$S_{\text{YM}} = -\frac{1}{4g^2} \int \text{Tr}(F^{\mu\nu} F_{\mu\nu}) d^4x \quad \text{where,}$$

$$F_{\mu\nu} = [D_\mu, D_\nu] \quad ; \quad D_\mu = \mathbb{I}\partial_\mu + igA_\mu \equiv \text{a matrix valued object}$$

with $A_\mu = A_\mu^a T^a$, a = think of it as unit vectors which are a basis of a Lie algebra.

This follows the identity:

$$\text{Tr}(T^a T^b) = \delta^{ab},$$

It is the positive definite trace of the Lie algebra. Here, the group must be:

1. Semi-Simple
2. Compact

22.1 Hamiltonian Analysis of YM Theory

Preliminary $\rightarrow \text{U}(1) = \text{Maxwell}$

$$[D_\mu, D_\nu] = ieF_{\mu\nu} = ie(\partial_\mu A_\nu - \partial_\nu A_\mu)$$

Where, the covariant derivative is defined as $D_\mu = \partial_\mu + ieA_\mu$

The canonical momentum conjugate to A_μ is

$$\pi_\mu = \frac{\partial \mathcal{L}}{\partial \dot{A}^\mu} = \frac{\partial \mathcal{L}}{\partial (\partial_0 A_\mu)}$$

So the Lagrangian density,

$$\begin{aligned} \mathcal{L} &= -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \\ &= \frac{1}{2} F_{0i} F^{0i} - \frac{1}{4} F_{ij} F^{ij} \end{aligned}$$

Now,

$$\begin{aligned} F_{0i} &= \partial_0 A_i - \partial_i A_0 \\ \Rightarrow F_{0i} &= A_i - \nabla \phi \\ \Rightarrow F_{0i} &= -A^i - \nabla \phi \quad [A_i = -A^i] \\ \therefore F_{0i} &= +E^i \end{aligned}$$

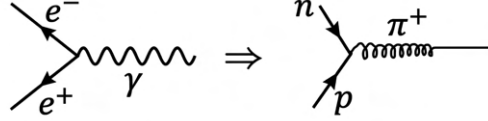
For Yang–Mills, the non-abelian electric and magnetic fields (in component notation) are,

$$E^{i,a} = F^{0i,a} = \dot{A}^{i,a} - D^i A^{0,a}$$

$$B^{i,a} = -\frac{1}{2}\epsilon^{ijk}F_{jk}^a.$$

Hence, the Lagrangian density for the non-abelian theory becomes:

$$\mathcal{L} = \frac{1}{2}(\mathbf{E}^a \cdot \mathbf{E}^a - \mathbf{B}^a \cdot \mathbf{B}^a).$$



22.2 Canonical Momenta and Primary Constraint

Compute the momentum conjugate to A_0^a and A_i^a :

$$\pi^{0,a} = \frac{\partial \mathcal{L}}{\partial \dot{A}_0^a} = 0, \quad (\text{primary constraint})$$

$$\pi^{i,a} = \frac{\partial \mathcal{L}}{\partial \dot{A}_i^a} = F^{0i,a} \equiv E^{i,a}.$$

Thus A_0^a has no conjugate momentum and is not dynamical; instead, it enforces a constraint (Gauss' law).

The canonical Hamiltonian density (Legendre transform) is,

$$\mathcal{H}_{\text{can}} = \pi^{i,a} \dot{A}_i^a - \mathcal{L} = \frac{1}{2}(\mathbf{E}^a \cdot \mathbf{E}^a + \mathbf{B}^a \cdot \mathbf{B}^a) - A_0^a (\mathcal{G}^a),$$

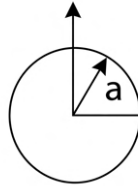
where $\mathcal{G}^a$ is the Gauss law constraint (density)

$$\mathcal{G}^a = D_i^{ab} \pi^{i,b} = \partial_i \pi^{i,a} + g f^{abc} A_i^b \pi^{i,c}.$$

Here we have used the structure constants $[T^a, T^b] = i f^{abc} T^c$.

Constraint structure and Lagrange multipliers

Because $\pi^{0,a} = 0$ (primary constraint), A_0^a appears as a Lagrange multiplier in the action enforcing Gauss' law. Here is a **mechanical example** of how a Lagrange multiplier works:



Consider a particle constrained to the circle of radius a with Lagrangian:

$$L = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) - \lambda(x^2 + y^2 - a^2).$$

Here λ is the Lagrange multiplier. Since λ has no momentum conjugate, it enforces the constraint $x^2 + y^2 = a^2$ (the same logic applies to A_0 in Yang–Mills).

Partition function (path integral) with constraints

The partition function with integration over the Lagrange multiplier(s) is,

$$Z = \int \mathcal{D}x \mathcal{D}y \mathcal{D}\lambda \exp \left[i \int \mu \left(\dot{x}^2 + \dot{y}^2 - \lambda F(x, y) \right) \right],$$

Or, integrating out the momenta in suitable gauges,

$$Z = \int \mathcal{D}x \mathcal{D}y \delta[F(x, y)] e^{i \int dt (\dot{x}^2 + \dot{y}^2)}$$

Where $F(x, y) = 0$ is the constraint surface and the δ enforces it.

This lecture performs the canonical (Hamiltonian) analysis of Yang—Mills theory, revealing Gauss's law as a constraint and A_0^a as a Lagrange multiplier, leading to the constrained path integral.

Gauge symmetry $\Rightarrow$ extra field components that are not physical $\Rightarrow$ become constraints in the Hamiltonian formulation.

(Class-23) Lecture 20 : Sat, Sep 27, 2025

Renormalizability

Renormalizability is tied to perturbation theory. It has nothing to do with momentum. It is the method of developing a bridge between classical and quantum theories order-by-order in the powers of $\hbar$.

23.1 Dimensional analysis in D spacetime dimensions

We take the action to be dimensionless (in units $\hbar = 1$), and the Lagrangian density has mass dimension

$$[S] = 0 \quad , \quad [\mathcal{L}] = D$$

Spatial/temporal coordinate has dimension

$$[x] = -1 \quad (\text{i.e. length has inverse mass dimension}).$$

1. Dimension of the scalar field

Start from the kinetic term for a scalar field,

$$\begin{aligned} \mathcal{L}_{\text{kin}} &= \frac{1}{2}(\partial\phi)^2 \\ \Rightarrow [\mathcal{L}_{\text{kin}}] &= [(\partial_\mu\phi)^2] = D \\ \Rightarrow [\partial_\mu\phi] &= \frac{D}{2} \\ \therefore [\phi] &= \frac{D}{2} - 1 = \frac{D-2}{2} \end{aligned}$$

2. Dimension of g

For an interaction term of the form $g\phi^3$ we require,

$$\begin{aligned} [g\phi^3] &= D \\ \Rightarrow [g] + [\phi^3] &= D \\ \Rightarrow [g] + 3 \cdot \frac{D-2}{2} &= D \\ \Rightarrow [g] &= D - 3 \cdot \frac{D-2}{2} = \frac{2D - 3D + 6}{2} \\ \therefore [g] &= \frac{6-D}{2} \end{aligned}$$

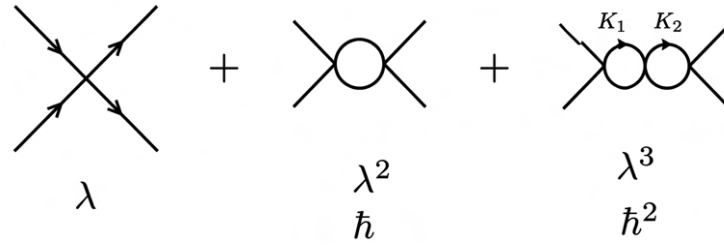
3. Dimension of λ

For ϕ^4 interaction ($n = 4$):

$$\begin{aligned} \mathcal{L}_\lambda &= \lambda\phi^4 \\ \Rightarrow [\mathcal{L}_\lambda] &= [\lambda\phi^4] = D \\ \Rightarrow [\lambda] + 4 \cdot \frac{D-2}{2} &= D \\ \therefore [\lambda] &= D - 2(D-2) = 4 - D \end{aligned}$$

Dyson series $\longrightarrow$ Non-Path integral formalism.

We sketch the leading diagram topologies for the $2 \rightarrow 2 : 4$ -point interaction.



All of them must be of the same order, meaning diagrams with the same overall power of the coupling should be compared when discussing divergences at that order of perturbation theory.

The second (1-loop) interaction is logarithmically divergent (in $D = 4$) while the third (higher-loop) diagram may be UV convergent depending on the number of loops and propagators.

For each of these loops in the third term:

$$\sim \int \frac{d^D k_{\perp}}{(K_{\perp}^2)^2}$$

It is UV convergent if,

$$\begin{aligned} &\sim \frac{dk}{k^{1+\epsilon}} \quad , \quad \epsilon > 0 \\ &\sim \frac{1}{\epsilon} \frac{1}{k^{\epsilon}} \end{aligned}$$

Since the number of external *log* (each *log* represents a particle) is fixed, the dimension of the contributions of each diagram:

$$\begin{aligned} \text{X} &= \text{dim} = M \sim E^B, [M] \sim \beta \\ \text{X} \bigcirc \text{X} & \end{aligned}$$

$\lambda^2(\hbar)$, $p \rightarrow$ momentum decreases; in higher-order perturbation theory, the momentum contribution will **decrease**.

Punchline

If the coupling constant has a positive dimension (i.e., the energy dimension of the coupling is positive), then the theory will be better behaved for higher orders in the perturbation expansion.

If the coupling constant λ decreases, momentum increases. Then the theory will be ill-behaved.

- ϕ^3 theory is well-behaved in $D < 6$ dimensions.
- ϕ^4 theory is well-behaved in $D < 4$ dimensions.

We'll always focus on behaviour in the UV range. In condensed matter, we focus on the IR range.

Critical cases:

- $\phi^4 \rightarrow 4D$
- $\phi^6 \rightarrow 3D$
- $\phi^3 \rightarrow 6D$

23.2 ϕ^6 theory

It's *not* good in 4D.

$$\begin{array}{c}
 \frac{\lambda \phi^6}{6!} \quad \times \quad \text{[Diagram: 8 external legs meeting at a central point]} \quad \sim \phi^8 \text{ Not in the original theory} \\
 \text{[Diagram: 2-loop diagram with 4 external legs]} \quad \text{2-loop} \quad \begin{array}{l} \text{8 external} \\ \text{leg (1-loop)} \\ \text{will be divergent} \end{array} \\
 \sim \int \frac{d^4 k}{\underbrace{k^4}_{\text{2 propagator}}} \sim \ln \Lambda \quad (k^2)^n \\
 \text{logarithmic divergent}
 \end{array}$$

To cancel this divergence, we need to include the corresponding divergent terms in the main theory,

- Starting with 'bare' Lagrangian spinor QED:

Consider loop corrections that generate an effective ϕ^8 interaction (8 external legs) in the original theory — such contributions may be divergent.

- Example: a typical one-loop diagram with two propagators contributes an integral of the form

$$\int \frac{d^4 k}{k^4} \sim \ln \Lambda,$$

i.e., a logarithmic divergence in four dimensions.

$$\infty - \infty = \text{finite terms}$$

23.3 Renormalization Strategies

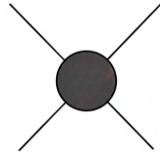
1. Start off with a *bare* Lagrangian.

$$\mathcal{L} = \bar{\psi} (i\not{\partial} - qA_\mu) \psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - (B\psi\bar{\psi} - C\bar{\psi}\not{\partial}\psi - DF_{\mu\nu}F^{\mu\nu})$$

$B, C, D, E, \dots$ must be cutoff dependent, included by hand to cancel divergence.

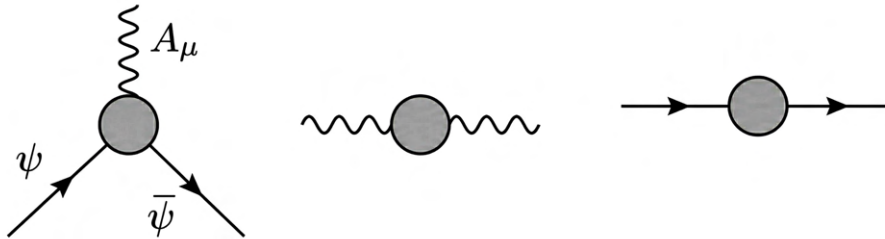
We can't have a term like $-E\bar{\psi}\psi\sigma^{\mu\nu}F_{\mu\nu}\psi$, because this term is not present in the original theory.

2. Identify diagrams with superficial degree of divergences.



If the number of external legs increases, the interaction will be less divergent.

3. One has to introduce counterterms.



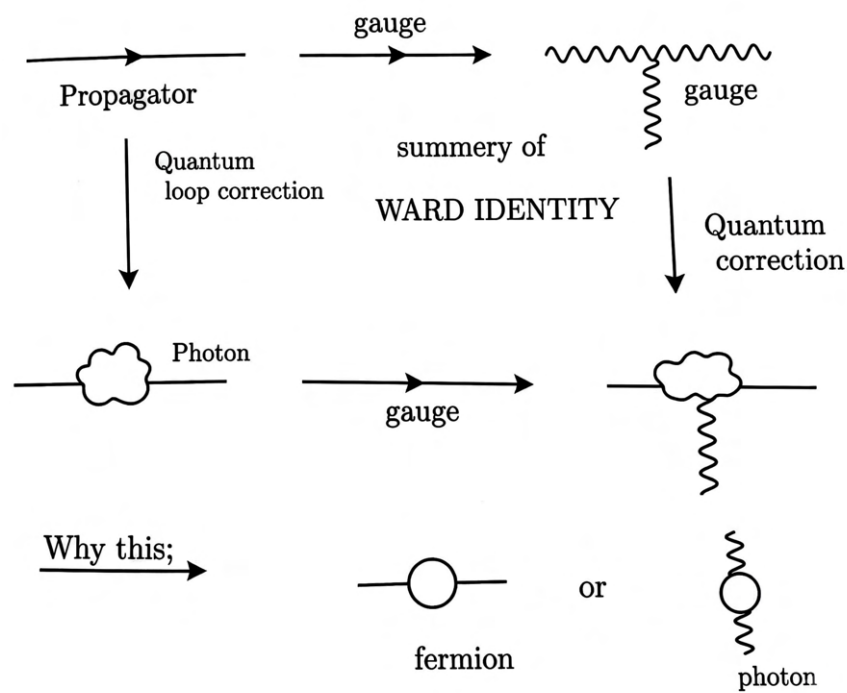
Now, introduce

$$\mathcal{L} = (1 + C) \bar{\psi} i \not{\partial} \psi$$

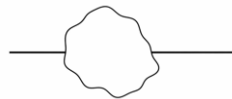
$$\psi \rightarrow \sqrt{1 + C} \psi = \text{wavefunction renormalization}$$

$$\text{and, } \frac{1}{4}(1 + D)F_{\mu\nu}F^{\mu\nu}$$

Summary



Fermion lines can't be added here. Because fermions won't be conserved if its like this:



Hence, the legitimate diagram is:



So, if gauging in classical and quantum mechanics doesn't match, an anomaly arises.

(Class-24) Lecture 21 : Thrs, Oct 16, 2025

Chiral Transformation

- Vector interaction doesn't flip chirality.
- Scalar interaction flips chirality.
- Mass term = $m\bar{\psi}\psi \rightarrow$ We get it from Yukawa theory.

We can write a mass term,

$$m\bar{\psi}\phi_0\psi = m\bar{\psi}_L\phi_0\psi_R \quad [\text{Rank of } \phi_0 = 1]$$

And,

$$\bar{\psi}\gamma^\mu\psi \rightarrow (\bar{\psi}_L\gamma^\mu\psi_L + \bar{\psi}_R\gamma^\mu\psi_R)$$

This means vector coupling won't swap the left and right handed chiralities. They couple differently.

(IA) different	same (IB)
0	1
2	3
4	5

Under a chiral transformation:

$$\psi_R \rightarrow U\psi_R, \quad \psi_L \rightarrow V\psi_L,$$

In order to make $\bar{\psi}_L\phi\psi_R$ to be invariant, we must write,

$$\Phi \rightarrow V\Phi U^\dagger,$$

U and V are independent of the group.

If the theory is of fermions, then Φ behaves like a VEV of $\langle 0|\psi_L\bar{\psi}_R|0\rangle$.

$$S_{EM} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \theta F_{\mu\nu}\tilde{F}^{\mu\nu}$$

In the second term, $\vec{E} \cdot \vec{B}$ breaks the time reversal. This term can be written as,

$$\theta F_{\mu\nu}\tilde{F}^{\mu\nu} = \epsilon^{\mu\nu\rho\lambda} \theta_\mu A_\nu F_{\rho\lambda}$$

This is like the *Chern-Simons* form. At the boundary,

$$\int \eta_\rho \epsilon^{\mu\nu\lambda\rho}$$

Now,

$$\begin{aligned} \Phi \rightarrow \langle 0|\psi_L\bar{\psi}_R|0\rangle \rightarrow \mathcal{L} = & Tr[\partial_\mu\Phi\partial_\mu\Phi^\dagger] + Tr[\Phi^\dagger\partial_\mu\Phi\Phi^\dagger\partial^\mu\Phi] \\ & + Tr[\Phi^\dagger\Phi\partial_\mu\Phi\partial^\mu\Phi] + \dots \end{aligned}$$

Exercise:

The Lagrangian density is given as follows:

$$\mathcal{L}_{EM} = -\frac{1}{4}F^{\mu\nu}F_{\mu\nu} + \theta F_{\mu\nu}\tilde{F}^{\mu\nu}$$

- Find the canonical momenta of $EM(\pi_\mu)$.
- Find the Hamiltonian density.
- Construct the charge operator by coupling A_μ field to an external source j_μ .

In the quantum context:

$$J_\mu = \frac{\delta \Gamma}{\delta A^\mu}$$

$$\text{Gauge Theory} \rightarrow A^\mu \rightarrow A^\mu + \partial_\mu x^\mu$$

vs

$\uparrow$ (Subset)

$$\text{Differential Variation} \rightarrow A^\mu \rightarrow A^\mu + \delta A^\mu$$

In Non-Abelian theory,

$$A_\mu \rightarrow U A_\mu U^\dagger + \frac{1}{ig} U \partial_\mu U^\dagger$$

Also,

$$U = e^{i\bar{\theta}} \cong 1 + i\bar{\theta} + \dots = \theta_a T^a$$

and,

$$A_\mu \rightarrow A'_\mu = A_\mu + i[\bar{\theta}, A_\mu] + \frac{1}{ig} i \partial_\mu \bar{\theta}$$

Thus, the Lagrangian is,

$$\mathcal{L}_{\text{QCD+EM}} = \text{Tr} \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \right) + \text{Tr} (J_\mu A^\mu)$$

Trace gives algebra to numbers. The variation in action is then:

$$\delta S = \text{Tr} [J_\mu D^\mu \bar{\theta}] = -\text{Tr} [D_\mu J^\mu \bar{\theta}]$$

Where,

$$D^\mu J_\mu = \partial^\mu J_\mu + ig[A^\mu, J_\mu] = 0 \rightarrow \text{Covariant conservation law}$$

(Class-25) Lecture 22 : Sat, Oct 18, 2025

Fermionic Path Integral & Chiral Anomaly**Outline**Fermionic Path Integral $\rightarrow$ QED $\rightarrow$ Effective Action $\rightarrow$ Chiral Anomaly

The Lagrangian is,

$$\mathcal{L} = \bar{\psi}(i\not{D} - m)\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} \quad , \quad D_\mu = \partial_\mu - ieA_\mu.$$

Therefore, in path integral,

$$Z[J] = \int \mathcal{D}A_\mu \mathcal{D}\bar{\psi} \mathcal{D}\psi e^{iS_{\text{QED}} + \text{source terms}}.$$

Here,

- Grassmann valued objects $\rightarrow \mathcal{D}A_\mu \mathcal{D}\bar{\psi} \mathcal{D}\psi$
- Source terms $\rightarrow JA + \bar{\psi}\bar{\eta} + \psi\eta$

We know that,

$$\int \mathcal{D}\bar{\psi} \mathcal{D}\psi e^{\bar{\psi}A\psi} = \det A = e^{\ln \det A}$$

And,

$$\int \mathcal{D}\bar{\phi} \mathcal{D}\phi e^{-i\phi^T A \phi} = \frac{N}{\sqrt{\det A}}$$

If it is complex, then,

$$\int \mathcal{D}\bar{\phi} \mathcal{D}\phi e^{-i\phi^T A \phi} = \frac{N}{\sqrt{\det A}} = N e^{-\ln \det A}$$

[No dependency on the coupling constants]

Analogous to:

$$\begin{aligned} \int dx e^{-ax^2} &= \sqrt{\frac{\pi}{a}} \\ \int dx dy e^{-a(x^2+y^2)} &= \frac{\pi}{a} \end{aligned}$$

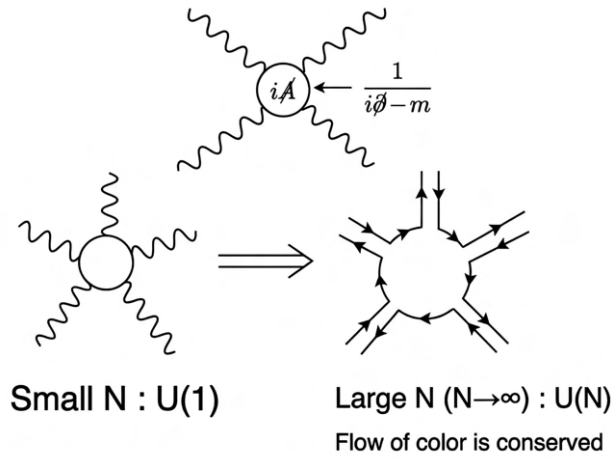
- Fermions do not add any quantum corrections. For free fermions,

$$\text{Classical Theory} = \text{Quantum Theory}$$

- Fermionic fields are **heavy**, hence they are not observed at low energies. This implies that we integrate out the corresponding fermionic degrees of freedom. The effective action is obtained by integrating out the unobservable degrees of freedom.

Effective action,

$$\begin{aligned}
 \Gamma[A] &= -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \ln \det(i\not{D} - m) \\
 &= -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \text{Tr} \ln(i\not{D} - m + e\not{A}) \\
 &= -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \text{Tr} \ln \left[(i\not{D} - m) \left\{ \mathbb{I} + \frac{i}{i\not{D} - m} e\not{A} \right\} \right] \\
 &= -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \text{Tr} \ln(i\not{D} - m) + \text{Tr} \ln \left[\mathbb{I} + \frac{i}{i\not{D} - m} e\not{A} \right] \\
 &= -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \text{Tr} \ln(i\not{D} - m) + \sum_{n=1}^{\infty} \text{Tr} \left[\left(\frac{1}{i\not{D} - m} e\not{A} \right)^n \right]
 \end{aligned}$$



Question:

How many coupling constants and parameters are present in the Standard Model?

25.1 OZI Rule & Furry's Theorem

How many diagrams are suppressed in large N ?

$$\text{wavy line} \text{---} \text{circle} = \text{tr} \left[\frac{(i\not{D} - m)eA}{\partial^2 + m^2} \right] \neq 0$$

Furry's Theorem: [Not valid for all dimensions.]

Any diagram with only photons in external legs will vanish if the number of these legs happens to be odd.

Question:

For all charge conjugation in 4 dimensions, prove Furry's theorem.



Vacuum Polarization

$$\Gamma[A] = \bullet + \frac{\hbar}{2i} \left[\underbrace{\text{diagram}}_{\Gamma'' \text{ (2nd derivative)}} + \text{diagram} \right] \approx \Gamma[A] = \sum_{n=p}^{\infty}$$

Hence, in 4 dimensions, charge-conjugation symmetry implies Furry's theorem.

25.2 Chiral Transformations

The separate transformation rule in the right and left handed wavefunctions is called the *Chiral transformation*.

$$\begin{aligned}\psi &\rightarrow e^{iq\theta} \psi \\ \psi_L &\rightarrow e^{iq\theta} \psi_L \\ \psi_R &\rightarrow e^{iq\theta} \psi_R\end{aligned}$$

$$\begin{aligned}\{\gamma^a, \gamma^b\} &= 2\eta^{ab} \mathbb{I} \\ \{\gamma^5, \gamma^\mu\} &= 0 \quad [\gamma^5 \equiv \bar{\gamma}] \\ (\gamma^5)^2 &= \mathbb{I}\end{aligned}$$

Projection Operator:

$$P_L = \frac{1 + \gamma_5}{2}, \quad P_R = \frac{1 - \gamma_5}{2}$$

Identity:

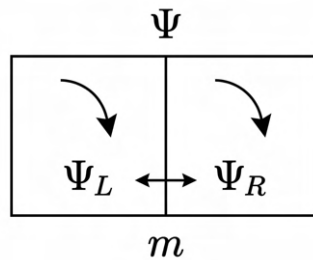
$$\begin{aligned}P_L &= \frac{1 + \bar{\gamma}}{2} \\ P_R &= \frac{1 - \bar{\gamma}}{2}\end{aligned}$$

Their identities,

$$P_L^2 = P_L, \quad P_R^2 = P_R, \quad \psi = \psi_L + \psi_R$$

Any observation is a projection operator.

$$\begin{aligned}m \bar{\psi} \psi &= m \bar{\psi}_L \psi_R + m \bar{\psi}_R \psi_L \\ \bar{\psi} \gamma^\mu \psi &= \bar{\psi}_L \gamma^\mu \psi_L + \bar{\psi}_R \gamma^\mu \psi_R\end{aligned}$$



ψ_L and ψ_R act differently, but the mass term $m\bar{\psi}\psi$ couples them.

Under Space Inversion,

$$\begin{aligned} a &\rightarrow -a && \text{(pseudoscalar)} \\ (a, b) &\rightarrow (a, b) && \text{(pseudovector)} \end{aligned}$$

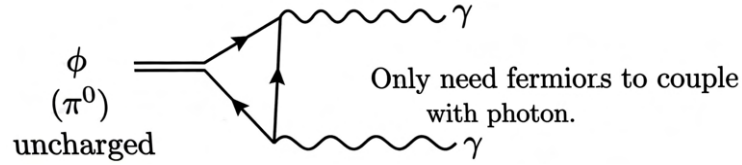
Refined Lagrangian is,

$$\mathcal{L} = g \epsilon^{\mu\nu\lambda\rho} F_{\mu\nu} F_{\lambda\rho} \phi$$

$$\pi^0 \rightarrow 2\gamma$$

Pseudo-scalar particle $\rightarrow$ vector particle

How is it possible? It can't couple to photons, unless a fermion acts as an intermediate particle.



Assumed:

$$\phi \bar{\psi} \gamma^5 \gamma^\mu \psi \phi \longrightarrow A_\mu (\bar{\psi} \gamma^\mu \psi_R - \bar{\psi}_R \gamma^\mu \psi_L)$$

In one loop, fermions have 3 colors. Hence, the number of degrees of freedom is equal to the number of loops. We write:

$$\begin{aligned} \psi &= \psi_L + \psi_R \\ &= e^{i\alpha} \psi_L + e^{i\beta} \psi_R \\ &= e^{i\left[\left(\frac{\alpha+\beta}{2} + \frac{\alpha-\beta}{2}\right)\right]} \psi_L + e^{i\left[\left(\frac{\alpha+\beta}{2} - \frac{\alpha-\beta}{2}\right)\right]} \psi_R \\ &= e^{i\frac{\alpha+\beta}{2}} \left[e^{i\frac{\alpha-\beta}{2}} \psi_L + e^{-i\frac{\alpha-\beta}{2}} \psi_R \right] \\ &= e^{i\mu} \left[e^{ik} \psi_L + e^{-ik} \psi_R \right] \quad \left[k = \frac{\alpha-\beta}{2}, \mu = \frac{\alpha+\beta}{2} \right] \\ &= e^{i\mu} \left[e^{i\gamma_5 k} \psi_L + e^{-i\gamma_5 k} \psi_R \right] \\ &= e^{i\mu} e^{i\gamma_5 k} \psi \end{aligned}$$

If there is no γ_5 dependence, it is a $U(1)$ transformation.

$$\gamma_5 = \tilde{\gamma} \quad \Rightarrow \quad \psi_L = \left(\frac{1 - \tilde{\gamma}}{2} \right) \psi, \quad \psi_R = \left(\frac{1 + \tilde{\gamma}}{2} \right) \psi$$

If the left transformation $\neq$ the right transformation, it is not an axial transformation.

Under Chiral Transformation

$$\psi \rightarrow e^{i\theta\gamma_5} \psi$$

$$\psi^\dagger \rightarrow \psi^\dagger e^{-i\theta\gamma_5}$$

$$\bar{\psi} = \psi^\dagger \gamma^0 = \psi^\dagger e^{-i\theta\gamma_5} \gamma^0 = \bar{\psi} e^{i\theta\gamma_5}$$

Hence, the integral measure, $\mathcal{D}\psi \mathcal{D}\bar{\psi}$ does not remain invariant under chiral transformations, which remain invariant classically.

This non-invariance of the integration measure gives rise to an *anomaly*.

$$\bar{\gamma}\psi_L = -\psi_L \quad , \quad \bar{\gamma}\psi_R = +\psi_R$$

Anomaly calculation is always exact, just like *WKB* approximation, which is “almost exact”. It is exactly 1-loop, calculated up to the 1st order.

Symmetry must exist classically.



Exercise:

Find $\Gamma^{(2)}(A)$ and evaluate it.

(Class-26) Lecture 23 : Thrs, Oct 23, 2025

Chiral Anomaly Calculation (Intuition)**Massless QED**

$$U(1)_L \times U(1)_R \rightarrow \text{Chiral symmetry}$$

$$\psi_L \rightarrow e^{i\alpha_L} \psi_L$$

$$\psi_R \rightarrow e^{i\alpha_R} \psi_R$$

$$\therefore \psi \rightarrow e^{i(\mu\gamma_5 + \lambda)} \psi$$

If $m \neq 0$, $(\bar{\psi}_L \psi_R + h.c.)$ remains invariant only when,

$$\alpha_L = \alpha_R \quad (\text{i.e., the vector transformation})$$

In the path integral language, the functional measure is not invariant under chiral transformation,

$$\begin{aligned} d\psi d\bar{\psi} &\rightarrow e^{2i\mu\gamma_5} d\psi d\bar{\psi} \\ \Rightarrow \mathcal{D}\psi \mathcal{D}\bar{\psi} &\rightarrow \mathcal{D}\psi \mathcal{D}\bar{\psi} \prod e^{2i\mu(x)\gamma_5} \end{aligned}$$

(This is an infinite product)

26.1 Fujikawa's Method**Outside Fujikawa's Method:**

There is a debate on anomaly:

$$\begin{aligned} \partial_\mu j^\mu &= 0 \Leftrightarrow \text{Classical result} \\ \Downarrow \\ J^\mu &\rightarrow \hat{j}^\mu \quad \text{2D} \quad \text{Diagram: wavy line with a circle and cross} \\ <\partial_\mu \hat{j}^\mu> &\neq 0 \quad \text{4D} \quad \text{Diagram: wavy line with a triangle and arrow} \end{aligned}$$

Connects higher and lower order anomalies $\rightarrow$ descent equation.

$$j^\mu = \bar{\psi}(x) \gamma^\mu \psi(y)$$

$$P(x, y) = P_1(x) P_2(y) \quad , \quad P(x) \neq P_1(x) P_2(x)$$

Anomaly appears to be a short-distance phenomenon.

$$\begin{array}{c} \text{Short scale} = \text{UV} \\ \text{Long scale} = \text{IR} \end{array}$$

Fujikawa's Method:

$$Z = e^{i\Gamma} \rightarrow \int \mathcal{D}\psi \mathcal{D}\bar{\psi} \prod e^{i\alpha(x)\gamma_5} e^{iS[\bar{\psi}, \psi, A]}$$

$e^{i\alpha(x)\gamma_5} e^{iS[\bar{\psi}, \psi, A]}$ – is invariant under global $U(1)$, i.e. $\alpha(x) = \alpha = \text{constant}$.

Otherwise,

$$\Delta S \Big|_{\alpha \neq \text{constant}} = \partial_\mu \alpha(x) j^\mu_{\text{Noether}} \rightarrow \alpha(x) \partial_\mu j^\mu(x) \quad [\alpha \rightarrow \alpha(x)]$$

Noether Procedure

$$\begin{aligned}\bar{\psi}\psi &= \bar{\psi}_L\psi_R + \bar{\psi}_R\psi_L \\ \bar{\psi}\gamma^\mu\psi &= \bar{\psi}_L\gamma^\mu\psi_L + \bar{\psi}_R\gamma^\mu\psi_R\end{aligned}$$

$$\begin{aligned}\psi &\rightarrow e^{i\theta}\psi = e^{i\theta}\psi_R + e^{i\theta}\psi_L \\ e^{i\theta\gamma_5}\psi &\rightarrow e^{i\theta}\bar{\psi}_R + e^{-i\theta}\bar{\psi}_L\end{aligned}$$

Axial charges are exactly opposite.

$$\begin{aligned}\begin{pmatrix} p \\ n \end{pmatrix} &\sim \begin{pmatrix} p e^{i\theta} \\ n e^{i\theta} \end{pmatrix} \quad [U \sim e^{i\theta}] \\ \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} &\sim \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{i\theta'} \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}\end{aligned}$$

$$\begin{aligned}\psi &\longrightarrow U\psi \quad , \quad U = U(1) \otimes U(1) \\ \Delta S|_{\alpha \neq \text{constant}} &= \partial_\mu \theta(x) \bar{\psi} \gamma^5 \gamma_\mu \psi \quad [\bar{\psi} \gamma^5 \gamma_\mu \psi = \text{axial constant}]\end{aligned}$$

26.2 Physics Behind Anomaly

$$(\square + m^2)\phi = 0$$

$$\text{Solution (static): } \phi \sim \frac{e^{-m\Gamma}}{\Gamma}$$

$$\begin{array}{ccc} \partial_\mu & \xrightarrow{\text{gauging}} & \partial_\mu + iqA_\mu \\ \updownarrow & \boxed{\text{velocity invariant}} & \downarrow \\ P & \xrightarrow{G} & P - qA \end{array}$$

$$\begin{aligned}p &\longrightarrow (p - qA) \\ H = \frac{p^2}{2m} &\longrightarrow H = \frac{(p - qA)^2}{2m} \\ v = \frac{\partial H}{\partial p} = \frac{p}{m} &\longrightarrow v = \frac{(p - qA)}{m}\end{aligned}$$

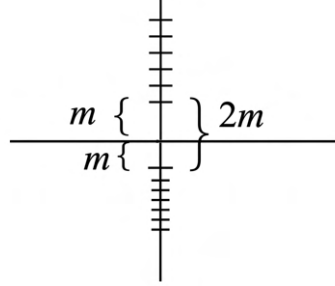
$$Z = e^{i\Gamma} \longrightarrow \int \mathcal{D}\psi \mathcal{D}\bar{\psi} \prod e^{i\alpha(x)\gamma_5} e^{iS[\bar{\psi}, \psi, A]}$$

Contributions are coming from all different momentum modes. To calculate this infinite product, one introduces a “gauge invariant” on fermionic sector.

We insert,

$$e^{-\frac{p^2}{m^2}} \quad (\text{a regulating mass})$$

$$\langle \partial_\mu j^\mu \rangle \neq \frac{\epsilon^{\mu\nu\lambda\rho}}{(2\pi)^2} \cdot F_{\mu\nu} F_{\lambda\rho} \quad [F_{\mu\nu} F_{\lambda\rho} = \vec{E} \cdot \vec{B}]$$



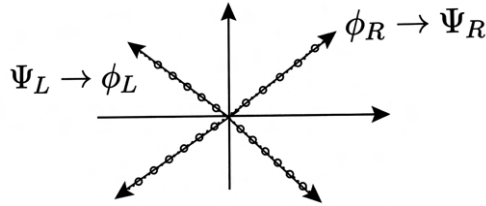
There is no spin in 1D, hence fermion and boson are exactly the same.

Jordan–Wigner (CMP) and Mandelstam (QFT)

26.3 Bosonic Picture (1+1)D

$$\square\phi = 0 \quad \text{massless scalar}$$

$$\phi(t, x) = \phi_R(t - x) + \phi_L(t + x)$$



$$q = Q_L + Q_R \quad , \quad A = Q_L - Q_R$$

Application: E -field

These two symmetries cannot be true at the same time. Only one is true; which one we choose. This is the origin of the anomaly in quantum mechanics (in the presence of interaction).

(Class-27 (Online)) Lecture 24 : Wed, Oct 29, 2025

Exterior Derivatives & WZW Model $df \longrightarrow$ equivalent of ∇ but with a twist**27.1 Exterior Algebra** $V^* \longleftarrow$ dual to a vector space

$$\begin{aligned}
\tilde{w} \in V^* : \quad \tilde{w}(v) &\longrightarrow \text{1-form, } v \in V \\
\tilde{F}(u, v) = -\tilde{F}(v, u) &\longrightarrow \text{2-form, } u, v \in V \\
\tilde{F} \longleftrightarrow \tilde{F}_{ij} u^i v^j &\longrightarrow \text{changes sign under exchange} \\
&\Rightarrow \tilde{F}_{ij} = -\tilde{F}_{ji}
\end{aligned}$$

Exterior derivative $\longrightarrow d$

$$\begin{array}{ccc}
d\tilde{w}_p & \longrightarrow & \tilde{\Omega}_{p+1} \\
\uparrow & & \uparrow \\
p\text{-form} & \longrightarrow & (p+1)\text{-form}
\end{array}$$

The basis of all forms is provided by $dx^\alpha \rightarrow$ (1-form). And,

$$\begin{aligned}
dx^b \wedge dx^a &= -dx^a \wedge dx^b &\Rightarrow \text{2-form} \\
dx^a \wedge dx^b \wedge dx^c &&\Rightarrow \text{3-form} \\
dx^a \wedge dx^b \wedge \cdots \wedge dx^n &&\Rightarrow \text{maximal-form (top form)}
\end{aligned}$$

Then,

$$\begin{array}{ccc}
df = \left(\frac{\partial f}{\partial x^\alpha} \right) dx^\alpha & & \\
\downarrow & & \downarrow \\
\text{0-form} & & \text{1-form}
\end{array}$$

Now,

$$\begin{aligned}
A &= A_j dx^j \\
\Rightarrow dA &= d(A_j dx^j) = \frac{\partial A_j}{\partial x^i} dx^i \wedge dx^j \\
&= \frac{1}{2!} (\partial_i A_j - \partial_j A_i) dx^i \wedge dx^j
\end{aligned}$$

Then,

$$\begin{aligned}
dw_p &= d(w_{i_1 i_2 \dots i_p} dx^{i_1} \wedge dx^{i_2} \wedge \cdots \wedge dx^{i_p}) \\
&= \partial_j w_{i_1 \dots i_p} dx^{i_1} \wedge dx^{i_2} \wedge \cdots \wedge dx^{i_p} \\
d^2 &= 0 \longrightarrow \begin{cases} \nabla \cdot (\nabla \times \vec{A}) = 0 \\ \nabla \times \nabla \phi = 0 \end{cases}
\end{aligned}$$

$$\begin{aligned}
d^2 w_p &= d(dw_p) = d(\partial_j w_{i_1 \dots i_p} dx^{i_1} \wedge dx^{i_2} \wedge \dots \wedge dx^{i_p}) \\
&= \partial_k \partial_j w_{i_1 \dots i_p} dx^k \wedge dx^j \wedge dx^{i_1} \wedge dx^{i_2} \wedge \dots \wedge dx^{i_p} \\
&\therefore d^2 w_p = 0
\end{aligned}$$

a)

$$d(A_p \wedge B_q) = dA_p \wedge B_q + (-1)^p A_p \wedge dB_q \quad [(p+q+1)\text{-form}]$$

b)

$$A_p \wedge B_q = (-1)^{pq} B_q \wedge A_p$$

Exercise:

Prove the identity (a).

Rough Proof:

$$\begin{aligned}
A_p &= A_{i_1 \dots i_p} dx^{i_1} \wedge dx^{i_2} \wedge \dots \wedge dx^{i_p} \\
B_q &= B_{j_1 \dots j_q} dx^{j_1} \wedge dx^{j_2} \wedge \dots \wedge dx^{j_q} \\
\therefore A_p \wedge B_q &= A_{i_1 \dots i_p} B_{j_1 \dots j_q} (dx^{i_1} \wedge dx^{i_2} \wedge \dots \wedge dx^{i_p}) \wedge (dx^{j_1} \wedge dx^{j_2} \wedge \dots \wedge dx^{j_q}) \\
&= B_{j_1 \dots j_q} A_{i_1 \dots i_p} (-1)^p (dx^{i_1} \wedge dx^{i_2} \wedge \dots \wedge dx^{i_p}) \wedge (dx^{j_1} \wedge dx^{j_2} \wedge \dots \wedge dx^{j_q})
\end{aligned}$$

27.2 Stokes theorem

$$\begin{aligned}
\int dw_p &= \int_{\partial M} w \\
\downarrow \quad \downarrow \\
(p+1 \text{ form}) \quad M_p &= \partial M_{p+1} \rightarrow \text{boundary} \\
\langle M_{p+1}, dw_p \rangle &= \langle \partial M_{p+1}, w_p \rangle
\end{aligned}$$

Example:

$$\int_a^b df = f(b) - f(a) \quad [df = 1\text{-form}]$$

$$\begin{aligned}
\int (\nabla \times \vec{A}) \cdot d\vec{\Sigma} &= \int \vec{A} \cdot d\vec{l} \\
&= \int (A_1 dx^1 + A_2 dx^2 + A_3 dx^3)
\end{aligned}$$

$$\begin{aligned}
A &= A_1 dx^1 + A_2 dx^2 + A_3 dx^3 \\
dA &= \partial_2 A_1 dx^2 \wedge dx^1 + \partial_3 A_1 dx^3 \wedge dx^1 + \partial_1 A_2 dx^1 \wedge dx^2 \\
&\quad + \partial_3 A_2 dx^3 \wedge dx^2 + \partial_1 A_3 dx^1 \wedge dx^3 + \partial_2 A_3 dx^2 \wedge dx^3
\end{aligned}$$

Basis:

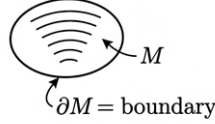
$$\begin{aligned}
&dx^1 \wedge dx^2, \quad dx^2 \wedge dx^3, \quad dx^3 \wedge dx^1 \\
&= (\partial_1 A_2 - \partial_2 A_1) dx^1 \wedge dx^2 + \dots
\end{aligned}$$

27.3 WZW model

Consider a map from a group G to $2D$,

$$g(x, t) : M \rightarrow G$$

Spacetime M which is
boundary of 3 manifold $\mathcal{M}$



Now,

$$g^{-1}dg = dg g^{-1} \Rightarrow g \rightarrow gh'$$

$g^{-1}dg$ is invariant under,

$$\begin{aligned} M = q_L \bar{q}_R &\longrightarrow M = U_L M U_R^\dagger \\ q_L &\rightarrow U_L q_L \Rightarrow q_R \rightarrow U_R q_R \end{aligned}$$

$$S = \lambda S_0 + \beta S_{\text{WZW}}$$

$$S_0 = \text{Tr} \int_{M=\partial\mathcal{M}} \underbrace{\langle g^{-1}\partial g, g^{-1}\partial g \rangle}_{\text{(no wedge!)}} \quad (\text{Maurer-Cartan})$$

$$S_{\text{WZW}} = \left(\frac{1}{2\pi}\right)^3 \int_{\mathcal{M}} \text{Tr} \left(\underbrace{g^{-1}dg \wedge g^{-1}dg \wedge g^{-1}dg}_{\text{winding number}} \right) \neq \text{Tr} \int_M dw_2$$

What's the variation of S_{WZW} ?

$$\delta S_{\text{WZW}} = \int_M w_2$$

$$\begin{aligned} \delta g &= g' - g \\ \Rightarrow g^{-1}g &= \mathbb{I} \\ \Rightarrow \delta g^{-1}g + g^{-1}\delta g &= 0 \\ \therefore \delta g^{-1} &= -g^{-1}\delta g g^{-1} \end{aligned}$$

$$g' = (1 + \epsilon)g = g + \epsilon g \Rightarrow \delta g = \epsilon g$$

- Check:

$$(\theta^{-1}dg)^{2m} = -d((g^{-1}dg)^{2m-1})$$

- Find:

$$\langle g^{-1}dg \wedge g^{-1}dg \wedge g^{-1}dg \rangle$$

Because,

$$g^{-1}d\delta g \wedge g^{-1}dg = -dg^{-1} \wedge dg = -d(g^{-1}dg)$$

$$\begin{aligned}
S_{\text{WZW}} &= - \int \langle g^{-1} dg, d(g^{-1} dg) \rangle \\
\Rightarrow \delta S_{\text{WZW}} &= \beta^3 \int \text{Tr} \left(\delta(g^{-1} dg) \wedge g^{-1} dg \wedge g^{-1} dg \right)
\end{aligned}$$

Now,

$$\begin{aligned}
\delta(g^{-1} dg) &= \delta g^{-1} dg + g^{-1} d(\delta g) \\
&= (-g^{-1} \delta g g^{-1}) dg + g^{-1} d(\epsilon g) \\
&= -g^{-1} \epsilon g g^{-1} dg + g^{-1} (d\epsilon g + \epsilon dg) \\
&= -g^{-1} \epsilon dg + g^{-1} d\epsilon g + g^{-1} \epsilon dg \\
\therefore \delta(g^{-1} dg) &= g^{-1} d\epsilon g
\end{aligned}$$

Back to the variation of WZW action,

$$\begin{aligned}
\delta S_{\text{WZW}} &= 3\beta \int_M \text{Tr} \left[g^{-1} d\epsilon g \wedge g^{-1} dg \wedge g^{-1} dg \right] \\
&= 3\beta \int_M \text{Tr} \left[d\epsilon \wedge dg g^{-1} \wedge dg g^{-1} \right] \quad \left[df = \frac{\partial f}{\partial x^i} dx^i \right] \\
&= -3\beta \int_M \text{Tr} \left[d\epsilon \wedge d(g dg^{-1}) \right] \quad [\because -dg dg^{-1} = -d(g dg^{-1})] \\
&= 3\beta \int_{M=\partial\mathcal{M}} \text{Tr} \left[d(d\epsilon \wedge g dg^{-1}) \right] \\
&= 3\beta \int_M \text{Tr} \left[\epsilon dg dg^{-1} \right] \\
\therefore \delta S_{\text{WZW}} &= 3\beta \int_M \text{Tr} \left[d\epsilon \wedge g dg^{-1} \right]
\end{aligned}$$

(Class-28) Lecture 25 : Thrs, Nov 06, 2025

The Wess Zumino Witten (WZW) Model

From the previous class, we worked on the zero curvature,

$$F(A) = 0 \\ \Rightarrow dA + A \wedge A = 0$$

For one form,

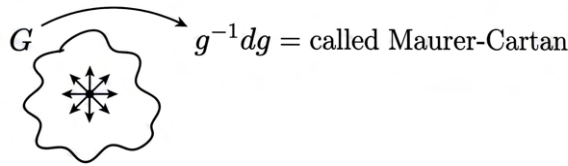
$$A = A_\mu dx^\mu \\ \Rightarrow dA = \partial_\mu A_\nu dx^\mu \wedge dx^\nu \\ \Rightarrow dA = \frac{1}{2} (\partial_\mu A_\nu - \partial_\nu A_\mu) dx^\mu \wedge dx^\nu$$

For the non-Abelian case:

$$A = A_\mu^a T^a dx^\mu \\ \Rightarrow A \wedge A = (A_\mu^a T^a dx^\mu) (A_\nu^b T^b dx^\nu) \\ = A_\mu^a A_\nu^b T^a T^b dx^\mu \wedge dx^\nu \\ = \frac{1}{2} A_\mu^a A_\nu^b [T^a, T^b] dx^\mu \wedge dx^\nu \\ = \frac{1}{2} A_\mu^a A_\nu^b f^{abc} T^c dx^\mu \wedge dx^\nu$$

Now volume is defined as:

$$\vec{A} \cdot (\vec{B} \times \vec{C}) = \epsilon_{ijk} A^i B^j C^k$$



When,

$$g = e^A \Rightarrow dg = e^A dA$$

In a 3D Lie group, volume is defined as:

$$\int g^{-1} dg \wedge g^{-1} dg \wedge g^{-1} dg$$

And $U(1) \rightarrow U(1) : \theta \rightarrow e^{in\theta}$, Where,

$$\theta : 0, 2\pi, \quad \phi : 0, \frac{2\pi}{n}$$

In $SU(2)$, the volume is equal to the volume of S^3 .

$$ds^2 = d\theta^2 + \sin^2 \theta d\phi^2 \\ ds^3 = d\psi^2 + \sin^2 \psi [d\theta^2 + \sin^2 \theta d\phi^2]$$

$$\text{and, } \sqrt{g} = \sin^2 \psi \sin \theta d\psi d\theta d\phi$$

28.1 Application of the background field method

$$S = \frac{1}{2\lambda^2} \int_{M=\partial M} d^2x \langle \partial_\mu g^{-1} \partial^\mu g \rangle + \beta \int_M d^3y \langle g^{-1} dg \wedge g^{-1} dg \wedge g^{-1} dg \rangle \sim 2\pi^2$$

Under symmetry transformation:

$$g(x) \rightarrow hg(x) \Rightarrow g^{-1}dg = g^{-1}h^{-1}dg$$

Here, it is a multiplicative transformation, unlike

$$\phi \rightarrow \phi + \delta\phi$$

Which is an additive transformation. And,

$$g = e^{i\phi}g(x) \quad \text{classical solution which satisfies EOM}$$

Exercise: Substitute this in the classical action around the classical solution and expand up to 2^{nd} order in χ .

From Taylor series expansion:

$$f(x) = f(x_0) + \left. \frac{\delta f}{\delta x} \right|_{x=x_0} (x - x_0) + \frac{1}{2!} \left. \frac{\delta^2 f}{\delta x^2} \right|_{x=x_0} (x - x_0)^2 + \dots$$

$$\begin{aligned} g^{-1}dg &= g_0^{-1}e^{-i\Phi} [d(e^{i\Phi}g_0)] \\ &= g_0^{-1}dg_0 + g_0^{-1}e^{-i\Phi}d(e^{i\Phi})g_0 \end{aligned}$$

$$e^{-x}d(e^x) = ?$$

$$e^x = 1 + x + \frac{1}{2!}xx + \dots$$

$$\Rightarrow d(e^x) = dx + \frac{1}{2!}(dx x + x dx)$$

In $SU(2)$,

$$x = (\sigma \cdot \pi)$$

$$dx = (\sigma \cdot d\pi)$$

$$= (\sigma \cdot \pi)(\sigma \cdot d\pi) + (\sigma \cdot d\pi)(\sigma \cdot \pi)$$

$$= 2(\pi \cdot d\pi) + i\sigma \cdot (\pi \times d\pi)$$

We used,

$$(\vec{\sigma} \cdot \vec{A})(\vec{\sigma} \cdot \vec{B}) = \vec{A} \cdot \vec{B} \mathbb{I} + i\vec{\sigma} \cdot (\vec{A} \times \vec{B})$$

$$\pi^+ \rightarrow u\bar{d}$$

$$\pi^- \rightarrow d\bar{u}$$

$$\pi^0 \rightarrow \frac{1}{\sqrt{2}}(u\bar{u} - d\bar{d})$$

$$\pi^i = \bar{q}\sigma^i q = \text{field}$$

Exercise:

Consider the Chern-Simons action (Effective Field Theory for Quantum Hall Effect),

$$S_{CS}[A] = \frac{K}{4\pi} \int \text{Tr} \left(A dA + \frac{2}{3} A \wedge A \wedge A \right)$$

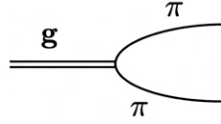
Find under a gauge transformation,

$$A \rightarrow A^g = g^{-1}(A + d)g$$

How the CS action transforms:

$$S_{CS}[A^g] = S[A] + \frac{K}{12\pi} \int_M \text{Tr} (g^{-1} dg \wedge g^{-1} dg \wedge g^{-1} dg) + \frac{K}{4\pi} \int \text{Tr} (dg g^{-1} \wedge A)$$

$$S[e^{i\pi}g] = S[g] + \frac{1}{\lambda^2} \int \langle \partial_\mu \pi \partial^\mu \pi \rangle + \frac{1}{4} \langle g_0^{-1} \partial_\mu g [\pi, \partial_\mu \pi] \rangle + \frac{B}{\lambda^2} \int \text{Tr} (g_0^{-1} \partial_\mu g [\pi, \partial_\mu \pi] e^{\mu\nu})$$



(Class-29) Lecture 26 : Sat, Nov 08, 2025

Gauge Field Quantization**29.1 Problem with Gauge Field Quantization**

(Short Discussion)

Recall the EOM for the Maxwell field:

$$\begin{aligned}
\partial_\mu F^{\mu\nu} &= j^\nu \\
\Rightarrow \partial_\mu (\partial^\mu A^\nu - \partial^\nu A^\mu) &= j^\nu \\
\Rightarrow [\eta^{\mu\nu} \square - \partial_\mu \partial_\lambda] A^\lambda &= j^\mu
\end{aligned}$$

Does this exist?

Say,

$$A_\mu = D_{\mu\nu} j^\nu$$

Formally,

$$P_{\mu\lambda} D^{\lambda\sigma} = \delta_\mu^\sigma$$

However,

$$\begin{aligned}
P_{\mu\lambda} P^{\lambda\sigma} &= (\eta_{\mu\lambda} \square - \partial_\mu \partial_\lambda) (\eta^{\lambda\sigma} \square - \partial^\lambda \partial^\sigma) \\
&= \delta_\mu^\sigma \square^2 - \partial_\mu \partial^\sigma \square - \partial_\mu \partial^\sigma \square + \partial_\mu \partial^\sigma \square \\
&= \square (\delta_\mu^\sigma \square - \partial_\mu \partial^\sigma) \\
&= \square P_\mu^\sigma
\end{aligned}$$

$$\begin{aligned}
\therefore K_{\mu\nu} K^{\nu\sigma} &= K_\mu^\sigma \\
\Rightarrow \frac{1}{\square} P_{\mu\lambda} &= K_{\mu\lambda}
\end{aligned}$$

Recall, $\frac{1}{\square}$ is the propagator for a massless scalar.

We know that the projection operator is defined as:

$$\begin{aligned}
P^2 &= P \\
\Rightarrow \lambda^2 &= \lambda \\
\Rightarrow \lambda(\lambda - 1) &= 0 \\
\therefore \lambda &= 0, 1
\end{aligned}$$

So, there is no inverse of projection operators.

Now,

$$[\eta_{\mu\lambda} \square - \partial_\mu \partial_\lambda] A^\lambda = j_\mu$$

We modify this to get around:

$$K_{\mu\lambda} K^{\lambda\sigma} = K_\mu^\sigma$$

This tells us that there are d.o.f which must be treated as *zero modes*.

Also,

$$\begin{aligned} [\sigma_i, \sigma_j] &= 2i\epsilon_{ijk} \sigma_k \quad \text{is not useful, but} \\ \left[\frac{\sigma_i}{2}, \frac{\sigma_j}{2} \right] &= i\epsilon_{ijk} \frac{\sigma_k}{2} \quad \text{is usefull} \end{aligned}$$

One way to circumvent:

$$\mathcal{L}_\lambda = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{\lambda}{2}(\partial_\mu A^\mu)^2$$

29.2 Finding the EOM

Variation of the 2nd term:

$$\begin{aligned} &\frac{\lambda}{2} \times 2(\partial_\mu A^\mu) \delta(\partial_\mu A) \\ &= \lambda(\partial_\mu A^\mu \partial_\alpha) \delta A^\alpha \\ &= \lambda(\partial^\nu \partial_x A^x) \delta A^\nu \end{aligned}$$

Now, the 1st term:

$$\begin{aligned} \partial_\mu F^{\mu\nu} &= j^\nu - \lambda \partial^\nu (\partial_\mu A^\mu) \\ \Rightarrow \partial_\mu (\partial^\mu A^\nu - \partial^\nu A^\mu) + \lambda \partial^\nu (\partial_\mu A^\mu) &= j^\nu \\ \Rightarrow \square A^\nu - \partial^\nu \partial_\mu A^\mu + \lambda \partial^\nu \partial_\mu A^\mu &= j^\nu \\ \Rightarrow \square A^\nu - (1 - \lambda) \partial^\nu \partial_\mu A^\mu &= j^\nu \end{aligned}$$

The choice $\lambda = 1$ is known as the *Feynman gauge*. If $\lambda \neq 0$, the operator is now invertible.

$$\lambda \rightarrow \infty \quad \text{Landau gauge}$$

Exercise: Find the propagator when $\lambda \neq 0$ or, 1.

29.3 Ward Identity

$$\begin{aligned} \Gamma[A] &\rightarrow \Gamma[A + D_\mu A] \\ &= \Gamma[A] + \frac{\delta \Gamma}{\delta A^\mu} D_\mu \epsilon + \dots \end{aligned}$$

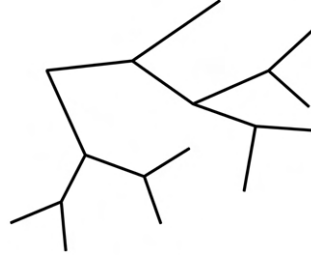
This must be zero if effective action is invariant under gauge transformation.

$$\Gamma[A] = \Gamma[A] - D_\mu \left[\frac{\delta \Gamma}{\delta A^\mu} \right] \epsilon = j^\mu = 0$$

- In 1st quantization, particle number is fixed.
- The 2nd quantization = number representation

Their difference is the loop. In a loop, the particle number is not fixed.

$$(\square + m^2)\phi = g\phi^3 \quad \text{and} \quad \phi = \phi_0 + k\phi^3$$

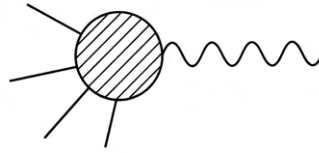


This is a tree diagram. No loop is present.

This type of diagram is calculated by **MADGRAPH**.

29.4 Scattering Amplitude

$$M \sim \epsilon^\mu T_{\mu\nu\lambda}$$



$$A^\mu = \sum \epsilon_\mu e^{ik \cdot x} \hat{a}(\mathbf{k})$$

$$\phi(x) = \sum e^{i\vec{k} \cdot \vec{x}} \hat{a}(\mathbf{k})$$

(Gauge transformation)

$$\varepsilon_\mu \rightarrow \varepsilon_\mu + k_\mu \Lambda$$

$$A_\mu \rightarrow A_\mu + \partial_\mu \Lambda$$

(FT)

$$\boxed{k^\mu T_{\mu\nu\lambda} = 0}$$

Transversality is one of the consequences of Ward Identity.

If we want to write a propagator, we need to choose a gauge:

λ – is a Lagrange multiplier

Lesson

We need to add a gauge-fixing term to the action.

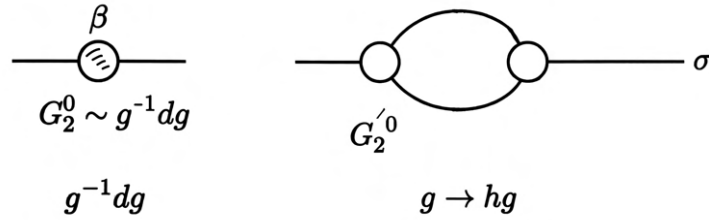
Naive path-integral:

$$Z = \int [\mathcal{D}A_\mu] e^{iS[A]}$$

$$V_A \equiv U A U^\dagger + \frac{1}{ig} U d U^\dagger$$

But under a gauge transformation: $A \rightarrow A^\mathcal{U}$

$$S[A^\mathcal{U}] = S[A]$$

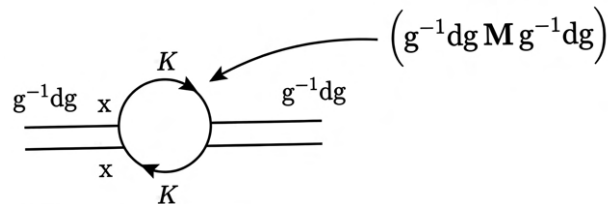


$$g^{-1} h^{-1} d(hg) = g^{-1} (h^{-1} dh) g + g^{-1} dg = g^{-1} [h^{-1} dh + dg g^{-1}] g$$

$$\begin{aligned} \text{Tr} \left(\partial_\mu g \partial^\mu g^{-1} \right) &= -\text{Tr} \left(g^{-1} dg \cdot g^{-1} dg \right) \\ &= -\text{Tr} \left[(dg g^{-1} + h^{-1} dh) (dg g^{-1} + h^{-1} dh) \right] \\ &= -\text{Tr} \left[dg g^{-1} \cdot dg g^{-1} \right] + \text{Tr} \left[h^{-1} dh \cdot h^{-1} dh \right] \\ &\quad + 2 \text{Tr} \left[dg g^{-1} \cdot h^{-1} dh \right] \end{aligned}$$

$$\begin{aligned} \text{Tr} \left(g^{-1} dg \right)^3 &= \text{Tr} \left(g_0^{-1} dg_0 \right)^3 + 3 \text{Tr} \left[dg g_0^{-1} \cdot h^{-1} dh \cdot h^{-1} dh \right] \\ &\quad + 3 \text{Tr} \left[dg_0 g_0^{-1} dg_0 g_0^{-1} h^{-1} dh \right] + \text{Tr} (h^{-1} dh)^3 \end{aligned}$$

$$h^{-1} dh = \left(1 - ix - \frac{i^2}{2} x^2 \right) (dx + x^2)$$

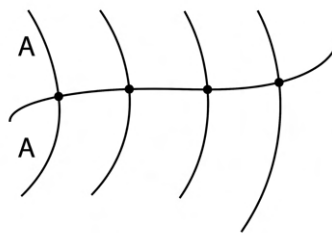


Find this:

$$\int d^2 k \frac{1}{k^2 (k+p)^2}$$

$$g^{-1} dg \quad \text{is Sukawa's Original Construction}$$

$$Z = \int [\mathcal{D}A_\mu] e^{iS[A] + \lambda F[A]} \quad [\lambda F[A] = \text{gauge fixing}]$$



fiber bundles

$$\int \delta(\quad) dy = 1$$

$\uparrow$
 $\det M$

$$\det M = \int d\eta \, d\bar{\eta} \, e^{\bar{\eta}^m \eta} \quad [\text{Ghost Field}]$$

(Class-30 (Online)) Lecture 27 : Fri, Nov 14, 2025

The Faddeev-Popov Procedure

We needed gauge fixing to obtain the propagator.

$$S(A) \longrightarrow S(A) + \lambda F[A]$$

$$Z(A) = \int \mathcal{D}A_\mu e^{iS[A]} \longrightarrow \int \mathcal{D}A_\mu \mathcal{D}\lambda e^{i(S[A] + \lambda F[A])}$$

$$Z(A) = \int \mathcal{D}A_\mu \delta(F[A]) e^{iS[A]}$$

This process removes the original manifest gauge invariance we initially had.

Fixing the gauge forces one to **pick** one element from the gauge-orbit.

Question: Can we do this insertion by retaining the gauge invariance? What is the problem?

$$A \rightarrow UAU^\dagger + \frac{1}{ig} U dU^\dagger = A^U$$

$$\begin{aligned} Z(A^U) &= \int \mathcal{D}(A^U) \delta(F[A^U]) e^{iS[A^U]} \\ &= \int \mathcal{D}A \delta(F[A^U]) e^{iS[A]} \neq Z(A) \end{aligned}$$

Here, $F[A]$ is not invariant. Then, how to resolve this?

Demand

$$\Delta_{\text{FP}}(A) \int \mathcal{D}U \delta(F[A^U]) = 1 \quad (\text{FP})$$

↑

(Faddeev-Popov Factor)

$$U = U(x)$$



**for compact
groups:**

$$D(u) = D(UU')$$

Fact:

$$\Delta_{\text{FP}}(A^U) = \Delta_{\text{FP}}(A)$$

Perform $A \rightarrow A^U$ in (FP):

$$\begin{aligned}
 1 &= \Delta_{\text{FP}}(A^U) \int \mathcal{D}U \delta(F[(A^U)^U]) \\
 &= \Delta_{\text{FP}}(A^U) \int \mathcal{D}U \delta(F[A^{UU'}]) \\
 &= \Delta_{\text{FP}}(A^U) \underbrace{\int \mathcal{D}(UU') \delta(F[A^{UU'}])}_{[\Delta_{\text{FP}}(A)]^{-1}} \\
 \therefore \Delta_{\text{FP}}(A^U) &= \Delta_{\text{FP}}(A)
 \end{aligned}$$

So, $\Delta_{\text{FP}}(A)$ is independent of the choice of U .

$$\begin{aligned}
 Z(A) &= \int \mathcal{D}A_\mu \underbrace{\Delta_{\text{FP}}(A) \int \mathcal{D}U \delta(F[A^U])}_{=1} e^{iS[A]} \\
 &= \int \mathcal{D}(A_\mu^{U^{-1}}) \Delta_{\text{FP}}(A^{U^{-1}}) \int \mathcal{D}U \delta(F[A^{U^{-1}U}]) e^{iS[A^{U^{-1}}]} \\
 &= \int \mathcal{D}A_\mu \underbrace{\Delta_{\text{FP}}(A)}_{\text{decoupled from the rest}} \int \mathcal{D}U \delta(F[A]) e^{iS[A]} \\
 &\quad \text{(This allows one to decouple the action of } U\text{)} \\
 &= \underbrace{\int \mathcal{D}U}_{\text{decoupled from the rest}} \int \mathcal{D}A_\mu \Delta_{\text{FP}}(A) e^{iS[A]} \delta(F[A])
 \end{aligned}$$

Question: How do we compute $\Delta_{\text{FP}}(A)$?

Let us consider an infinitesimal transformation, $U = 1 + i\epsilon$. Then,

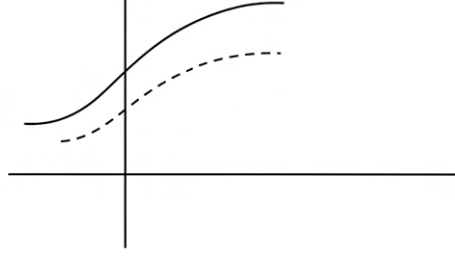
$$\begin{aligned}
 A \rightarrow A^U &= UA_\mu U^\dagger + \frac{1}{ig} U \partial_\mu U^\dagger \\
 &= A_\mu + i[\epsilon, A_\mu] - \frac{1}{ig} \partial_\mu \epsilon \\
 \Rightarrow \delta_\mu A &= -\frac{1}{g} (\partial_\mu \epsilon + ig[A_\mu, \epsilon]) = -\frac{1}{g} D_\mu \epsilon
 \end{aligned}$$

$$\begin{aligned}
 \Delta_{\text{FP}}(A) \int \mathcal{D}U \delta(F[A]) &= \Delta_{\text{FP}}(A) \int \mathcal{D}\epsilon \delta\left(F[A] + \frac{\delta F}{\delta A_\mu} \delta A_\mu\right) \\
 &= \Delta_{\text{FP}}(A) \int \mathcal{D}\epsilon \delta\left[\frac{\delta F}{\delta A_\mu} \left(-\frac{1}{g} D_\mu \epsilon\right)\right] \\
 &= \Delta_{\text{FP}}(A) \int \mathcal{D}\epsilon \delta\left[D_\mu \left(\frac{\delta F}{\delta A_\mu}\right) \epsilon\right] \\
 &= \Delta_{\text{FP}}(A) \frac{1}{\det\left[D_\mu \left(\frac{\delta F}{\delta A_\mu}\right)\right]} = 1
 \end{aligned}$$

For infinitesimal changes,

$$dU = i d\epsilon \quad \Rightarrow \quad d\mathcal{D}U = \mathcal{D}\epsilon$$

Original gauge choice: $F(A) = 0$



Recall:

$$\int \delta(\alpha x) dx = \frac{1}{\alpha}$$

$$\begin{aligned} \Delta_{\text{FP}}(A) &= \det \left(D_\mu \frac{\delta F}{\delta A_\mu} \right) \\ \Rightarrow Z_{\text{FP}}(A) &= \int \mathcal{D}A_\mu \det \left(D_\mu \frac{\delta F}{\delta A_\mu} \right) e^{iS[A]} \quad \left[M = D_\mu \frac{\delta F}{\delta A_\mu} \right] \\ &= \int \mathcal{D}A_\mu \mathcal{D}\bar{\eta} \mathcal{D}\eta e^{i[S(A) + \int \bar{\eta} M \eta]} \end{aligned}$$

$\eta, \bar{\eta}$ are Grassmann valued, they satisfy Fermi statistics.

Question: Why for QED ghosts do not matter?

Say, we pick the Lorentz gauge:

$$\partial_\mu A^\mu = 0 = F(A)$$

$$\begin{aligned} D_\mu \epsilon &= \partial_\mu \epsilon + ig[A_\mu, \epsilon] = \partial_\mu \epsilon \quad \left[\frac{\delta F}{\delta A_\mu} = \partial^\mu \right] \\ \therefore D_\mu \left(\frac{\delta F}{\delta A_\mu} \right) &= \partial_\mu \partial^\mu = \square \end{aligned}$$

Thus,

$$\begin{aligned} \bar{\eta} M \eta &= \bar{\eta} \square \eta \\ \Rightarrow \int \mathcal{D}A_\mu \mathcal{D}\eta \mathcal{D}\bar{\eta} e^{(iS[A] + \bar{\eta} M \eta)} &= \int \mathcal{D}A_\mu e^{iS[A]} \underbrace{\int \mathcal{D}\eta \mathcal{D}\bar{\eta} e^{i\bar{\eta} \square \eta}}_{\text{(independent of } A)} \end{aligned}$$

The η 's are spinless but anticommuting (Grassmann-valued).

There are gauges where even in non-Abelian theories were ghosts decouple. Example:

$$F(A) = \eta_\mu A^\mu = 0 \Rightarrow \text{Axial gauge} \quad \eta^\mu = \text{null vectors}$$

$$\frac{\delta F}{\delta A_\mu} = \eta^\mu 0 \Rightarrow M = 0$$

Anomalies in the Functional Integral

$$Z[\eta] = \int \mathcal{D}\psi \mathcal{D}\bar{\psi} e^{-S_E}$$

$$S_E = -\frac{1}{4}F^2 + \bar{\psi} (i\mathcal{D}) \psi$$

Under a chiral transformation,

$$\begin{aligned} \psi &\rightarrow \psi' \equiv e^{i\alpha\gamma_5}\psi \\ \bar{\psi} &\rightarrow \bar{\psi}' \equiv \bar{\psi} e^{i\alpha\gamma_5} \end{aligned}$$

Since $\psi, \bar{\psi}$ are Grassmann valued,

$$\mathcal{D}\psi \mathcal{D}\bar{\psi} \rightarrow \underbrace{\frac{1}{\det(e^{2i\alpha\gamma_5})}}_{\mathcal{F}} \mathcal{D}\psi \mathcal{D}\bar{\psi}$$

$\mathcal{F} \rightarrow$ to include action, need an exponential

i.e.

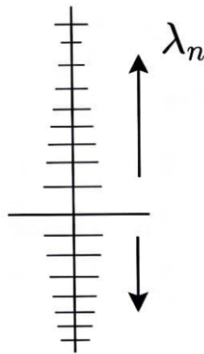
$$\mathcal{F} = e^{-\ln \det(e^{2i\alpha\gamma_5})} = e^{-\text{Tr} \ln(e^{2i\alpha\gamma_5})} \cong e^{-\text{Tr} \ln(1+2i\alpha\gamma_5+\dots)}$$

Let's try to evaluate this expression in the 1st order term of $\alpha(x)$ as we allow for de-
pending,

$$\text{Tr}(0) = \int \mathcal{D}^4x \langle x|0|x \rangle = \int d^4x \langle x|\eta' \rangle \langle n|0|m \rangle \langle m|x \rangle$$

The $\text{Tr}(\gamma_5)$ term is divergent / ill-defined

$$\gamma_5 \text{ eigenvalues} = \pm 1$$



$$\mathcal{D}\psi_n = \lambda_n \psi_n \quad \leftarrow \text{unbounded}$$

$\therefore \mathcal{D}$ also contains the chiral gauge field

Under a gauge transformation, the eigenvalues of the Dirac operator change (“flow”).

We need to regulate this. One way to do it is the *heat kernel* method. It was employed by Fujikawa to get the right answer.

$$\text{Tr} \left(\gamma_5 e^{-\mathcal{D}^2/\Lambda^2} \right) \rightarrow \text{symmetric in the +ve and -ve eigenvalues}$$

But,

$$\begin{aligned}\not{D} &= \not{\partial} - ie\not{A} = \gamma^\mu D_\mu \\ \Rightarrow \not{D}\not{D} &= \gamma^\mu \gamma^\nu D_\mu D_\nu\end{aligned}$$

We know,

$$\begin{aligned}[\gamma^\mu, \gamma^\nu] &= 2i\sigma^{\mu\nu} \\ \{\gamma^\mu, \gamma^\nu\} &= 2\eta^{\mu\nu} \\ \therefore \gamma^\mu \gamma^\nu &= \eta^{\mu\nu} + i\sigma^{\mu\nu} \\ \therefore \not{D}\not{D} &= (\eta^{\mu\nu} + i\sigma^{\mu\nu}) D_\mu D_\nu \\ &= D^2 + \frac{i}{2}\sigma^{\mu\nu}[D_\mu, D_\nu] \\ &= D^2 + \frac{i}{2}\sigma^{\mu\nu}F_{\mu\nu}(-ie) \\ \therefore \not{D}^2 &= D^2 + \frac{e}{2}\sigma^{\mu\nu}F_{\mu\nu}\end{aligned}$$

Also,

$$D_\mu D_\nu = \frac{1}{2}\{D_\mu, D_\nu\} + \frac{1}{2}[D_\mu, D_\nu]$$

$$\begin{aligned}\text{Tr}\left(\gamma_5 e^{-\not{D}^2/\Lambda^2}\right) &= \text{Tr}\left[\gamma_5 e^{-\frac{1}{\Lambda^2}(D^2 + \frac{e}{2}\sigma^{\mu\nu}F_{\mu\nu})}\right] \\ &= \text{Tr}\left[\gamma_5 e^{-\frac{D^2}{\Lambda^2}} e^{-\frac{e}{2\Lambda^2}\sigma^{\mu\nu}F_{\mu\nu}}\right] \\ &= \text{Tr}\left[\gamma_5 e^{-\frac{D^2}{\Lambda^2}} \left\{1 - \frac{e}{2\Lambda^2}\sigma^{\mu\nu}F_{\mu\nu} + \frac{1}{2!}\left(-\frac{e}{2\Lambda^2}\right)^2(\sigma^{\mu\nu}F_{\mu\nu})(\sigma^{\alpha\beta}F_{\alpha\beta}) + \dots\right\}\right] \\ &= \text{Tr}(\gamma_5 e^{-D^2/\Lambda^2}) - \frac{e}{2\Lambda^2}\text{Tr}\left(\gamma_5 e^{-D^2/\Lambda^2}\sigma^{\mu\nu}F_{\mu\nu}\right) \\ &\quad + \frac{1}{2!}\text{Tr}\left[\gamma_5 e^{-D^2/\Lambda^2}\sigma^{\mu\nu}\sigma^{\alpha\beta}F_{\mu\nu}F_{\alpha\beta}\right] + \dots\end{aligned}$$

In 4D, $\text{Tr}(\gamma_5) = 0$: any non-vanishing answer for trace,

$$\text{Tr}(\gamma_5 \gamma^\mu \gamma^\nu \gamma^\alpha \gamma^\beta) = 4i \epsilon^{\mu\nu\alpha\beta}$$

From the infinite series, we pick up only one term. Thus, in 4D, only contributions (Abelian) come from,

$$\begin{aligned}&\frac{1}{2!}\left(\frac{e}{2\Lambda^2}\right)^2 \text{Tr}\left(\gamma_5 e^{-D^2/\Lambda^2}\sigma^{\mu\nu}\sigma^{\alpha\beta}\right)F_{\mu\nu}F_{\alpha\beta} \\ &= \frac{e^2}{8\Lambda^4} \int d^4x e^{-D^2/\Lambda^2} \text{Tr}\left(\gamma_5 \sigma^{\mu\nu}\sigma^{\alpha\beta}\right)F_{\mu\nu}(x)F_{\alpha\beta}(x) \\ &= \frac{e^2}{8\Lambda^4} \int d^4x e^{-(\partial - i\bar{e}A)^2} \left(-\frac{1}{4}i\epsilon^{\mu\nu\alpha\beta}\right)F_{\mu\nu}(x)F_{\alpha\beta}(x)\end{aligned}$$

The lowest non-zero contribution, as we can see, is already at the $\bar{e}^2$ level. At this level,

$$-i\bar{e}^2 \int d^4x e^{-\partial^2/\Lambda^2} \epsilon^{\mu\nu\alpha\beta} F_{\alpha\beta}(x)F_{\mu\nu} F_{\alpha\beta}$$

Since we are calculating a trace, it is independent of the choice of basis. In that case, in Fourier mode,

$$\begin{aligned} & \frac{\bar{e}^2}{8\lambda^4} \frac{1}{(2\pi)^4} \int d^4k e^{-k^2/\Lambda^2} \quad \left[\frac{k}{\lambda} \text{ is a dimensionless quantity} \right] \\ &= \frac{(2\pi)^2}{(2\lambda)^4} \int_0^\infty k^3 dk e^{-k^2/\Lambda^2} \end{aligned}$$

$$\text{Let, } k^2 = y$$

$$\Rightarrow k dk = \frac{dy}{2}$$

$$\Rightarrow d^4k = k^3 dk d(2\pi)^2$$

$$\text{And, } \frac{y}{\Lambda^2} = t \Rightarrow t = y\Lambda^2$$

$$\begin{aligned} \therefore & \frac{(2\pi)^2}{(2\lambda)^4} \int_0^\infty k^3 dk e^{-k^2/\Lambda^2} \\ &= \frac{(2\pi)^2}{(2\lambda)^4} \frac{1}{2} \int_0^\infty y dy e^{-y/\Lambda^2} \\ &= \Lambda^4 + \int_0^\infty t dt e^{-t} \\ &= \Lambda^4 \end{aligned}$$

The factor Λ^4 from the integral cancels the expansion parameter Λ in the denominator! This shows that the result is independent of the regulator.

The end result is:

$$\begin{aligned} Z[A] &= \int \mathcal{D}\psi \mathcal{D}\bar{\psi} e^{-S} \quad \text{is unchanged by} \\ \psi &\rightarrow e^{i\alpha\gamma_5}\psi \end{aligned} \tag{C1}$$

This action picks up a factor:

$$\begin{aligned} S &\rightarrow S + \partial_\mu \alpha j^\mu \rightarrow S + \alpha \underbrace{\partial_\mu j^\mu} \\ &= 0 \quad [\text{if we have invariance under (C1)}] \end{aligned}$$

But we just saw that,

$$Z \rightarrow \int \mathcal{D}\psi \mathcal{D}\bar{\psi} e^{i\alpha(F \wedge F)} e^{i\alpha(\partial J)}$$

So,

$$\partial \cdot J \neq 0 \quad \text{but } \alpha \text{ to (C)} \quad F \wedge F \text{ is independent of } \Lambda$$